

High-Resolution Spectroscopy of Metastable-State Transitions in Ytterbium

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中央研究院
ACADEMIA SINICA

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Preface

Precision spectroscopy plays an important role in modern atomic physics and provides essential tools for laser cooling, optical frequency standards, and precision measurements. Owing to its rich electronic structure and multiple stable isotopes, neutral ytterbium offers an ideal platform for investigating atomic transitions, isotope shifts, and hyperfine interactions.

The work presented in this thesis was motivated by the development of a metastable-manifold cooling platform for ytterbium atoms and the need for reliable spectroscopic references. Several optical transitions were investigated using modulation transfer spectroscopy and wavelength-controlled frequency measurements. Through isotope-selective optical pumping and hyperfine-structure analysis, isotope shifts and hyperfine transition frequencies were systematically characterized.

In addition to experimental spectroscopy, King-plot analyses were performed to examine the consistency among isotope-shift measurements from different transitions. Interactive optical-system modeling tools and AI-assisted spectroscopic analysis methods were also developed to facilitate experimental design and spectral interpretation.

The remainder of this thesis is organized as follows. Chapter 1 introduces the role of precision spectroscopy in the experimental platform. Chapters 2 and 3 present the theoretical background and spectroscopy principles. Chapters 4 through 8 describe the experimental investigations and isotope-shift analyses. Chapters 9 through 10 introduce the optical-system modeling platform, and AI-assisted spectroscopy tools. Finally, Chapter 11 summarizes the main conclusions and discusses future prospects.

Hao-Rong, Yang
IAMS Academia Sinica, July 2026

Acknowledgments

Looking back, I feel fortunate that my first encounter with atomic physics began in Lab110 at the Institute of Atomic and Molecular Sciences (IAMS), Academia Sinica. Over 2 years, I have learned not only experimental techniques and methods of analysis, but also a new way of seeing the world and appreciating the beauty of nature.

Whenever I look up at the night sky, I cannot help but feel how small we are in the vastness of the universe. Yet what fascinates me most is that physicists devote themselves to studying atoms and quantum systems, objects roughly ten orders of magnitude smaller than ourselves. Invisible to the naked eye, these tiny building blocks nevertheless govern the world around us. There is something beautiful about the fact that the grandeur of the cosmos and the subtlety of the quantum world are both ruled by the same laws of nature. Perhaps humanity will never fully understand quantum mechanics within my lifetime. Nevertheless, being able to touch the quantum world through my own experiments and understand nature little by little has already brought me immense joy.

I would like to express my sincere gratitude to Dr. Chun-Chia Chen and Dr. Shayne Bennetts. When I first joined the laboratory, I had almost no background in atomic physics. Yet, you welcomed me with patience and trust, guided me into this fascinating field, and inspired in me an even deeper passion for physics. I would also like to thank all the members of the laboratory for creating such a pleasant and supportive environment throughout these years. My gratitude also goes to my family, especially my beloved cats, whose unconditional companionship and support have always been my greatest source of comfort and strength.

In a universe that is 13.8 billion years old and spans billions of light-years, atoms come together to form molecules, molecules give rise to life, and life somehow becomes capable of contemplating the very laws that brought it into existence. That, in itself, is something close to a miracle. Perhaps we will never find all the answers within our lifetimes. Yet we continue to ask questions, to explore, and to seek understanding. To have been able to take part in this journey, however briefly, is already a tremendous blessing, and for that, I am profoundly grateful.

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1

Precision Spectroscopy in the Experimental Platform

1.0.1 In This Experiment

In the present experiment, precision laser spectroscopy is employed to establish a consistent spectroscopic framework for ytterbium atoms across several optical transitions. By performing high-resolution measurements of multiple transitions, we determine the transition center frequencies and resolve the hyperfine structures of fermionic isotopes. From these measurements, the hyperfine interaction constants (A and B coefficients) and isotope shifts are extracted in a self-consistent manner.

These spectroscopic parameters provide the quantitative basis required for constructing an experimentally validated energy-level diagram. Such a diagram is essential for identifying the correct resonance conditions of the cooling, repumping, and state-manipulation transitions involved in the experimental platform. In particular, precise knowledge of the transition frequencies ensures reliable laser frequency stabilization and allows consistent calibration among different laser systems used in the experiment.

Furthermore, accurate isotope-shift measurements enable cross-validation between different transitions through King-plot analysis. This provides an additional internal consistency check for the measured transition frequencies and helps confirm the reliability of the extracted spectroscopic parameters. The resulting spectroscopic dataset therefore forms a coherent frequency reference system for the entire experimental platform, enabling stable operation and reproducible

control of the atomic states.

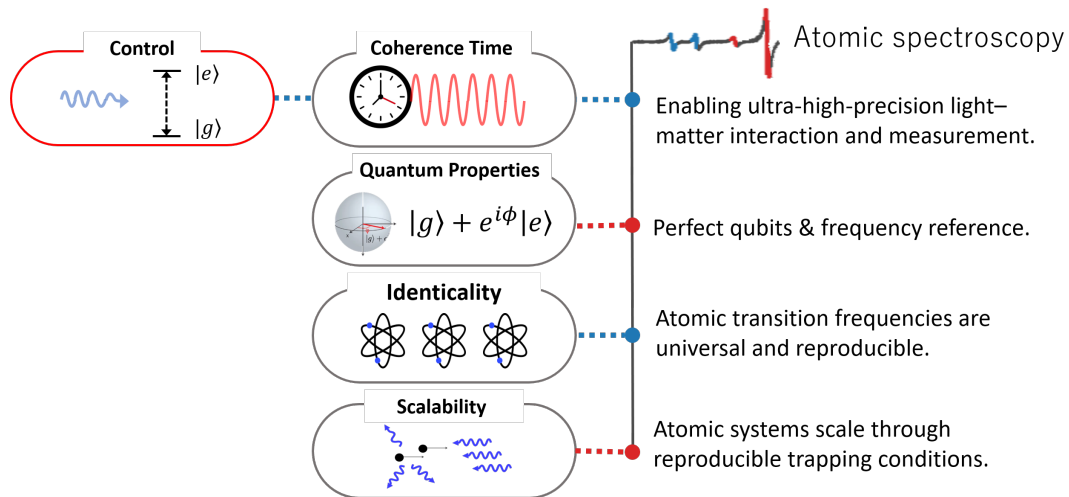


Figure 1.1: Conceptual diagram of the performance criteria for a controllable quantum two-level platform. A control field drives coherent transitions between g and e , enabling state preparation and manipulation. Long coherence time is required to preserve the relative phase ϕ in the superposition state $g + e^{i\phi}e$. Identical quantum systems ensure uniform transition frequencies and dynamical responses across particles, supporting reproducible operations. Scalability emphasizes the ability to extend coherent control from a single qubit to larger ensembles or multi-qubit architectures.

1.0.2 Other Semiconductor-Related Applications

Beyond atomic physics experiments, spectroscopic techniques also play an important role in modern semiconductor manufacturing processes. In particular, optical emission spectroscopy (OES) is widely used for monitoring plasma-assisted processes such as reactive ion etching and plasma-enhanced deposition.

During plasma etching, energetic electrons excite atoms, ions, and radicals within the plasma, causing them to emit characteristic spectral lines. By analyzing the spectral distribution and intensity of this emission, information about the chemical composition, plasma density, and reaction dynamics inside the chamber can be obtained in real time. An example of an OES spectrum used for plasma species identification is shown in Fig. 1.2(b), where characteristic emission peaks correspond to different atomic or molecular species present in the plasma.

Spectroscopy is also commonly applied for endpoint detection in etching processes. As the material being etched changes—for example when a dielectric layer is completely removed and the underlying silicon layer becomes exposed—the emission intensity of specific spectral lines can exhibit distinct temporal variations. Monitoring the time evolution of the OES signal therefore enables accurate identification of the etching endpoint, thereby improving fabrication precision and preventing over-etching, as illustrated in Fig. 1.2(a).

These examples illustrate that spectroscopy serves not only as a fundamental research tool in atomic physics but also as a practical diagnostic technique in advanced technological industries.

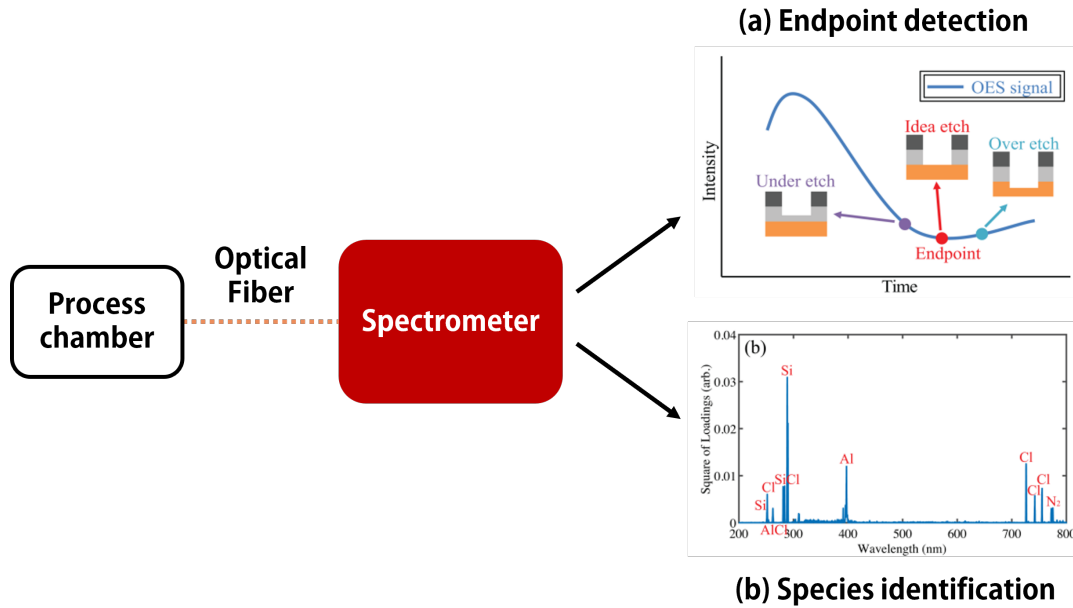


Figure 1.2: Schematic illustration of optical emission spectroscopy (OES) used in plasma-assisted semiconductor processing. Light emitted from the plasma in the process chamber is collected through an optical fiber and analyzed by a spectrometer. The spectroscopic information can be used for (a) endpoint detection by monitoring the temporal evolution of the OES signal during etching, and (b) plasma species identification through characteristic emission lines in the emission spectrum. Subfigure (a) is adapted from Fig. 1 of Ref. [1], and subfigure (b) is adapted from Fig. 5(b) of Ref. [1].

1.1 Metastable-State Cooling and Continuous Operation Scheme

Building upon the precision spectroscopic framework established in the previous section, we have obtained accurate transition center frequencies, isotope shifts, and hyperfine constants that define a practically operable energy-level map and frequency coordinate system. It is precisely this experimentally validated spectroscopic foundation that enables the design and implementation of a multi-transition cooling and manipulation architecture. Without reliable knowledge of resonance conditions and hyperfine splittings, population transfer across different manifolds and frequency stabilization of multiple lasers would lack predictability and reproducibility. The operational scheme presented below therefore represents a direct extension and application of the precision spectroscopic results.

The operational concept of the platform is illustrated in Fig. 1.3. In the first stage,

atoms are captured and precooled on the strong $^1S_0 \rightarrow ^1P_1$ transition, which provides a large scattering rate and efficient initial cooling. Population is subsequently transferred out of the ground-state cycling transition through optical pumping processes involving intermediate states, leading to accumulation in the long-lived 3P_2 metastable manifold.

The 3P_2 state is chosen as the basis for further manipulation because of its extended lifetime, well-resolved hyperfine structure, and distinct transition pathways to the 3D_J manifold. These properties make it a suitable platform for implementing additional cooling and trapping mechanisms that are not strictly dependent on repeated decay back to the ground state. By coupling 3P_2 to appropriate 3D_J states, a second-stage cooling configuration can be realized within the metastable manifold.

This architecture reduces the exclusive reliance on ground-state cycling transitions and opens a pathway toward continuous or quasi-continuous operation of the atomic system. The physical feasibility of this scheme, the criteria for selecting specific transitions, and the associated measurement strategies required to support it will be discussed in detail in the following chapters.

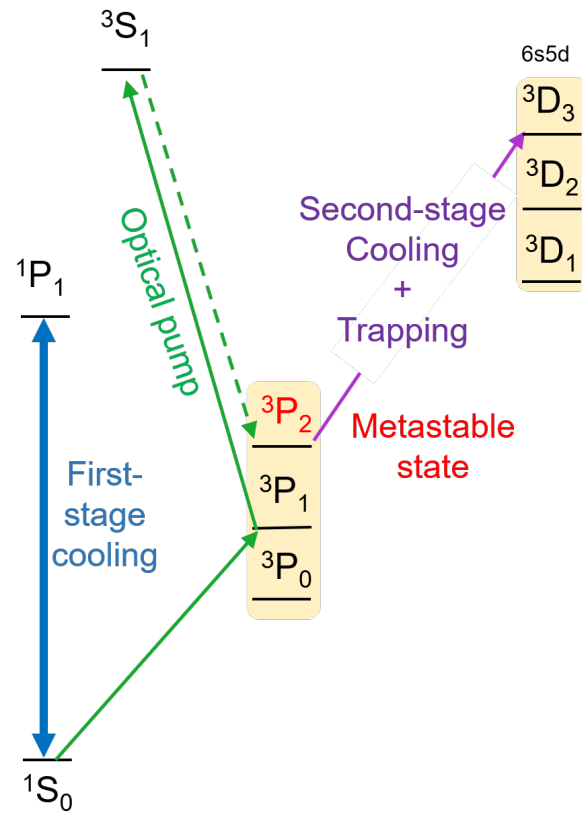


Figure 1.3: Energy-level structure and operational concept of the metastable-state cooling scheme. Atoms are first cooled on the $1S_0 \rightarrow 1P_1$ transition. Optical pumping via intermediate states transfers population to the long-lived $3P_2$ metastable state. Coupling between the $3P_2$ and $3D_J$ manifolds enables second-stage cooling and trapping within the metastable configuration, forming the basis for continuous or extended atomic operation.

2

Why Ytterbium?

2.1 A Comparative Analysis with Strontium

Ytterbium (Yb) and strontium (Sr) are two widely used atomic species in laser cooling and precision measurement experiments. Both elements possess ultranarrow clock transitions and are therefore suitable candidates for optical frequency standards. Despite these similarities, their cooling transitions and atomic structures exhibit several important differences that motivate our choice of ytterbium.

2.1.1 Atomic and Spectroscopic Advantages

Higher Scattering Rate and Capture Efficiency A key practical difference appears already in the narrow-line cooling stage. The natural linewidth Γ of the $^1S_0 \rightarrow ^3P_1$ intercombination transition is approximately 180 kHz in Yb, whereas it is much narrower (~ 7 kHz) in Sr. Under comparable saturation conditions, the larger linewidth in ytterbium permits a higher maximum photon scattering rate and hence a stronger attainable radiative force. This facilitates a broader effective velocity capture range in the narrow-line MOT and improves the overall capture efficiency.

Atomic Mass and Recoil Considerations Atomic mass also plays an important role in laser cooling. Ytterbium is heavier (170–176 u) than strontium (84–88 u). The recoil velocity per scattered photon is therefore smaller in Yb, leading to reduced momentum diffusion and potentially more stable cooling dynamics.

Magnetic-Field Sensitivity and Zeeman Structure The Zeeman shift is given by

$$\Delta E = g_F \mu_B B m_F,$$

where ΔE is the Zeeman energy shift, g_F is the hyperfine Landé g -factor, μ_B is the Bohr magneton, B is the applied magnetic field, and m_F is the magnetic quantum number corresponding to the projection of the total angular momentum F along the quantization axis.

The hyperfine Landé g -factor is given by

$$g_F = g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)},$$

where g_J is the Landé g -factor of the electronic state, F is the total angular momentum quantum number, J is the total electronic angular momentum quantum number, and I is the nuclear spin quantum number.

For ^{171}Yb ($J = 1$, $I = 1/2$), the allowed hyperfine states are $F = 3/2$ and $1/2$. For ^{87}Sr ($J = 1$, $I = 9/2$), the hyperfine states are $F = 11/2$, $9/2$, and $7/2$. The smaller number of hyperfine manifolds in ^{171}Yb simplifies operation in residual magnetic fields and reduces magnetic sensitivity in precision experiments.

Electronic Configuration and Inner-Shell Opportunities The electronic configurations are

- Yb: $[\text{Xe}] 4f^{14}6s^2$
- Sr: $[\text{Kr}] 5s^2$

The filled $4f$ shell in Yb enables access to inner-shell excitations such as $4f^{13}5d6s^2$ configurations. These transitions can exhibit narrow linewidths and high quality factors Q , which are advantageous for precision spectroscopy and timekeeping.

2.1.2 Isotopic Properties and Hyperfine Structure

Isotopic Abundance and Suitability for Isotope-Shift Studies Yb possesses seven stable isotopes with relatively balanced natural abundances, including both bosonic and fermionic species. This rich isotopic diversity provides an ideal platform for isotope-shift measurements and multidimensional King-plot analyses.

In contrast, Sr contains only four stable isotopes, with the majority population concentrated in ^{88}Sr .

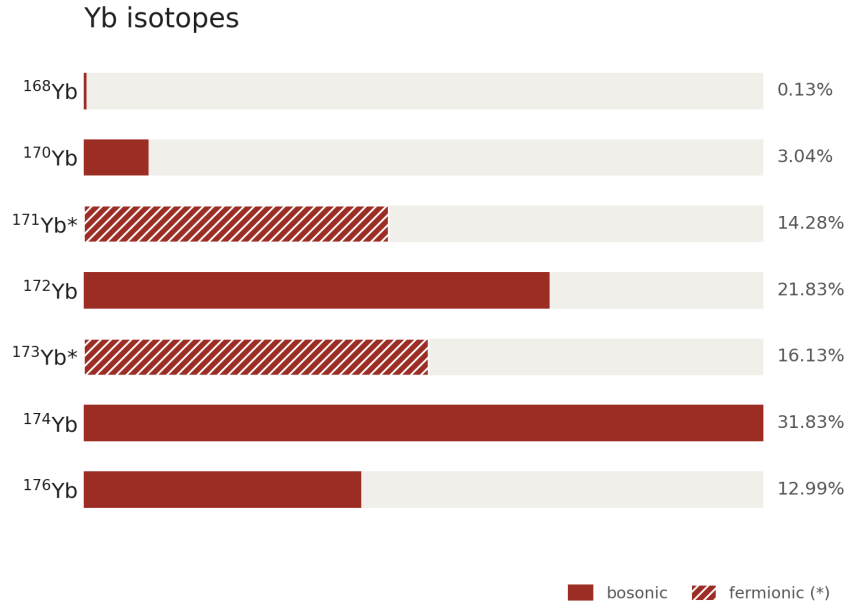


Figure 2.1: Natural abundance distribution of stable Yb isotopes. The relatively balanced abundance distribution allows multiple isotope pairs to be simultaneously observed in spectroscopy experiments.

The broader isotope distribution of Yb significantly improves the robustness of isotope-shift analyses and provides more isotope combinations for King-plot investigations compared with Sr.

Fermionic Isotopes and Hyperfine Simplicity Hyperfine structure plays a central role in state preparation and quantum manipulation. Strontium offers only one stable fermionic isotope, ^{87}Sr ($I = 9/2$), whose relatively large nuclear spin produces many hyperfine and Zeeman sublevels.

Ytterbium, on the other hand, provides two fermionic isotopes, ^{171}Yb ($I = 1/2$) and ^{173}Yb ($I = 5/2$). In particular, the simple nuclear spin of ^{171}Yb results in significantly fewer hyperfine levels and optical transition channels, greatly simplifying optical pumping, state preparation, and coherent manipulation.

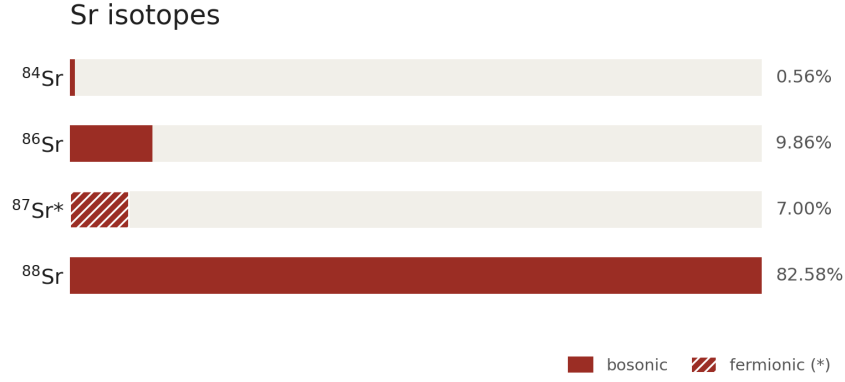


Figure 2.2: Natural abundance distribution of stable Sr isotopes. Compared with Yb, the isotope population is strongly dominated by ^{88}Sr .

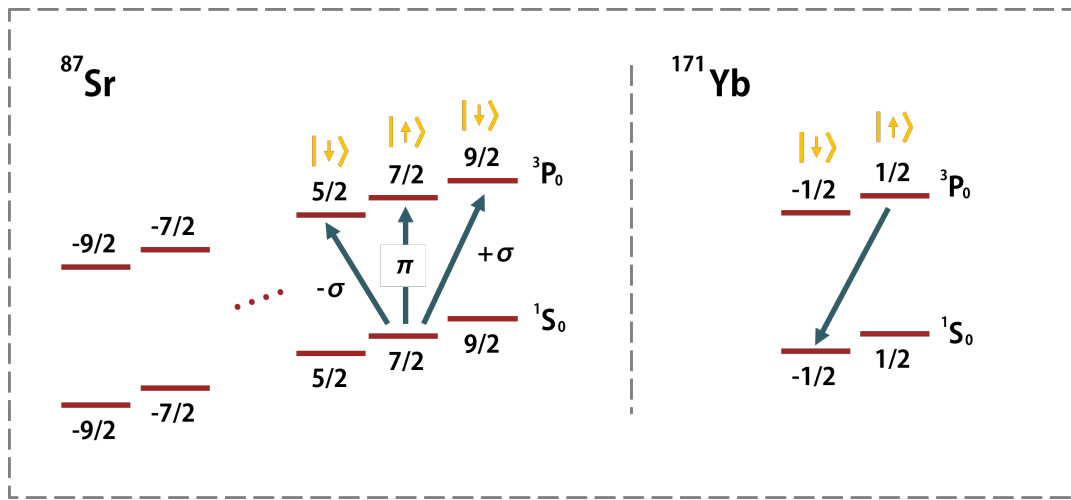


Figure 2.3: Comparison of hyperfine and Zeeman sublevel structures relevant for quantum-state control. Atoms with large nuclear spin, such as ^{87}Sr ($I = 9/2$), exhibit many hyperfine and Zeeman sublevels, resulting in multiple optical transition pathways and increased complexity in state preparation. In contrast, atoms such as ^{171}Yb with nuclear spin $I = 1/2$ possess a much simpler level structure, enabling more controllable optical transitions and facilitating high-fidelity quantum-state manipulation.

Relative Transition Strengths Besides the hyperfine energy structure, the relative strengths of the allowed hyperfine transitions are also of practical importance. These transition probabilities are determined by the Wigner-Eckart theorem and are proportional to the square of the corresponding Wigner 6- j coefficients.

Figure 2.4 summarizes the calculated relative transition strengths for the three optical transitions investigated in this thesis, namely the 493 nm ($^3\text{P}_2 \rightarrow 6s6d^3\text{D}_3$), 680 nm ($^3\text{P}_1 \rightarrow ^3\text{S}_1$), and 1798 nm ($^3\text{P}_2 \rightarrow 6s5d^3\text{D}_3$) transitions. The listed fractions represent the normalized transition strengths among all allowed hyperfine components for ^{171}Yb and ^{173}Yb .

These calculated strengths provide useful guidance for assigning experimen-

tally observed hyperfine spectra. Stronger transitions are generally expected to produce larger signal amplitudes in spectroscopy measurements, while weaker components may be difficult to resolve because of limited signal-to-noise ratio or spectral overlap. Consequently, the Wigner-coefficient calculations serve as an important reference throughout the hyperfine analyses presented in the following chapters.

493 nm ($^1S_0 - ^1P_1$) $J = 2 \rightarrow J' = 3$			680 nm ($^3P_1 - ^3S_1$) $J = 1 \rightarrow J' = 1$			1798 nm ($^3P_2 - ^3D_3$) $J = 2 \rightarrow J' = 3$		
^{173}Yb ($I = 5/2$)			^{173}Yb ($I = 5/2$)			^{173}Yb ($I = 5/2$)		
F	F'	Fraction (%)	F	F'	Fraction (%)	F	F'	Fraction (%)
0.5	0.5	3.7037	1.5	1.5	6.6667	0.5	0.5	3.7037
0.5	1.5	2.9630	1.5	2.5	15.5556	0.5	1.5	2.9630
1.5	0.5	1.0582	2.5	1.5	15.5556	1.5	0.5	1.0582
1.5	1.5	5.4180	2.5	2.5	1.9048	1.5	1.5	5.4180
1.5	2.5	6.8571	2.5	3.5	15.8730	1.5	2.5	6.8571
2.5	1.5	1.1429	3.5	2.5	15.8730	2.5	1.5	1.1429
2.5	2.5	6.6122	3.5	3.5	28.5714	2.5	2.5	6.6122
2.5	3.5	12.2449				2.5	3.5	12.2449
3.5	2.5	0.8163				3.5	2.5	0.8163
3.5	3.5	6.4500				3.5	3.5	6.4500
3.5	4.5	19.4004				3.5	4.5	19.4004
4.5	3.5	0.3527				4.5	3.5	0.3527
4.5	4.5	4.4092				4.5	4.5	4.4092
4.5	5.5	28.5714				4.5	5.5	28.5714
^{171}Yb ($I = 1/2$)			^{171}Yb ($I = 1/2$)			^{171}Yb ($I = 1/2$)		
F	F'	Fraction (%)	F	F'	Fraction (%)	F	F'	Fraction (%)
1.5	2.5	40.0000	0.5	0.5	22.2222	1.5	2.5	40.0000
2.5	2.5	2.8571	0.5	1.5	11.1111	2.5	2.5	2.8571
2.5	3.5	57.1429	1.5	0.5	11.1111	2.5	3.5	57.1429
			1.5	1.5	55.5556			

Figure 2.4: Calculated relative hyperfine transition strengths for the 493 nm, 680 nm, and 1798 nm transitions in ^{171}Yb and ^{173}Yb . The transition fractions are normalized according to the squared Wigner 6- j coefficients and indicate the expected relative intensities of the allowed hyperfine transitions.

The overall isotope properties of Yb and Sr are summarized in Table 2.1.

Table 2.1: Comparison of isotope properties between Yb and Sr.

Isotope Property	Yb	Sr
Stable isotopes	7	4
Fermionic species	2	1
Bosonic species	5	3
Most abundant isotope	^{174}Yb (31.83%)	^{88}Sr (82.58%)
Suitability for isotope-shift studies	Excellent	Moderate

2.1.3 Summary and Motivation for This Work

Strontium retains certain advantages. Its extremely narrow $^1S_0 \rightarrow ^3P_1$ linewidth enables lower Doppler temperatures,

$$T_D = \frac{\hbar\Gamma}{2k_B},$$

and the larger nuclear spin of ^{87}Sr provides additional internal degrees of freedom for certain quantum-control protocols. Nevertheless, when considering scattering efficiency, isotopic richness, hyperfine simplicity, and the versatile electronic structure of ytterbium, Yb offers a balanced combination of practical and spectroscopic advantages. Its larger scattering rate facilitates efficient laser cooling. Furthermore, the rich isotopic composition enables extensive isotope-shift and King-plot studies, while the relatively simple hyperfine structure of ^{171}Yb simplifies state preparation and coherent manipulation. For these reasons, ytterbium was selected as the atomic species employed throughout this work.

Figure 2.5 illustrates the laser-cooling and spectroscopy system developed in our laboratory. The system incorporates several optical transitions serving different experimental purposes, including the 556 nm narrow-line cooling transition, the 578 nm optical clock transition, the 680 nm and 770 nm repumping transitions, the 1798 nm transition for the metastable magneto-optical trap (MOT), and the 493 nm imaging transition. Together, these transitions provide a versatile platform for laser cooling, precision spectroscopy, and isotope-selective manipulation of ytterbium atoms.

Among these transitions, the primary focus of this thesis is the development and spectroscopic investigation of the 680 nm, 1798 nm, and 493 nm systems. The 680 nm transition is employed as the repumping transition that transfers atoms into the metastable 3P_2 state, the 1798 nm transition provides the cooling transition for the metastable MOT, and the 493 nm transition serves as the imaging transition for 3P_2 state fluorescence detection. These three transitions constitute the core of the metastable cooling scheme and form the basis of the isotope-shift measurements, hyperfine spectroscopy, and King-plot studies presented in the following chapters.

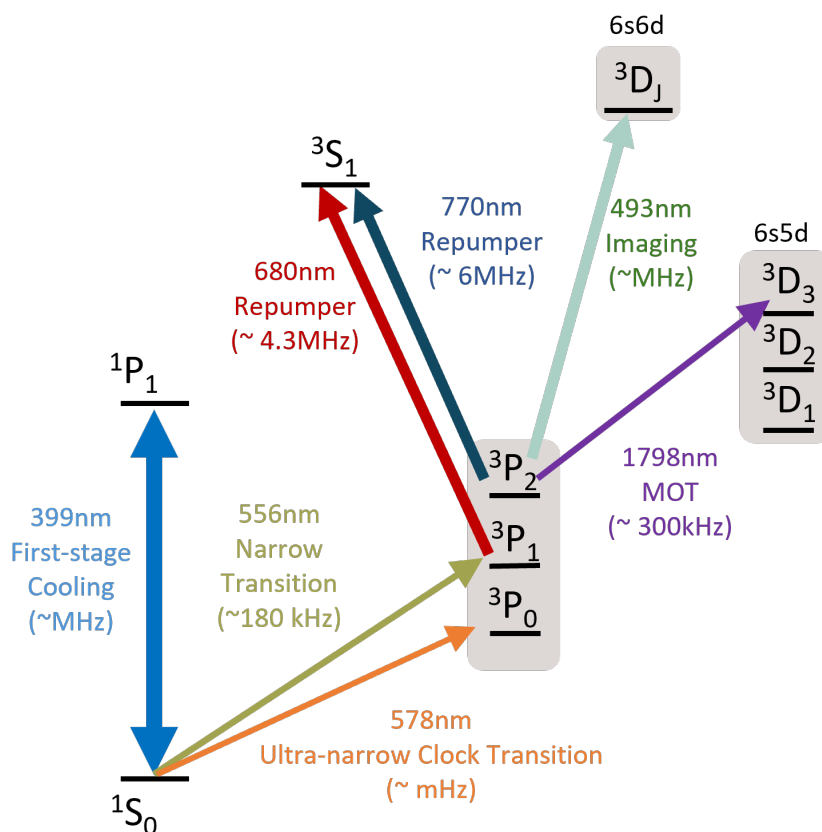


Figure 2.5: Simplified energy-level diagram of the ytterbium laser-cooling and spectroscopy system implemented in our laboratory.

3

Modulation Transfer Spectroscopy (MTS)

3.1 Introduction

High-resolution laser spectroscopy requires an error signal that is suitable for laser frequency stabilization. Ideally, such a signal should provide Doppler-free spectral resolution, a high signal-to-noise ratio (SNR), a well-defined zero crossing, and minimal background offset. However, no single spectroscopic technique simultaneously satisfies all of these requirements. As a result, several Doppler-free spectroscopic methods have been developed, each offering different advantages.

Saturated absorption spectroscopy (SAS) is one of the most widely used Doppler-free techniques. By employing counter-propagating pump and probe beams, SAS selectively interacts with atoms whose longitudinal velocity is close to zero, thereby eliminating Doppler broadening and producing narrow sub-Doppler resonances. However, the resulting signal is superimposed on a Doppler background and exhibits an absorption profile, making it less suitable for direct laser frequency stabilization.

To improve the signal quality, frequency modulation spectroscopy (FMS) introduces phase modulation using an electro-optic modulator (EOM). The modulated optical signal is subsequently demodulated to generate a dispersive error signal with a significantly improved SNR. Since the modulation is applied before the laser interacts with the atoms, the photodetector continuously receives an RF-modulated signal, even when the laser is far from resonance. Consequently, residual amplitude modulation (RAM), optical interference, and electronic offsets may

introduce a non-zero background into the demodulated signal.

To overcome these limitations, this work employs modulation transfer spectroscopy (MTS). Similar to SAS, MTS provides Doppler-free spectroscopy through counter-propagating pump and probe beams. At the same time, it adopts the phase modulation and synchronous demodulation techniques used in FMS to obtain a dispersive error signal with a high SNR. More importantly, unlike FMS, only the pump beam is phase-modulated, while the probe beam remains unmodulated before interacting with the atomic medium. As a result, a demodulated signal is generated only in the vicinity of the atomic resonance, producing an intrinsically background-free error signal. These characteristics make MTS particularly well suited for precision laser frequency stabilization. The operating principle of MTS is described in the following section.

3.2 Principle of Modulation Transfer Spectroscopy

The experimental configuration of modulation transfer spectroscopy used in this work is illustrated in Fig. 3.1. The laser beam is divided into a pump beam and a probe beam. Unlike FMS, only the pump beam passes through an electro-optic modulator (EOM) and is phase-modulated before entering the hollow cathode lamp (HCL), whereas the probe beam remains unmodulated. After interacting with the atomic medium, only the probe beam is detected by the photodetector. The detected signal is subsequently demodulated to generate the error signal used for laser frequency stabilization.

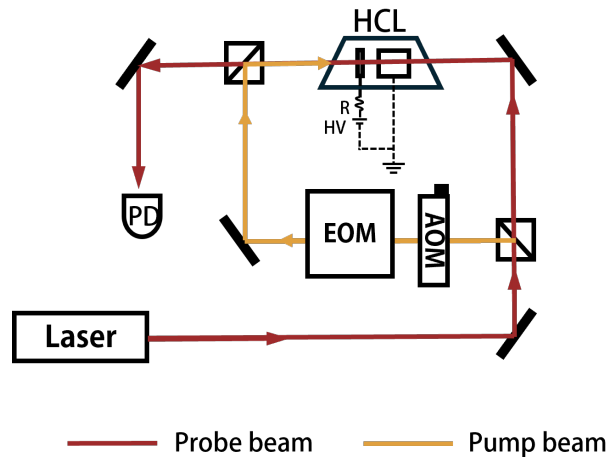


Figure 3.1: Schematic diagram of the modulation transfer spectroscopy (MTS) setup. The laser beam is divided into a pump beam and a probe beam. The pump beam passes through an electro-optic modulator (EOM) to generate phase modulation before entering the hollow cathode lamp (HCL), whereas the probe beam remains unmodulated. After the interaction, only the probe beam is detected by the photodetector (PD).

The key difference between FMS and MTS lies in the origin of the detected modulation. In FMS, the optical modulation is generated by the EOM before the light interacts with the atomic medium. Consequently, the photodetector continuously receives an RF-modulated signal regardless of whether the laser is resonant with the atomic transition. In contrast, the probe beam in MTS is initially unmodulated. Therefore, if the probe beam is detected without interacting with the atomic medium, no RF modulation is present and no demodulated signal can be obtained. When the laser frequency approaches an atomic resonance, the modulation carried by the pump beam is transferred to the probe beam through its interaction with the atoms. As a result, RF sidebands are generated on the probe beam, allowing the photodetector to detect an RF signal that can subsequently be demodulated.

Since this modulation transfer occurs only in the vicinity of the atomic resonance, the probe beam remains essentially unmodulated away from resonance. Consequently, no RF signal is available for demodulation, and the output remains close to zero. This behavior is the fundamental reason why MTS produces an intrinsically background-free spectrum. The detected RF signal is mixed with the reference modulation signal generated by the same RF source that drives the EOM. After passing through a low-pass filter (LPF), the high-frequency components are removed, leaving a dispersive error signal with a well-defined zero crossing at the atomic resonance. The steep slope around the zero crossing provides excellent frequency discrimination, making the signal highly suitable for precision laser frequency stabilization.

For these reasons, MTS was adopted throughout this work as the primary spectroscopic technique. The following chapters describe its implementation in the spectroscopy of the 680 nm, 1798 nm, 399 nm and 493 nm ytterbium transitions, together with the corresponding experimental procedures and measurement results.

4

Characterization of the Yb $^3P_1 \rightarrow ^3S_1$ Transition at 680 nm

4.1 Introduction

The $6s6p\ ^3P_1 \rightarrow 6s7s\ ^3S_1$ transition in neutral ytterbium (Yb) at 680 nm plays an important role in state manipulation within the metastable triplet manifold. It provides an essential pathway for controlling population transfer among long-lived excited states, which are widely used in precision spectroscopy and quantum-state engineering. Accurate characterization of this transition is also relevant for continuous optical atomic clock operation, where preparation and control of the 3P_2 state constitute important intermediate steps. Precise knowledge of the transition frequencies and isotopic structure therefore supports reliable frequency stabilization and the optimization of repumping schemes.

Several triplet-triplet transitions in ytterbium, such as the $6s6p\ ^3P_2 \rightarrow 6s7s\ ^3S_1$ transition at 770 nm and the $6s6p\ ^3P_0 \rightarrow 6s7s\ ^3S_1$ transition at 649 nm, have been investigated in the context of metastable-state manipulation and precision spectroscopy. In contrast, systematic high-resolution data on the $6s6p\ ^3P_1 \rightarrow 6s7s\ ^3S_1$ transition at 680 nm remain scarce. To date, a comprehensive Doppler-free characterization of this line with isotope-resolved accuracy has not been reported.

The objectives of this work are as follows:

1. To establish a robust Doppler-free spectroscopy setup capable of resolving the hyperfine and isotopic components of the 680 nm line with sufficient signal-to-noise ratio (SNR) for laser frequency stabilization.

2. To characterize operational systematic frequency shifts associated with the frequency-stabilization scheme.
3. To extract isotope-resolved transition frequencies suitable for subsequent King-plot analysis.

4.2 Experimental Setup

The experimental arrangement for modulation transfer spectroscopy (MTS) of the $^3P_1 \rightarrow ^3S_1$ transition in ytterbium at 680 nm is shown schematically in Fig. 4.1.

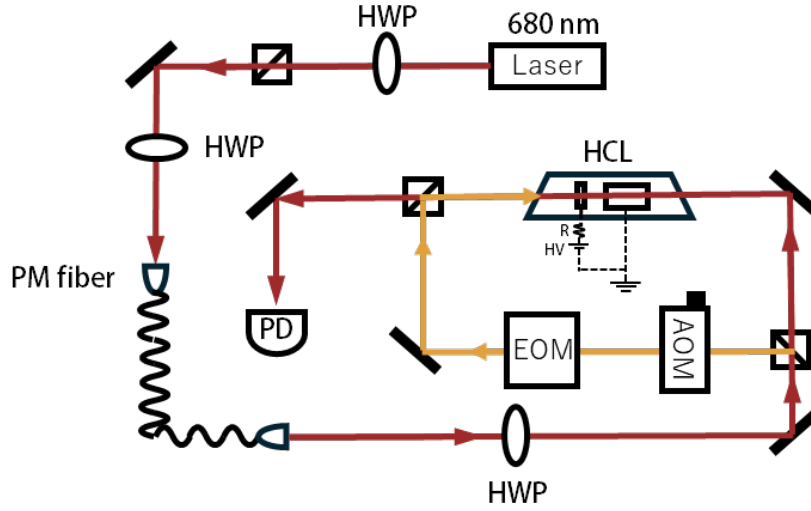


Figure 4.1: Schematic diagram of the experimental setup for modulation transfer spectroscopy of the $^3P_1 \rightarrow ^3S_1$ transition in ytterbium at 680 nm. Light from the ECDL is split into pump and probe beams by a PBS. The pump beam is sent through an AOM and phase-modulated by an EOM before being recombined with the counter-propagating probe beam in the HCL.

4.2.1 Laser Source and Optical Layout

The fiber output was collimated using a fiber collimator (Thorlabs A230TM, focal length $f = 4.51$ mm, NA = 0.55). The fiber fundamental mode is modeled as a Gaussian beam with waist $w_f \approx \text{MFD}/2$ at the fiber facet. When the facet is placed at the front focal plane of the collimator, the collimated $1/e^2$ radius is

$$w \approx \frac{f\lambda}{\pi w_f}, \quad (4.1)$$

where λ is the wavelength. Using $f = 4.51$ mm at $\lambda = 680$ nm and a representative single-mode fiber with $\text{MFD} \approx 4.3 \mu\text{m}$, we obtain $w \approx 0.454$ mm ($2w \approx 0.908$ mm). The corresponding Rayleigh range is

$$z_R = \frac{\pi w^2}{\lambda} \approx 0.95 \text{ m.} \quad (4.2)$$

4.2.2 Atomic Source

Ytterbium atoms were generated using a see-through hollow cathode lamp (HCL, Hamamatsu L2783-70Ne-Yb) operated with neon buffer gas. Within the discharge plasma, energetic electron-atom and ion-atom collisions lead to redistribution of population among various excited states [2].

In particular, the $6s6p\ ^3P_1$ state, although not a true metastable state (in contrast to 3P_0) and possessing a relatively short lifetime on the order of a few microseconds, can acquire a steady-state population under discharge conditions. As a result, absorption transitions originating from this level become observable. This collisional excitation mechanism enables the observation of the 680 nm $^3P_1 \rightarrow ^3S_1$ transition, even though the 3P_1 state is not thermally populated under equilibrium conditions.

4.2.3 Modulation Transfer Spectroscopy Configuration

The optical configuration of the modulation transfer spectroscopy (MTS) system is illustrated in Fig. 4.1. The laser beam was divided into pump and probe paths using a polarization beam splitter (PBS). The probe beam was sent directly through the hollow cathode lamp (HCL) and detected by a photodetector (PD), while the pump beam was first frequency-shifted by an acousto-optic modulator (AOM) and subsequently phase-modulated by an electro-optic modulator (EOM). The two beams were then recombined at the PBS to form collinear, counter-propagating beams within the HCL.

To minimize power broadening, the probe beam power was maintained at a low level, typically within the range of 500 μW to 1000 μW . In contrast, the pump beam power was adjusted during measurements to optimize the MTS signal for different isotopic and hyperfine transitions.

Frequency scans were carried out in segmented regions. The scan procedure initially targeted frequency intervals exhibiting strong absorption features and was subsequently extended to adjacent regions. Mode-hop behavior was continuously monitored using a wavelength meter, and the laser operating conditions were adjusted by tuning both the frequency and injection current to avoid operation near the threshold region. For each scan cycle, only one scan direction was retained to ensure consistency of the frequency axis.

To improve data quality, the valid scan range was determined using a custom al-

gorithm that excluded sections affected by noise spikes or distorted signals. Multiple scans over the same spectral region were then averaged to enhance the signal-to-noise ratio prior to further analysis, including hyperfine structure identification and isotope shift measurements.

Detection and demodulation were implemented using the phase-sensitive scheme described in Chp. 3. In practice, the probe signal detected by the PD was processed through RF amplification, mixing with a local oscillator, and low-pass filtering to extract the dispersive error signal used for spectroscopy.

4.2.4 Detection and Signal Processing

Following the optical interaction described above, the transmitted probe beam was converted into an electrical signal and processed using a phase-sensitive detection scheme.

The probe beam was detected using a fast silicon photodiode module (Thorlabs PDA10A; fixed gain, 200 nm to 1100 nm, 150 MHz bandwidth, effective detector area 0.8 mm²). The detector output was subsequently amplified in two stages using low-noise broadband amplifiers (Mini-Circuits ZFL-1000LN+, 0.1 MHz to 1000 MHz, ~20 dB gain; followed by ZFL-500LN+, 0.5 MHz to 500 MHz, ~20 dB gain). The amplified signal was then fed into a level-7 double-balanced mixer (Mini-Circuits ZAD-3+, 0.025 MHz to 200 MHz).

The local oscillator (LO) port of the mixer was driven by a split copy of the EOM driver signal (approximately 7 dBm), ensuring phase coherence with the pump-beam modulation. The resulting intermediate-frequency (IF) output, which carries the demodulated MTS signal, was recorded for analysis with a total RF pre-amplification of approximately 40 dB.

The IF signal was further low-pass filtered and amplified using a low-noise preamplifier (Stanford Research Systems SR560) operated in DC-coupling mode with a gain of 200 and a 100 Hz low-pass cutoff (6 dB roll-off)¹. This filtering stage suppresses high-frequency noise while preserving the characteristic MTS lineshape.

Finally, the processed dispersive signal was simultaneously monitored on an oscilloscope for real-time observation and recorded for subsequent analysis.

¹In practice, the cutoff frequency was typically set to 3 kHz.

4.2.5 Data Analysis Method

Spectra obtained from different scan ranges were combined onto a common frequency axis. For each scan segment, five consecutive oscilloscope traces were recorded and averaged to improve the signal-to-noise ratio. To establish an absolute frequency scale, both the wavemeter readout and the MTS signal were recorded simultaneously.

In the final analysis procedure, a complete scan cycle (forward and backward sweeps using a sawtooth modulation) was acquired within a single oscilloscope trace. The two halves of the scan were extracted separately and rescaled to equal length, enabling a point-by-point correspondence between the scan reference and the atomic signal. This reconstruction allows the frequency associated with each data index to be determined directly, thereby generating a reproducible detuning axis.

Compared with the initial implementation, in which the scan boundaries were identified manually from the oscilloscope trace, the refined reconstruction eliminates subjective judgment and reduces systematic frequency shifts arising from scan-range drift. Furthermore, after approximate resonance positions were identified, subsequent scans were localized to narrower frequency intervals centered on individual spectral features. This localized scanning strategy reduces distortions associated with wide-range sweeps and improves the reliability of resonance identification.

The reconstructed spectra were compared with theoretical stick spectra generated from the hyperfine-structure formula. A Python-based fitting interface was developed to adjust the magnetic dipole (A) and electric quadrupole (B) hyperfine coefficients, together with a global frequency offset, to align calculated transition frequencies with the measured features. This procedure was used primarily for transition identification and consistency checks with literature values, rather than for final precision determination.

4.3 Results and Discussion

4.3.1 Hyperfine Shift Identification and Fitting Procedure

The measured spectra were compared with theoretical transition frequencies calculated using the standard hyperfine Hamiltonian [3]. For a state with total electronic angular momentum J , nuclear spin I , and total angular momentum F , the hyperfine energy shift is expressed as

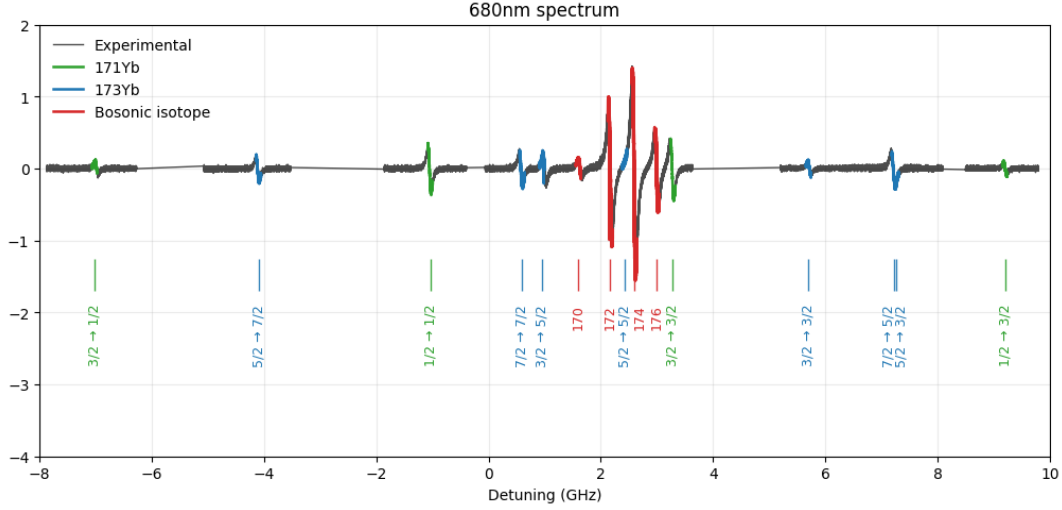


Figure 4.2: Modulation transfer spectroscopy (MTS) dispersive spectra of the Yb $^3P_1 \rightarrow ^3S_1$ transition at 680 nm, obtained using the refined scan-axis reconstruction procedure described in Sec. 3.2.5. Each trace corresponds to a localized frequency scan centered on a specific resonance, and five consecutive acquisitions were averaged to improve the signal-to-noise ratio. The frequency axis is converted into a common detuning axis referenced to the NIST database value. The reconstructed detuning axis was derived directly from synchronized oscilloscope traces, ensuring reproducible frequency calibration across all scan segments. Compared with earlier wide-range scans, the localized scanning strategy significantly improves baseline stability and reduces distortions associated with scan-range drift. The resulting spectra clearly resolve the isotopic components of ^{171}Yb and ^{173}Yb together with the bosonic isotopes, providing a reliable basis for hyperfine-structure analysis.

$$E_F = \frac{1}{2}AK + B \frac{\frac{3}{4}K(K+1) - I(I+1)J(J+1)}{2I(2I-1)2J(2J-1)}, \quad (4.3)$$

where

$$K = F(F+1) - I(I+1) - J(J+1). \quad (4.4)$$

Here, A and B denote the magnetic dipole and electric quadrupole hyperfine constants, respectively. For isotopes with nuclear spin $I = \frac{1}{2}$ (e.g., ^{171}Yb), the quadrupole term vanishes. The transition frequencies were calculated from the differences between upper-state (3P_1) and lower-state (3S_1) hyperfine energy shifts,

$$\nu_{F' \rightarrow F} = \nu_0 + \frac{E_{F'}^{(u)} - E_F^{(l)}}{h}, \quad (4.5)$$

together with a global frequency offset ν_0 . A Python-based fitting interface was developed to overlay calculated transition frequencies with the experimental MTS signals. The hyperfine constants A and B for both states, along with a global frequency offset, were treated as adjustable parameters to align predicted transitions with the measured dispersive features. This procedure was used primarily for tran-

sition identification and consistency checks with literature values, rather than for final precision extraction of hyperfine constants.

Hyperfine Structure of ^{171}Yb

For ^{171}Yb , manual adjustment of the fitting interface yielded approximate hyperfine constants of $A(^3S_1) \approx 6840$ MHz and $B = 0$, and $A(^3P_1) \approx 3985$ MHz with $B = 0$. These values are in good agreement with previously reported measurements, e.g., $A(^3S_1, ^{171}\text{Yb}) = 6837.83(19)$ MHz reported by Zhou *et al.* [4], and $A(^3P_1, ^{171}\text{Yb}) = 3957.833(28)$ MHz as determined by Jones *et al.* [5].

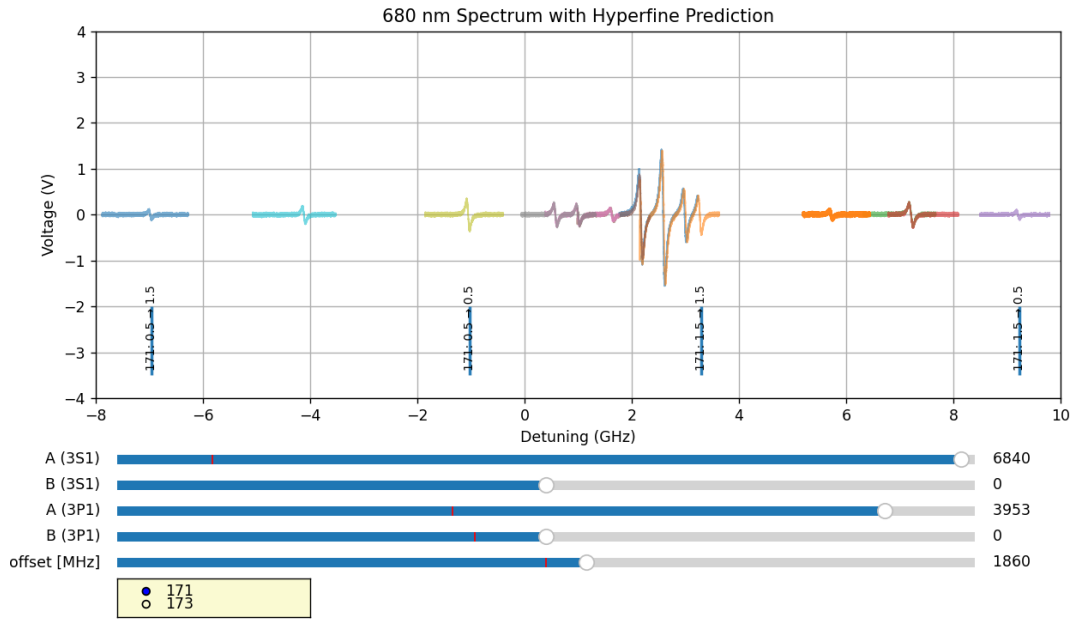


Figure 4.3: Preliminary fitting of ^{171}Yb hyperfine structure using the standard hyperfine splitting formula, with magnetic dipole (A) and electric quadrupole (B) coefficients treated as free parameters.

Hyperfine Structure Fitting - ^{173}Yb

Figure 4.4 shows the same fitting procedure applied to ^{173}Yb . The fitted values were $A(^3S_1) \approx -1897$ MHz and $B(^3S_1) \approx -66$ MHz, and $A(^3P_1) \approx -1094$ MHz and $B(^3P_1) \approx -827$ MHz. These results are in reasonable agreement with previously reported values. For example, Zhou *et al.* [4] reported $A(^3S_1, ^{173}\text{Yb}) = -1891.46(7)$ MHz and $B(^3S_1, ^{173}\text{Yb}) = -0.62(16)$ MHz. It is worth noting that the two rightmost predicted transitions, $^{173}\text{Yb} : F = \frac{5}{2} \rightarrow F' = \frac{7}{2}$ and $^{173}\text{Yb} : F = \frac{3}{2} \rightarrow F' = \frac{5}{2}$, are separated by only a small frequency interval. This close spacing makes them challenging to distinguish in both the measured absorption spectrum and the hyperfine-model fitting. In addition, the $F = \frac{5}{2} \rightarrow F' = \frac{5}{2}$ transition, located within

the bosonic isotope group, is embedded among nearby strong lines, further complicating its identification in the spectrum. These spectral overlaps motivate the more detailed quantum-interference analysis presented in the following section.

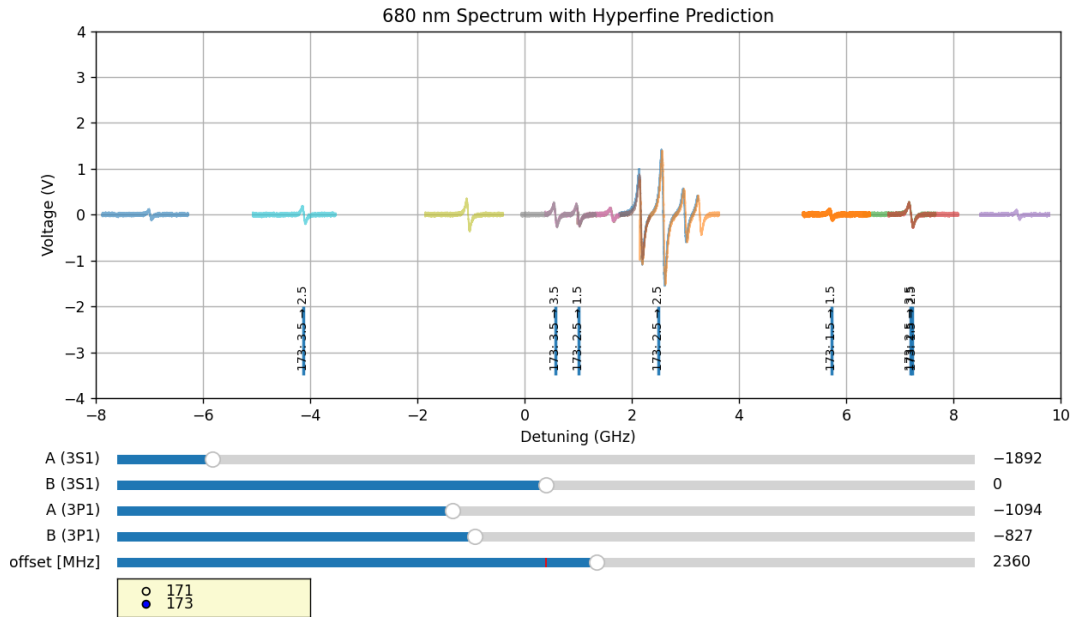


Figure 4.4: Preliminary fitting of ^{173}Yb hyperfine structure using the standard hyperfine splitting formula, with magnetic dipole (A) and electric quadrupole (B) coefficients treated as free parameters.

4.3.2 Challenges in Resolving the ^{173}Yb Transitions

The difficulty in resolving the closely spaced fermionic spectral features of ^{173}Yb in the present modulation transfer spectroscopy (MTS) setup arises from a combination of line-broadening mechanisms and spectral overlap with stronger bosonic isotope signals.

- **Pressure broadening** [6]

Collisions between atoms at finite vapor pressures perturb the atomic energy levels and increase the resonance linewidth. As a result, transitions separated by only a few tens of megahertz may become difficult to resolve experimentally.

- **Power broadening** [3]

Optical excitation at intensities above the saturation intensity enlarges the effective linewidth of the transition. Power broadening therefore increases the spectral overlap between neighboring resonances.

- **Spectral overlap with bosonic isotopes**

Several fermionic components lie within the dense cluster of strong bosonic resonances (^{172}Yb , ^{174}Yb , ^{176}Yb). Because bosonic isotopes have larger natural abundances and stronger signal amplitudes, the weaker fermionic contributions can be partially obscured by the wings of the bosonic lines.

- **Relative line strength**

The ^{173}Yb hyperfine transitions have intrinsically smaller amplitudes due to dipole matrix elements and statistical weights. In the electric-dipole approximation, the transition strength for a hyperfine component $F \rightarrow F'$ is governed by the Wigner–Eckart theorem [7] and can be expressed as

$$S_{F \rightarrow F'} \propto (2F + 1)(2F' + 1) \left\{ \begin{matrix} J' & F' & I \\ F & J & 1 \end{matrix} \right\}^2 |\langle \gamma J' \| d \| \gamma J \rangle|^2. \quad (4.6)$$

Here I denotes the nuclear spin and the curly brackets represent a Wigner- $6j$ symbol. The reduced dipole matrix element $\langle \gamma J' \| d \| \gamma J \rangle$ is common to all isotopes, while the angular momentum coupling coefficients redistribute the oscillator strength among the hyperfine components.

For ^{173}Yb ($I = \frac{5}{2}$), the total dipole strength of a fine-structure transition is distributed among seven hyperfine components. The strongest component carries only about one-third of the total intensity. In contrast, bosonic isotopes ($I = 0$) exhibit a single resonance that concentrates the full oscillator strength, resulting in a much larger observed signal.

Under the present operating conditions, effective linewidths broadened to approximately 15 MHz to 20 MHz combined with separations of only a few tens of megahertz between nearby fermionic lines and bosonic neighbors lead to significant spectral overlap. Consequently, fermionic contributions often appear only as shoulders or weak distortions rather than clearly resolved peaks.

To quantify this effect, the measured spectra were first fitted using derivative Voigt profiles with fermionic line centers constrained by hyperfine constants. However, including the fermionic components produced negligible improvement in the coefficient of determination (R^2) compared with a boson-only fit, indicating that the fermionic contributions are too strongly blended to be uniquely identified under the present experimental conditions.

4.3.3 Resolving Overlapping Fermionic Features via Quantum Interference Fitting

A prominent dispersive feature is observed in the ^{173}Yb spectrum near 7.2 GHz. Conventional derivative-Voigt fits to the central structure yield an apparent linewidth of

$$\text{FWHM} \approx 75 \text{ MHz}, \quad \text{HWHM} \approx 37 \text{ MHz}.$$

From hyperfine-structure estimates using the tabulated A and B constants for the $^3P_1 \rightarrow ^3S_1$ transition, the two relevant fermionic hyperfine components of interest, $F = \frac{7}{2} \rightarrow F' = \frac{5}{2}$ and $F = \frac{5}{2} \rightarrow F' = \frac{3}{2}$, are predicted to be separated by only

$$\Delta_{\text{hfs}} \approx 31.7 \text{ MHz}.$$

This separation is smaller than the observed linewidth and therefore places the system firmly in the strongly overlapping (unresolved) regime. In this regime the two transitions cannot be experimentally resolved as distinct peaks.

Empirically, an unconstrained two-component derivative-Voigt fit provides a convenient description of the overall spectral envelope but leaves structured residuals in the overlap region (Fig. 4.5). The fitted centers of these two *effective* components are 7.2397 GHz and 7.3010 GHz, corresponding to an *effective* separation

$$\Delta_{\text{eff}} \approx 61 \text{ MHz}.$$

We emphasize that Δ_{eff} obtained from a phenomenological two-component fit does not necessarily coincide with the true hyperfine separation Δ_{hfs} . Because MTS is a coherent phase-sensitive detection technique, the observed dispersive signal can be influenced by phase mixing, crossover-type contributions, and line-pulling effects from nearby resonances.

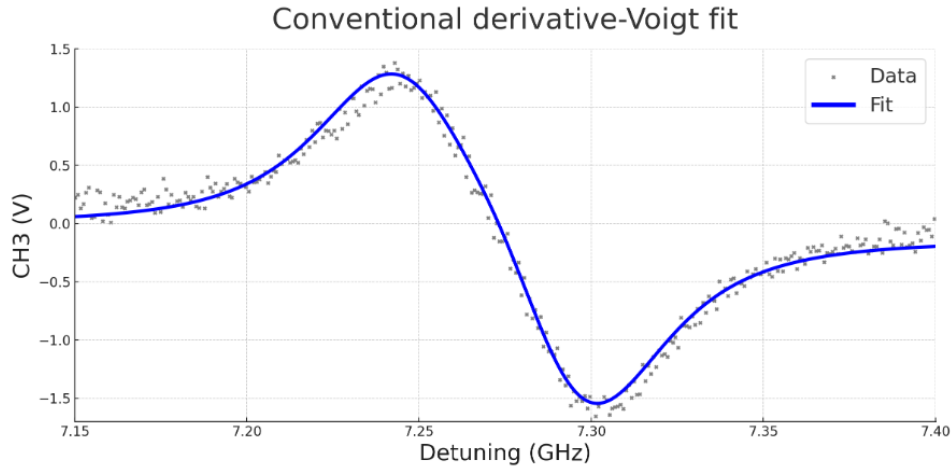


Figure 4.5: Derivative-Voigt fit to the ^{173}Yb spectral feature near 7.2 GHz. The model comprises two independent derivative-Voigt components with free centers, amplitudes, and shared widths. The fitted centers are 7.2397 GHz and 7.3010 GHz, and the effective linewidth is approximately 75 MHz (FWHM). Structured residuals remain in the overlap region, indicating that the conventional model does not fully capture the observed lineshape.

Motivation: Line Pulling in MTS

The primary purpose of the analysis presented in this section is to investigate the influence of line pulling on the modulation transfer spectroscopy signal. When several resonances are separated by frequencies comparable to their linewidths, the observed dispersive feature may not correspond exactly to the true transition frequency of any single component. Instead, neighboring resonances can distort the overall lineshape and shift the apparent zero-crossing position of the MTS signal.

Such line-pulling effects are particularly relevant for frequency stabilization, since the locking point of the laser is typically defined by the zero-crossing of the dispersive signal. Understanding the magnitude of this shift is therefore important for assessing the reliability of the derived transition frequencies.

Implications of Linewidth-to-Separation Ratio

When the separation between neighboring resonances becomes comparable to their linewidth, interference between excitation pathways can influence the detected spectral signal. In contrast, when the separation greatly exceeds the linewidth, the resonances behave as independent transitions and interference effects become negligible [8].

In the present experiment, the hyperfine-structure estimate $\Delta_{\text{hfs}} \approx 31.7 \text{ MHz}$ is already comparable to the homogeneous width scale under our conditions, while the experimentally fitted effective separation $\Delta_{\text{eff}} \approx 61 \text{ MHz}$ remains of the same

order as the observed HWHM (≈ 37 MHz). Therefore, the signal originates from strongly overlapped spectral contributions for which a simple incoherent sum of independent profiles may be insufficient.

Limitations of the Conventional Voigt Model

When the experimental data are modeled using independent derivative-Voigt components:

- systematic residuals appear in the overlap region,
- the extracted linewidths can become artificially broadened to accommodate slope distortions,
- the fitted centers may exhibit line pulling relative to expected spectral positions.

Such behavior is commonly encountered in fermionic spectra where multiple hyperfine transitions contribute to the detected signal [9, 8]. These observations motivate the use of a model that allows for coherent cross terms between nearby resonances.

4.3.4 Quantum-Interference Analysis

As discussed in the previous section, the observed fermionic spectral feature cannot be adequately described by two independent resonances because the separation between neighboring transitions is comparable to the effective linewidth. Under these conditions, the spectral responses strongly overlap, resulting in line-shape distortion and line-pulling effects.

Quantum interference between closely spaced hyperfine transitions has been shown to modify the observed spectral profile in precision spectroscopy. Brown *et al.* demonstrated that the coherent summation of excitation amplitudes associated with neighboring hyperfine levels introduces an interference contribution to the spectral lineshape, leading to asymmetric profiles and systematic shifts of the extracted line centers [10]. Subsequent studies on ytterbium spectroscopy further showed that these interference effects become increasingly important when the laser is simultaneously near resonance with multiple hyperfine transitions, and that the interference strength increases as the spacing between neighboring transitions decreases [11]. Motivated by these observations, we describe the measured MTS signal using a phenomenological quantum-interference (QI) model that incorporates coherent cross terms between neighboring resonances,

$$\begin{aligned}
S(\nu) = A & \left[S_1 \operatorname{Re}\{G(\nu - \omega_1, \gamma)\} + S_2 \operatorname{Re}\{G(\nu - \omega_2, \gamma)\} \right. \\
& + c_{\text{sym}} \operatorname{Re}\{G(\nu - \omega_1, \gamma)G^*(\nu - \omega_2, \gamma)\} \\
& \left. + c_{\text{asym}} \operatorname{Im}\{G(\nu - \omega_1, \gamma)G^*(\nu - \omega_2, \gamma)\} \right].
\end{aligned} \tag{4.7}$$

where $G(\nu - \omega_i, \gamma)$ denotes a complex Lorentzian lineshape with half-width at half maximum γ . The coefficients S_i represent the relative amplitudes of the two resonances, while c_{sym} and c_{asym} describe the symmetric and antisymmetric interference contributions, respectively. These additional terms allow the model to reproduce the asymmetric dispersive profile arising from the coherent overlap of neighboring transitions.

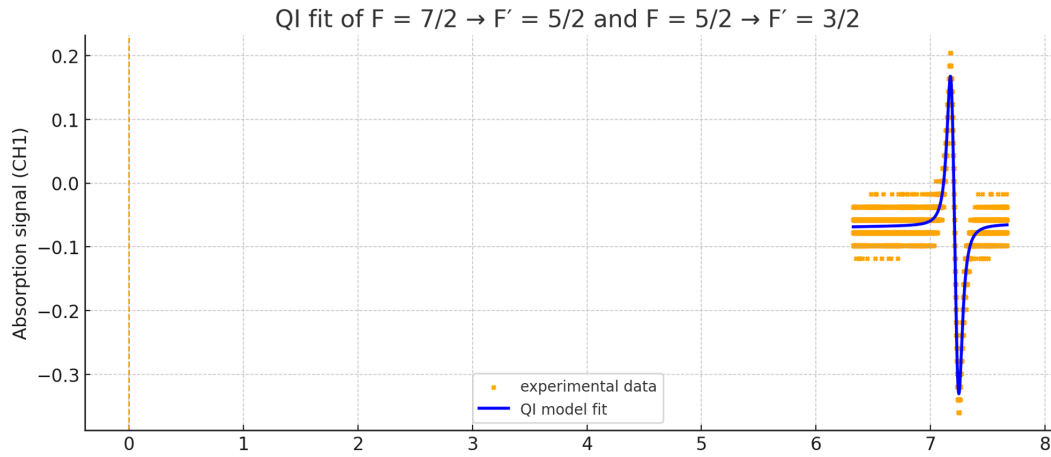


Figure 4.6: Quantum-interference (QI) fit to the overlapping ^{173}Yb hyperfine transitions. Orange points show the experimental modulation-transfer spectroscopy (MTS) signal, and the blue curve represents the fitted QI model. The fit reproduces the asymmetric dispersive profile arising from coherent interference between the $F = \frac{7}{2} \rightarrow F' = \frac{5}{2}$ and $F = \frac{5}{2} \rightarrow F' = \frac{3}{2}$ transitions, demonstrating that the observed resonance cannot be described as a simple superposition of two independent Lorentzian components.

Figure 4.6 presents the QI fit to the overlapping fermionic spectral feature of ^{173}Yb . The experimental data (orange points) are accurately reproduced by the fitted QI model (blue curve), demonstrating that the observed spectrum cannot be interpreted as a simple superposition of two independent resonances.

The fitted parameters are

$$\Delta = 61.72(63) \text{ MHz}, \quad \gamma = 0.0433(3) \text{ GHz}, \tag{4.8}$$

yielding

$$\frac{\Delta}{\gamma} \approx 1.4. \quad (4.9)$$

This ratio indicates that the transition separation is comparable to the effective linewidth, placing the spectrum in the strong-overlap regime where coherent interference cannot be neglected. Consequently, the neighboring transitions no longer behave as independent spectral features, and the measured MTS signal represents their collective coherent response. The interference terms naturally account for both the observed asymmetric lineshape and the associated line-pulling effects.

The excellent agreement between the experimental data and the fitted spectrum demonstrates that the phenomenological QI model provides a physically consistent description of the observed composite resonance. Although the individual hyperfine components remain unresolved, the fitted transition separation supports the interpretation that the measured asymmetry originates primarily from quantum-interference-induced line-shape distortion. This analysis therefore forms the basis for the transition assignments and frequency determination summarized in Table 4.1. In particular, it indicates that the weak $^{173}\text{Yb } F = \frac{5}{2} \rightarrow F' = \frac{5}{2}$ transition is embedded within the composite spectral feature and contributes predominantly through its interference with neighboring resonances rather than as an independently resolved peak.

4.3.5 Unresolved $^{173}\text{Yb } (F = \frac{5}{2} \rightarrow F' = \frac{5}{2})$ Transition

In addition to the resolved hyperfine components discussed above, the ^{173}Yb transition $F = \frac{5}{2} \rightarrow F' = \frac{5}{2}$ could not be observed as an isolated resonance in the present spectra. Its expected frequency lies within a crowded spectral region where neighboring bosonic isotopes and stronger fermionic transitions overlap. Because the separation between these resonances is comparable to the effective linewidth, the weak $F = \frac{5}{2} \rightarrow F' = \frac{5}{2}$ component is embedded within the composite lineshape and does not emerge as a distinct spectral peak.

Quantum-interference cross terms arise only among transitions belonging to the same isotope that share a common initial state and couple to closely spaced excited states with indistinguishable decay pathways. Consequently, overlap between the unresolved ^{173}Yb transition and neighboring bosonic isotopes contributes through incoherent addition rather than coherent interference. Nevertheless, such overlap can still modify the observed lineshape and introduce line-pulling effects that shift the apparent transition center.

The interpretation of this unresolved feature is supported by previous theoretical studies. Jayjong *et al.* [12] investigated isotope-overlap effects in the $^1S_0 \rightarrow ^3P_1$ transition of ytterbium at 556 nm using a rate-equation model. Their simulations showed that weak fermionic transitions need not appear as isolated resonances when overlapped with stronger neighboring features, but may instead manifest as shoulder-like distortions embedded within the overall spectral profile. Although the transition studied in Ref. [12] differs from the present 680 nm $^3P_1 \rightarrow ^3S_1$ system, the underlying consequence of unresolved isotope overlap is qualitatively similar. More recently, quantum-interference analyses (e.g., [13]) further demonstrated that when resonance separations become comparable to the linewidth, unresolved transitions primarily contribute through subtle lineshape distortions and line pulling rather than generating independently resolvable peaks.

Experimental Results for the Unresolved ^{173}Yb Transition ($F = \frac{5}{2} \rightarrow F' = \frac{5}{2}$)

Consistent with the considerations above, no isolated resonance is observed at the expected position of the ^{173}Yb $F = \frac{5}{2} \rightarrow F' = \frac{5}{2}$ transition in the measured hollow-cathode-lamp spectra. Instead, the global fit exhibits a small asymmetry and residual line pulling, suggesting that the present system operates in a regime where overlap effects cannot be neglected.

Although this transition cannot be directly resolved, its frequency can still be inferred indirectly from the hyperfine constants obtained from the resolved transitions. From the global fit, the magnetic-dipole and electric-quadrupole hyperfine constants are determined to be

$$A(^3S_1, ^{171}\text{Yb}) = 6.838 \text{ GHz}, \quad (4.10)$$

$$A(^3S_1, ^{173}\text{Yb}) = -1.892 \text{ GHz}, \quad (4.11)$$

$$B(^3S_1, ^{173}\text{Yb}) = 0.0134 \text{ GHz}. \quad (4.12)$$

Using these constants together with the standard hyperfine-structure Hamiltonian, the frequencies of all allowed hyperfine transitions can be calculated. The analysis predicts that the unresolved ^{173}Yb $F = \frac{5}{2} \rightarrow F' = \frac{5}{2}$ transition should occur at

$$\Delta\nu_{\text{pred}} \approx -0.0973 \text{ GHz}, \quad (4.13)$$

relative to the ^{174}Yb reference transition.

Although no independent peak is observed at this frequency, the predicted po-

sition is fully consistent with the region where lineshape distortions are observed experimentally. Together with the theoretical studies discussed above, the present analysis therefore provides supporting evidence for the existence of the $^{173}\text{Yb } F = \frac{5}{2} \rightarrow F' = \frac{5}{2}$ transition despite its absence as a spectrally resolvable feature.

Future measurements on an isotope-selective ^{173}Yb magneto-optical trap (MOT) will provide an independent platform for testing this prediction. Such experiments are expected to eliminate spectral crowding from bosonic isotopes while substantially reducing Doppler and discharge broadening. The resulting narrower linewidths and cleaner lineshapes should enable a direct verification of the inferred frequency of the unresolved $^{173}\text{Yb } F = \frac{5}{2} \rightarrow F' = \frac{5}{2}$ transition.

4.4 Frequency Stabilization and $6s6p\ ^3P_1 \rightarrow 6s7s\ ^3S_1$ isotope shifts

The 680 nm laser was generated by a Moglab external-cavity diode laser (ECDL) and frequency-stabilized using the built-in digital lock controller (DLC) with its internal PID servo. The error signal was derived from modulation-transfer spectroscopy (MTS) on the ytterbium atomic transition, providing a dispersion-shaped signal with a steep zero-crossing suitable for long-term frequency locking. The error signal was directly fed into the DLC, which controlled the diode laser current and piezo actuator to correct fast and slow frequency fluctuations, respectively.

To characterize the long-term frequency stability, the locked laser frequency was recorded over five consecutive days using a calibrated wavemeter while remaining locked to the same spectroscopic feature. Frequency readouts were logged at regular time intervals, and for each day the mean frequency and standard deviation were evaluated. An overall five-day average was then obtained, providing a stable optical frequency reference for subsequent hyperfine-resolved measurements of the $^3P_1 \rightarrow ^3S_1$ transition.

Using this frequency-stabilized laser, hyperfine-resolved spectroscopy of the 680 nm transition was performed for all naturally abundant ytterbium isotopes. For each resolved hyperfine component, the absolute optical frequency was obtained from the wavemeter while the laser remained locked to the zero-crossing of the corresponding MTS signal. The measured transition frequencies, averaged over five consecutive days of measurements under identical locking conditions, are summarized in Table 4.1.

Table 4.1: Isotope shifts of the $6s6p\ ^3P_1 \rightarrow ^3S_1$ transition lines. Frequency shifts are given in MHz relative to ^{174}Yb . Centroid values are not listed here. The fourth column lists the absolute frequency with respect to the wavemeter center frequency. Uncertainties are expressed as $\pm(\text{systematic})\pm(\text{stability})$.²

Isotope	$F \rightarrow F'$	Frequency shift / MHz	Wavemeter(WS8-10) / THz
^{170}Yb	–	–962.0(3.0)	$440.777061 \pm 10\text{ MHz} \pm 2.97\text{ MHz}$
^{172}Yb	–	–424.0(0.8)	$440.777599 \pm 10\text{ MHz} \pm 0.55\text{ MHz}$
^{174}Yb	–	0	$440.778023 \pm 10\text{ MHz} \pm 0.55\text{ MHz}$
^{176}Yb	–	+406.0(1.0)	$440.778429 \pm 10\text{ MHz} \pm 0.84\text{ MHz}$
^{171}Yb	$3/2 \rightarrow 1/2$	–9586.0(5.0)	$440.768437 \pm 10\text{ MHz} \pm 4.98\text{ MHz}$
	$1/2 \rightarrow 1/2$	–3648.0(1.4)	$440.774375 \pm 10\text{ MHz} \pm 1.30\text{ MHz}$
	$3/2 \rightarrow 3/2$	+672.0(1.6)	$440.778695 \pm 10\text{ MHz} \pm 1.48\text{ MHz}$
	$1/2 \rightarrow 3/2$	+6609.0(0.6)	$440.784632 \pm 10\text{ MHz} \pm 0.00\text{ MHz}$
^{173}Yb	$5/2 \rightarrow 7/2$	–6715.0(3.8)	$440.771308 \pm 10\text{ MHz} \pm 3.78\text{ MHz}$
	$7/2 \rightarrow 7/2$	–2019.0(1.4)	$440.776004 \pm 10\text{ MHz} \pm 1.30\text{ MHz}$
	$3/2 \rightarrow 5/2$	–1594.0(1.5)	$440.776429 \pm 10\text{ MHz} \pm 1.41\text{ MHz}$
	$3/2 \rightarrow 3/2$	+3136.0(3.3)	$440.781159 \pm 10\text{ MHz} \pm 3.21\text{ MHz}$
	$7/2 \rightarrow 5/2$	+4616.0(1.4)	$440.782639 \pm 10\text{ MHz} \pm 1.30\text{ MHz}$

The measured transition frequencies $\nu_{F \rightarrow F'}$ are related to the hyperfine-free transition frequency (centroid) ν_0 through the standard hyperfine Hamiltonian,

$$\nu_{F \rightarrow F'} = \nu_0 + \Delta E_u(F') - \Delta E_l(F), \quad (4.14)$$

where $\Delta E_{u,l}(F)$ denote the hyperfine energy shifts of the upper (3S_1) and lower (3P_1) states, respectively. Prior to centroid extraction, the observed spectral features were compared with calculated hyperfine transition patterns based on the standard hyperfine Hamiltonian [14], as presented in Secs. 3.2 and 3.3, to assist in the identification of individual hyperfine components. For the fermionic isotopes ^{171}Yb and ^{173}Yb , the centroid frequencies were determined from a least-squares fit to the observed hyperfine transitions.

In the case of ^{173}Yb , two hyperfine components ($F = \frac{5}{2} \rightarrow F' = \frac{5}{2}$ and $F = \frac{5}{2} \rightarrow F' = \frac{3}{2}$) were not spectrally resolved in the present hollow-cathode lamp measurement due to overlap with stronger bosonic resonances. A constrained least-squares procedure was therefore employed, in which the upper-state hyperfine constants of the 3S_1 level were fixed to literature values ($A_u = -1892\text{ MHz}$,

²The systematic uncertainty of $\pm 10\text{ MHz}$ corresponds to the absolute accuracy specification of the HighFinesse WS-8/WS-10 wavelength meter (see <https://www.highfinesse.com/en/wavelengthmeter/wavelengthmeter-ws-8-10.html>).

$B_u = -0.2$ MHz) reported by Yamaguchi [15]. The centroid frequency together with the lower-state (3P_1) hyperfine constants were then determined from the five observed transitions, yielding

$$\nu_0(^{173}\text{Yb}) = 440.777\,794\,439\text{ THz.} \quad (4.15)$$

The fitted lower-state hyperfine constants differ from the literature values reported by Atkinson *et al.* [16], particularly for the electric-quadrupole constant B_l . This discrepancy is attributed to the absence of transitions originating from the $F = \frac{5}{2}$ lower hyperfine level in the present dataset, which significantly reduces sensitivity to the quadrupole interaction and leads to strong correlations between A_l and B_l in the constrained fit. Consequently, these fitted constants should be regarded primarily as parameters providing a self-consistent description of the present dataset, whereas the centroid frequency ν_0 remains the more robust observable.

For ^{171}Yb , the upper-state hyperfine constant was fixed to the literature value $A_u = 6847.7$ MHz [16], and the electric-quadrupole term was set to zero. From the four observed hyperfine transitions, the centroid frequency was determined as

$$\nu_0(^{171}\text{Yb}) = 440.777\,257\,092\text{ THz.} \quad (4.16)$$

These centroid frequencies, derived from the same frequency-stabilized laser and wavemeter-referenced dataset, form the basis for the isotope-shift comparison and the subsequent King-plot analysis presented in the following section.

4.5 Conclusion

In this work, we have experimentally characterized the $^3P_1 \rightarrow ^3S_1$ transition in ytterbium at 680 nm using modulation transfer spectroscopy (MTS) with a hollow-cathode lamp source. To overcome systematic uncertainties present in the first measurement round, we implemented an improved scan-axis reconstruction method in the second round of experiments. This approach enabled robust frequency calibration and localized scans centered on individual resonances.

The refined strategy yielded dispersive spectra with improved baseline stability and partially resolved hyperfine features. A quantum interference (QI) fitting model was employed to interpret the overlapping ^{173}Yb transitions $F = 7/2 \rightarrow F' = 5/2$ and $F = 5/2 \rightarrow F' = 3/2$, and to account for the asymmetric lineshapes arising from interference effects in the overlap region. In particular, the QI analysis provides a physical explanation for why individual fermionic components cannot be cleanly

resolved as isolated resonances under the present experimental conditions.

Following this interpretation, the transition frequencies reported in Table 2 were determined using PID-stabilized laser locking to the corresponding dispersive features, rather than extracted directly from spectral fits. The resulting separation between the identified components is found to be broadly consistent, within experimental and model-dependent uncertainties, with the expected hyperfine splitting from theoretical calculations.

The measured transition frequencies summarized in Table 2 should therefore be regarded as experimentally stabilized values, informed by QI-assisted interpretation of unresolved spectral features, and provide a useful reference framework for subsequent comparison with hyperfine-structure calculations and isotope-shift analyses.

These results establish reliable data sets for stabilizing laser frequencies to address the 680 nm $^3P_1 \rightarrow ^3S_1$ transition. Looking forward, extending this approach to cold-atom platforms will suppress pressure- and discharge-induced broadening, enabling linewidths close to the natural limit and allowing more precise determinations of hyperfine constants and isotope shifts.

In addition, we demonstrated stable frequency locking of the 680 nm laser using the built-in digital lock controller (DLC) of a Moglab ECDL, referenced to the MTS error signal. Continuous operation over a five-day measurement period showed reliable long-term performance, with reproducible daily mean frequencies and no observable lock degradation.

5

Characterization of the Yb ${}^3P_2 \rightarrow 6s5d\,{}^3D_3$ Transition at 1798 nm

5.1 Introduction

Metastable states in alkaline-earth-like atoms (AEAs), such as ytterbium, provide a versatile platform for quantum information processing and precision measurement. The long-lived 3P_J manifold supports qubits with extended coherence times and enables magnetic trapping, Raman-driven rotations, and continuous cold-atom generation. In particular, the 3P_2 state, possessing a large magnetic moment, serves as an intermediate reservoir for magnetically trapped atoms and allows simultaneous optical cooling on the ${}^3P_2 \rightarrow {}^3D_3$ transition.

This pathway has been successfully exploited in other AEAs, including Ca, Sr, and Mg, to realize secondary Doppler-cooling stages, continuous atomic beams, and metastable-state magneto-optical traps (MOTs). Despite its significance, experimental data on the corresponding ytterbium transition remain scarce.

The $6s5p\,{}^3P_2 \rightarrow 6s5d\,{}^3D_3$ transition near 1798 nm offers a narrow linewidth and favorable branching ratios for metastable-state laser cooling, yet its isotope shifts and hyperfine structure have not been systematically measured. Such data are essential for designing efficient repumping schemes, modeling population dynamics in magnetically trapped Yb [17], and benchmarking configuration-interaction calculations for heavy AEAs [18].

In this work, we report high-resolution isotope-shift spectroscopy of the ytterbium $6s5p\,{}^3P_2 \rightarrow 6s5d\,{}^3D_3$ transition at 1798 nm. We employ modulation transfer spectroscopy (MTS) using a hollow-cathode lamp (HCL) to obtain Doppler-free

spectra. From the recorded lines, we identify all observable isotopic components—including the hyperfine-resolved transitions of the fermionic isotopes—and extract the isotope shifts and hyperfine constants of the $6s5d\ ^3D_3$ state. This measurement constitutes the first systematic spectroscopic characterization of the 1798 nm line in Yb and establishes a frequency-reference pathway for future laser cooling and continuous-atom-source applications.

5.2 Experimental Setup

The experimental arrangement for modulation transfer spectroscopy (MTS) of the $^3P_2 \rightarrow ^3D_3$ transition in ytterbium at 1798 nm is shown schematically in Fig. 1. Light from a fiber laser at 1798 nm is split into probe and pump beams by a beam-splitter. The probe travels unmodulated to the hollow-cathode lamp (HCL), while the pump is routed through a single electro-optic phase modulator (EOM) driven at a radio frequency Ω . The modulated pump and the counter-propagating probe are mode-matched and spatially overlapped inside the HCL to drive the nonlinear four-wave-mixing process that transfers phase modulation to the probe. The probe is detected on an InGaAs photodetector and the signal is demodulated to obtain a dispersive MTS error signal for laser-frequency stabilization. Beam diameters, polarization, and focusing are adjusted to match the elongated discharge region of the lamp and to maximize modulation-transfer efficiency. No frequency shifter is employed in this setup.

For absolute-frequency measurements, a small fraction of the 1798 nm laser output is frequency doubled to 899 nm using a second-harmonic generation (SHG) module before being sent to the wavemeter. This configuration is adopted because the operating wavelength of the fiber laser lies outside the optimal measurement range of the wavemeter, whereas the frequency-doubled light can be measured with significantly improved accuracy and stability.

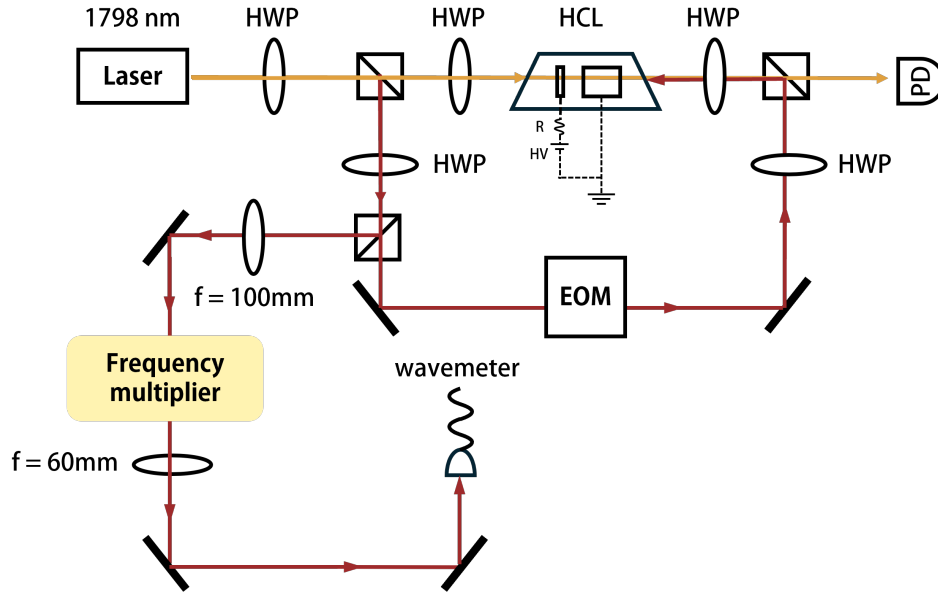


Figure 5.1: Schematic of the 1798 nm MTS setup. The ECDL output is split into an unmodulated probe and a pump that is phase-modulated by a single EOM before being recombined to counter-propagate in the HCL. The detected probe is demodulated at the EOM drive frequency to generate the MTS error signal.

5.3 Experimental Method for the $^3P_2 \rightarrow ^3D_3$ Transition

For the spectroscopic study of the 1798 nm $^3P_2 \rightarrow ^3D_3$ transition in ytterbium, two independent approaches were implemented to reconstruct the frequency axis and calibrate the laser scan. This dual-method strategy ensured both accuracy in frequency determination and robustness against systematic errors.

5.3.1 Method 1: Frequency-axis reconstruction

In the first approach, the laser frequency was scanned across the region of interest while both the scan reference signal and the atomic response were simultaneously recorded on a digital oscilloscope. Each complete scan cycle, consisting of the forward and backward sweeps, was extracted and analyzed separately. The two traces were resampled to the same number of data points, establishing a one-to-one correspondence between the scan reference (oscilloscope Channel 2) and the atomic absorption signal (oscilloscope Channel 3).

This procedure provided a direct mapping of each data index to its corresponding frequency. The frequency scale was determined by assigning the beginning and end of the scan reference to the nominal scan range and interpolating linearly

across all intermediate points. Once the frequency was known for each index, it was converted into detuning relative to a chosen center frequency, allowing precise comparison of absorption features. By reconstructing the scan axis directly from the recorded data, this method effectively removed uncertainties from manual identification of scan boundaries and minimized errors arising from slow temporal drifts of the laser scan. Furthermore, instead of performing a broad scan, the laser modulation amplitude and offset were tuned such that each spectral acquisition targeted only a limited frequency interval containing expected resonances. This localized scanning strategy reduced systematic distortions and improved the accuracy of resonance identification.

5.3.2 Method 2: Wavemeter-servoed controlled scanning

As an independent verification, we employed a wavemeter-driven scanning scheme. The PreciLaser was locked to the HighFinesse WS-8 wavemeter through its analog control port. A sawtooth waveform was applied to modulate the laser frequency according to

$$f(t) = f_0 + \Delta f [\text{sawtooth}(\alpha t) - 0.5], \quad (5.1)$$

where f_0 is the frequency center, Δf is the scan amplitude, and α is the repetition rate of the sawtooth modulation.

To ensure stable control, the wavemeter analog output (50 Ω source impedance) was passed through a two-stage RC low-pass filter ($R = 150 \Omega$, $C = 47 \mu\text{F}$; cutoff $\approx 22 \text{ Hz}$) before driving the laser servo PZT input (10 k Ω). This filtering suppressed high-frequency fluctuations and yielded a smooth, linear frequency ramp.

Three scan configurations were implemented:

- Run A (scope_0): $f_0 = 333.4007 \text{ THz}$, $\Delta f = 0.011 \text{ THz}$, $\alpha = 0.05 \text{ Hz}$; frequency range: 333.3952–333.4062 THz
- Run B (scope_1): $f_0 = 333.409 \text{ THz}$, $\Delta f = 0.011 \text{ THz}$, $\alpha = 0.055 \text{ Hz}$; frequency range: 333.4035–333.4145 THz
- Run C (scope_2): $f_0 = 333.393 \text{ THz}$, $\Delta f = 0.010 \text{ THz}$, $\alpha = 0.055 \text{ Hz}$; frequency range: 333.388–333.398 THz

During each scan, both the wavemeter readout and the atomic absorption signal were recorded synchronously. The frequency calibration was thus obtained directly from the wavemeter trace, without relying on the oscilloscope scan reference channel.

5.4 Data Processing and Frequency-Axis Mapping

The oscilloscope traces were analyzed using a custom Python script to reconstruct the frequency axis and align the MTS signals from multiple scans.

- Interpolation and smoothing of CH1 to reduce noise and stabilize extrema detection.
- Automatic scan-segment identification by locating the left-hand maximum and right-hand minimum of the smoothed CH1 trace.
- Frequency-axis reconstruction using a linear mapping from the experimental scan boundaries.
- Extraction of CH2 spectra within the identified scan segment, plotted as a function of detuning.
- Verification outputs, including diagnostic plots and printed scan parameters.

The resulting spectra from three separate scans are shown in Fig. 5.2. The signals from `scope_0`, `scope_1`, and `scope_2` are aligned on a common detuning axis, demonstrating the reproducibility of the frequency calibration. The overlapping resonance features confirm that the frequency-axis reconstruction is robust and provides a reliable basis for further fitting and comparison.

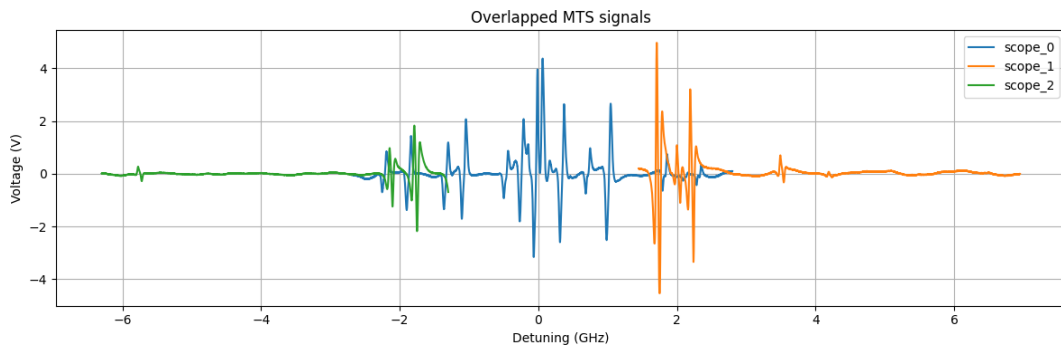


Figure 5.2: Overlapped MTS signals after frequency-axis reconstruction. Data from three independent scans (`scope_0`, `scope_1`, `scope_2`) are aligned on a common detuning axis, confirming the consistency of the calibration procedure.

5.5 Validation and Consistency

To evaluate the reliability of the frequency calibration, the reconstructed spectra obtained from the frequency-axis reconstruction method were compared against those obtained using the wavemeter-servo-controlled scanning approach. Both methods yielded overlapping resonance features within the measurement

uncertainty, demonstrating that the frequency axes established by the two procedures are mutually consistent.

The frequency-axis reconstruction method, based on oscilloscope traces, provides flexibility in dealing with scan drifts and does not rely on external hardware for calibration. In contrast, the wavemeter-driven scheme establishes an absolute frequency scale that is directly traceable to the calibrated wavemeter reference. The agreement between the two methods confirms that the frequency-axis reconstruction does not introduce systematic errors and that the detuning axis derived from the oscilloscope-based procedure is accurate.

Figure 5.3 presents a direct comparison of the two calibration methods. Data from the three oscilloscope scans (scope_0, scope_1, scope_2) were aligned using the frequency-axis reconstruction, while the corresponding wavemeter-driven scans (S3, S4, S9) were processed independently. The overlapping resonance features demonstrate excellent consistency between the two techniques, strengthening confidence in the robustness of the calibration procedure and enabling reliable identification of the hyperfine and isotopic resonances in the 1798 nm $^3P_2 \rightarrow ^3D$ transition.

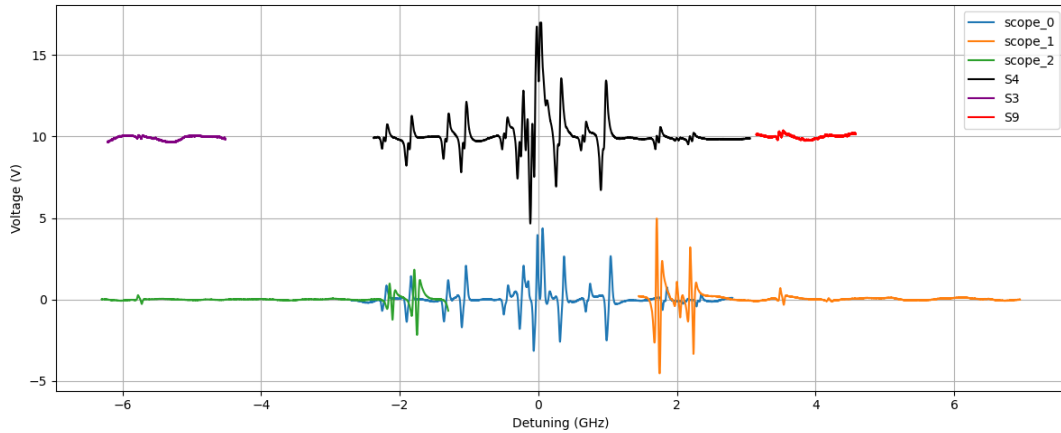


Figure 5.3: Comparison of overlapped MTS signals obtained from two calibration approaches. The traces labeled scope_0, scope_1, and scope_2 correspond to the frequency-axis reconstruction method, while S3, S4, and S9 were calibrated using the wavemeter-driven approach. The agreement of resonance positions across methods confirms the consistency and reliability of the calibration strategy.

5.6 Results and Discussion

5.6.1 Experimental MTS Spectra

Figure 5.4 shows the experimentally recorded modulation transfer spectroscopy (MTS) signal of the 1798 nm $^3P_2 \rightarrow ^3D_3$ transition in ytterbium. The

spectrum presented here serves as the experimental dataset for the analyses described in the following sections. The individual resonance features, isotope assignments, and hyperfine-structure analysis are presented in detail in the subsequent sections.

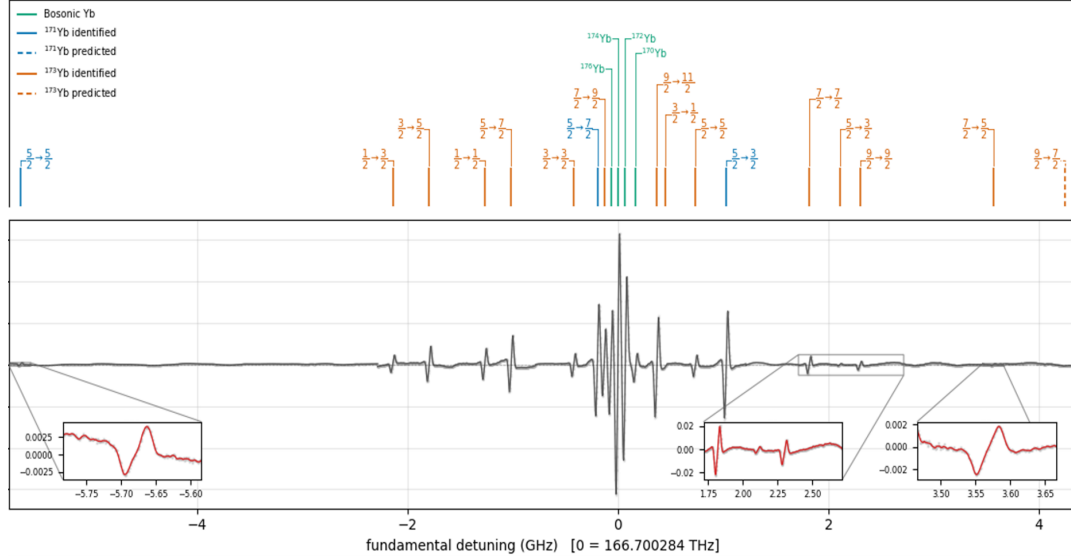


Figure 5.4: Experimental MTS spectrum of the 1798 nm $^3P_2 \rightarrow ^3D_3$ transition in ytterbium.

5.6.2 Preliminary Hyperfine Structure Estimation

To facilitate the identification of the resonance features observed in the experimental MTS spectrum, preliminary hyperfine-structure estimations were first carried out for the fermionic isotopes ^{171}Yb and ^{173}Yb . The calculated spectra provide an initial guide for assigning the observed transitions before a quantitative determination of the hyperfine constants based on stabilized spectroscopy measurements is performed.

As described in Sec. 2.5, the simulated stick spectra were generated using the standard hyperfine-splitting formula. During this preliminary analysis, only the upper-state (3D_3) hyperfine constants were treated as adjustable parameters, while the lower-state (3P_2) constants were fixed to their known values. A global frequency offset was also introduced to align the calculated transition frequencies with the experimentally observed spectral features. The fitted parameters presented in this section were used only as initial estimates to facilitate transition assignment and should not be regarded as the final experimentally determined hyperfine constants.

Figures 5.5 and 5.6 show the resulting simulations for ^{173}Yb and ^{171}Yb , respectively.

¹⁷³Yb

For ¹⁷³Yb, the observed transition pattern could be satisfactorily reproduced using

$$A(^3D_3) \approx -430 \text{ MHz}, \quad B(^3D_3) \approx 909 \text{ MHz}.$$

These values are close to the theoretical predictions reported in Ref. [19] and were adopted only for the purpose of preliminary transition identification. A more accurate determination of the hyperfine constants based on stabilized spectroscopy measurements is presented in section 5.8.

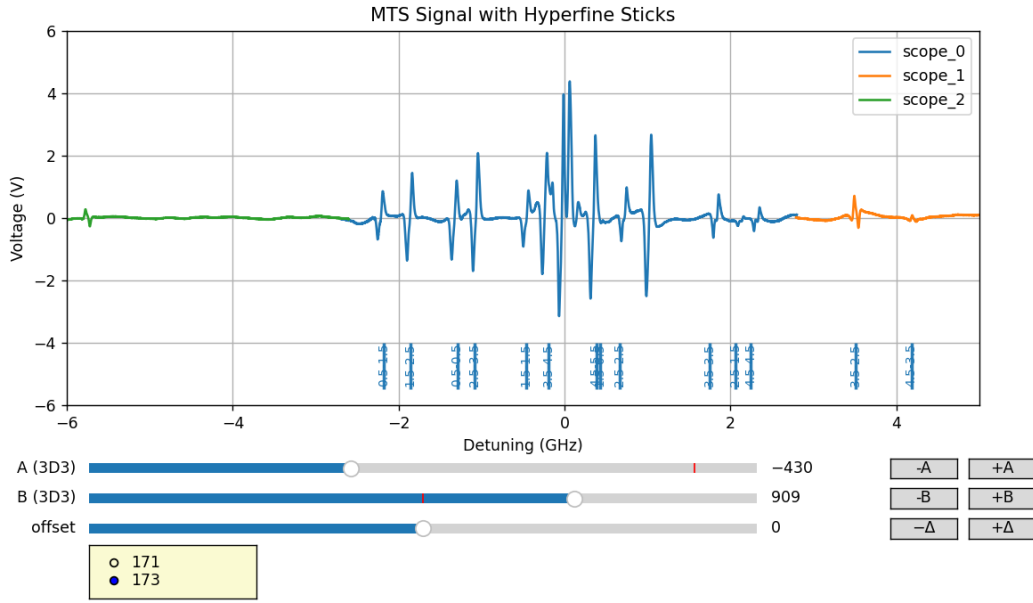


Figure 5.5: Preliminary simulation of the ¹⁷³Yb hyperfine structure at 1798 nm used for transition identification.

¹⁷¹Yb

For ¹⁷¹Yb, the observed transition pattern could be reasonably reproduced using

$$A(^3D_3) \approx 1627 \text{ MHz}.$$

This value is close to the theoretical prediction reported in Ref. [19] and was likewise used only as an initial estimate for assigning the observed transitions. The final value of the hyperfine constant is obtained in a later section using stabilized frequency measurements.

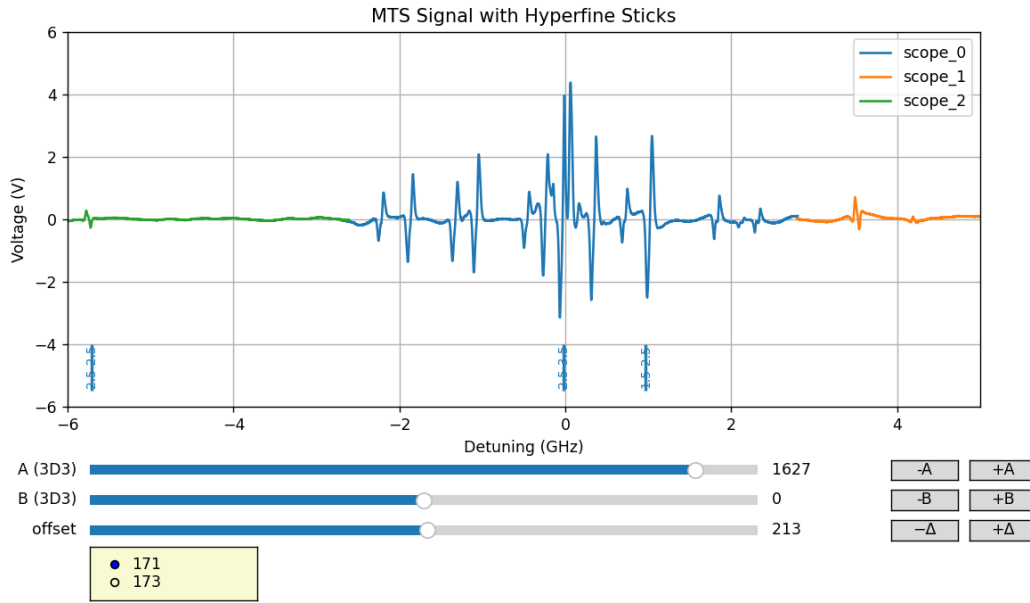


Figure 5.6: Preliminary simulation of the ^{171}Yb hyperfine structure at 1798 nm used for transition identification.

Although these preliminary simulations provide a useful guide for transition assignment, additional experimental verification is required because several resonance features remain partially overlapped. Such verification is provided by the isotope-selective optical pumping measurements described in the following section.

5.7 Isotope-Selective Optical Pumping Verification

Isotope-selective optical pumping was implemented using the 556 nm and 680 nm transitions to provide an independent experimental verification of the transition assignments. Atoms are first excited from the ground state to the 3P_1 state by the 556 nm transition and are subsequently transferred to the metastable 3P_2 state through the 680 nm transition. Since the 1798 nm spectroscopy probes the $^3P_2 \rightarrow ^3D_3$ transition, increasing the population in the metastable 3P_2 state directly enhances the corresponding 1798 nm MTS signal.

Because both the 556 nm and 680 nm transitions are isotope selective, tuning the pumping lasers to a specific isotope or hyperfine transition preferentially transfers only the addressed isotope into the metastable 3P_2 state. Consequently, the corresponding $^3P_2 \rightarrow ^3D_3$ transition exhibits a significant increase in signal amplitude, whereas other isotopes or hyperfine transitions remain largely unaffected. This selective enhancement provides a direct experimental method for verifying

the transition assignments.

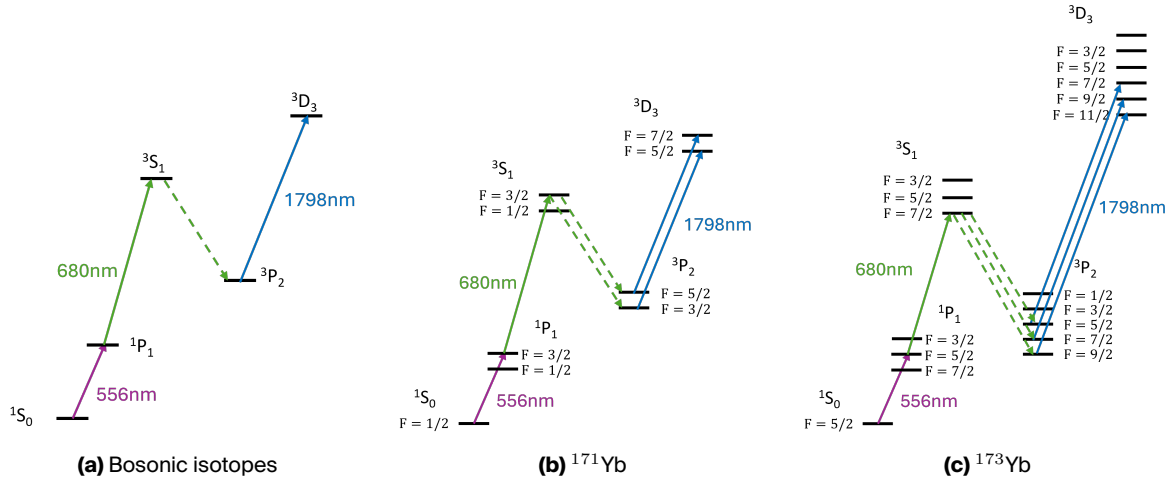


Figure 5.7: Schematic illustration of the isotope-selective optical pumping process used in this work. Atoms are transferred from the ground state to the metastable 3P_2 state through the 556 nm and 680 nm transitions. By tuning the pumping lasers to a selected isotope or hyperfine transition, the population of the corresponding metastable level is selectively increased. As a result, only the associated 1798 nm $^3P_2 \rightarrow ^3D_3$ transition exhibits a pronounced enhancement in the MTS signal, providing an independent verification of the isotope and hyperfine assignments.

To verify this behavior experimentally, 1798 nm modulation-transfer spectroscopy (MTS) spectra were recorded with and without isotope-selective optical pumping. The results are summarized in Fig. 5.8. The upper panel presents the original MTS spectra, while the lower panels show the residual spectra obtained by subtracting the spectrum recorded without optical pumping from that recorded with optical pumping.

For the bosonic isotopes, clear signal enhancement is observed for ^{172}Yb , ^{174}Yb , and ^{176}Yb , while the enhancement of ^{170}Yb is noticeably weaker owing to its relatively low natural abundance. The enhancement appears only at the expected resonance positions of the pumped isotopes, confirming the corresponding isotope assignments.

The advantage of isotope-selective optical pumping is even more pronounced for the fermionic isotopes. Several hyperfine transitions of ^{171}Yb and ^{173}Yb become significantly more distinguishable in the residual spectra after subtraction of the no-pumping signal. Since only the addressed hyperfine transition experiences an increase in population, the corresponding 1798 nm resonance is selectively enhanced, whereas neighboring transitions remain nearly unchanged. This selective enhancement substantially reduces ambiguities arising from spectral overlap and facilitates reliable identification of closely spaced hyperfine components.

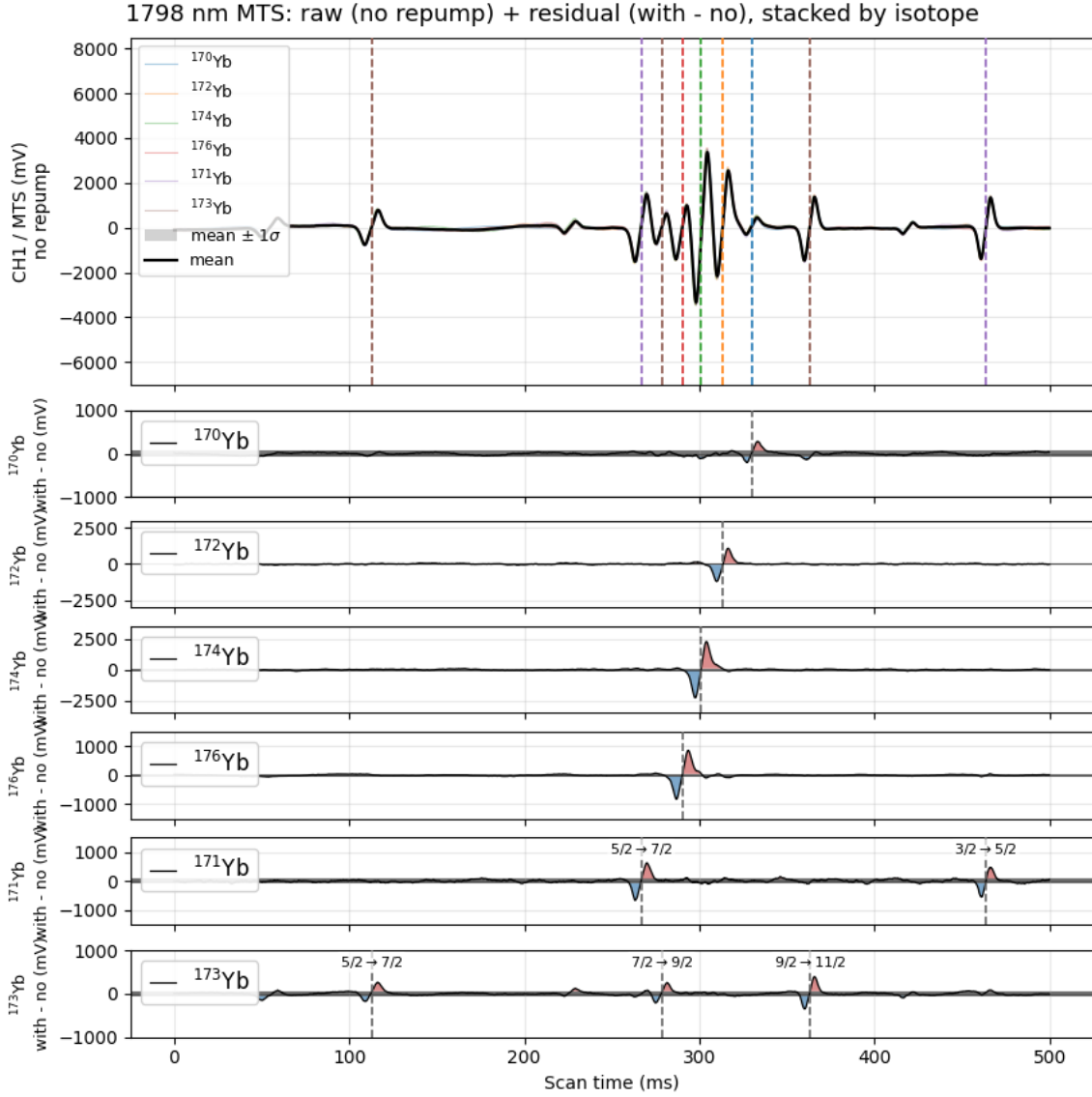


Figure 5.8 : Comparison of 1798 nm modulation-transfer spectroscopy signals measured with and without isotope-selective optical pumping. The residual spectra, obtained by subtracting the spectrum recorded without optical pumping from that recorded with optical pumping, clearly reveal the isotope-dependent enhancement produced by selective population transfer into the metastable 3P_2 state. The enhanced spectral features provide independent experimental verification of both isotope and hyperfine assignments.

Overall, isotope-selective optical pumping provides an independent experimental verification of the isotope and hyperfine states assignments. By selectively increasing the population of the addressed atoms in the metastable 3P_2 state, the corresponding 1798 nm transition is preferentially enhanced, yielding resonance features that are free from ambiguity and minimally affected by overlap with neighboring spectral lines. The resulting isotope-dependent signal enhancement is fully consistent with the preliminary transition assignments and establishes a reliable basis for the laser frequency stabilization and long-term measurements described

in the following sections.

5.7.1 Discussion of Weak Hyperfine Transitions in ^{173}Yb

Following the isotope-selective optical pumping measurements, the transition assignments of the observed 1798 nm spectrum can be established with high confidence. Nevertheless, the experimentally observed amplitudes of several weak hyperfine transitions deserve additional discussion, particularly the $F = 3/2 \rightarrow F' = 1/2$ component. For the $^{173}\text{Yb } ^3P_2 \rightarrow ^3D_3$ transition at 1798 nm, the relative line strengths are governed by the Wigner-Eckart theorem and the corresponding $6j$ coefficients. The calculated transition strengths are summarized in Fig. 2.4.

Among all allowed hyperfine components, the $F = 3/2 \rightarrow F' = 1/2$ transition carries only approximately 0.816% of the total transition strength, making it the weakest allowed transition. By comparison, the strongest transition, $F = 9/2 \rightarrow F' = 11/2$, contributes approximately 28.57%, while the nearby $F = 9/2 \rightarrow F' = 7/2$ transition contributes about 4.41%. Although the $F = 9/2 \rightarrow F' = 7/2$ transition is itself relatively weak, it is nevertheless expected to be considerably stronger than the $F = 3/2 \rightarrow F' = 1/2$ transition based on the calculated transition strengths.

Experimentally, the observable amplitude of the $F = 3/2 \rightarrow F' = 1/2$ transition is further reduced because the neighboring $F = 9/2 \rightarrow F' = 11/2$ resonance, the strongest hyperfine component, produces broad spectral tails that partially obscure the intrinsically weak transition. Consequently, while the $F = 9/2 \rightarrow F' = 7/2$ transition is also relatively weak and could not be reliably resolved in the present measurements, the $F = 3/2 \rightarrow F' = 1/2$ transition is even more difficult to observe owing to its substantially smaller intrinsic transition strength.

Overall, the experimentally observed weak-line behavior can be consistently explained by the combination of the intrinsically weak transition strengths predicted by the Wigner-Eckart theorem and spectral overlap with neighboring strong resonances. This discussion further supports the reliability of the transition assignments established in the previous sections. With the transition assignments confirmed, the identified resonances were subsequently used for laser frequency stabilization and long-term frequency measurements, as described in the following section.

5.8 Laser Frequency Stabilization and Long-Term Measurement

5.8.1 Long-Term Frequency Measurement

After establishing a stable modulation transfer spectroscopy (MTS) signal for the $^{173}\text{Yb } ^3P_2 \rightarrow ^3D_3$ transition and validating the frequency-axis calibration procedures described in Secs. 3–5, the dispersive MTS error signal was used to stabilize the laser frequency. The error signal was fed into a commercial analog laser servo controller (Vescent D2-125) for high-bandwidth noise suppression and long-term frequency stabilization.

The proportional and integral gains, as well as the corner frequencies of the servo, were optimized to achieve a stable lock with a steep error-signal slope while avoiding self-oscillation. Under optimized conditions, the zero crossing of the MTS signal coincided with the center of the selected hyperfine transition and provided a robust frequency reference for long-term operation.

With the locking parameters fixed, the laser was continuously locked to the selected hyperfine components of the 1798 nm transition for typically 1 h in each measurement session, and the measurements were repeated on consecutive sessions over a span of nine days. During this period, the absolute laser frequency was monitored using a wavemeter and recorded at regular intervals. Simultaneously, the error signal and control voltages were logged to verify that the system remained within the linear locking regime.

From this nine-day stabilized data set, the center frequencies of eight resolvable hyperfine components were extracted. These values, together with their relative frequency shifts and absolute frequencies referenced to the wavemeter, are summarized in Table 3. The listed uncertainties include both systematic contributions associated with wavemeter calibration and long-term drift, as well as statistical contributions arising from short-term fluctuations under locked conditions.

5.8.2 Hyperfine-Constant Fitting Based on Measured Hyperfine Transitions

Following the transition assignment and long-term frequency measurements, the experimentally determined hyperfine transition frequencies of ^{171}Yb and ^{173}Yb were used to determine the hyperfine constants of the 3P_2 and 3D_3 states. These hyperfine constants provide a quantitative description of the hyperfine structure

Table 5.1: Hyperfine constants of the 3P_2 and 3D_3 states obtained from the measured hyperfine transitions. For ^{171}Yb ($I = 1/2$), the electric-quadrupole interaction is absent.

Isotope	State	A (MHz)	B (MHz)
^{171}Yb	3P_2	+2601.8	0
	3D_3	+1509.3	0
^{173}Yb	3P_2	-742.18	+1346.7
	3D_3	-433.50	+826.3

of the 1798 nm transition and also enable the frequencies of unresolved hyperfine components to be calculated.

The level shifts were modeled using the standard hyperfine Hamiltonian,

$$E_F = \frac{1}{2}AK + B\frac{\frac{3}{4}K(K+1) - I(I+1)J(J+1)}{2I(2I-1)J(2J-1)}, \quad (5.2)$$

where

$$K = F(F+1) - I(I+1) - J(J+1),$$

and A and B denote the magnetic-dipole and electric-quadrupole hyperfine constants, respectively.

For ^{173}Yb ($I = 5/2$), both magnetic-dipole and electric-quadrupole interactions contribute to the hyperfine structure, whereas for ^{171}Yb ($I = 1/2$), the electric-quadrupole term vanishes and only the magnetic-dipole interaction remains. The measured hyperfine transitions were fitted to determine the hyperfine constants of the 3P_2 and 3D_3 states. The resulting values are summarized in Table 5.1.

The fitted hyperfine constants provide a consistent description of the measured hyperfine structure. Combined with the experimentally determined transition frequencies, they enable a complete set of hyperfine transition frequencies for the 1798 nm transition to be established. The complete set of bosonic resonances and hyperfine transitions employed in the present analysis is summarized in Table 5.2. In addition to the experimentally measured transitions, the table also includes unresolved hyperfine components whose frequencies were calculated from the fitted hyperfine constants. These predicted transitions are identified by “pred.”.

Table 5.2: Measured resonance frequencies of the $^{170-176}\text{Yb } ^3P_2 \rightarrow ^3D_3$ transition at 1798 nm. Frequency shifts are given relative to the ^{174}Yb bosonic resonance. Values marked as “pred.” are calculated from the fitted hyperfine constants.

Isotope	$F \rightarrow F'$	Frequency shift / GHz	Frequency / THz
^{170}Yb	–	+0.1700	166.7004530
^{172}Yb	–	+0.0700	166.7003530
^{174}Yb	–	0.0000	166.7002830
^{176}Yb	–	–0.0645	166.7002185
^{171}Yb	$5/2 \rightarrow 7/2$	–0.1940	166.7000890
	$5/2 \rightarrow 5/2$	–5.4765	166.6948065
	$3/2 \rightarrow 5/2$	+1.0280	166.7013110
^{173}Yb	$1/2 \rightarrow 3/2$	–2.1385	166.6981445
	$3/2 \rightarrow 5/2$	–1.7940	166.6984890
	$1/2 \rightarrow 1/2$	–1.2660	166.6990170
	$5/2 \rightarrow 7/2$	–1.0135	166.6992695
	$3/2 \rightarrow 3/2$	–0.4205	166.6998625
	$7/2 \rightarrow 9/2$	–0.1290	166.7001540
	$9/2 \rightarrow 11/2$	+0.5700	166.7008530
	$3/2 \rightarrow 1/2$	+0.4530 (pred.)	166.7007360 (pred.)
	$5/2 \rightarrow 5/2$	+0.7355	166.7010185
	$7/2 \rightarrow 7/2$	+1.8192	166.7021022
	$5/2 \rightarrow 3/2$	+2.1087	166.7023917
	$9/2 \rightarrow 9/2$	+2.2997	166.7025827
	$7/2 \rightarrow 5/2$	+3.5685	166.7038515
	$9/2 \rightarrow 7/2$	+4.2510 (pred.)	166.7045340 (pred.)

6

Characterization of the Yb $^1S_0 \rightarrow ^1P_1$ Transition at 399nm

6.1 Introduction

The 399 nm $^1S_0 \rightarrow ^1P_1$ transition in neutral ytterbium (Yb) is a strong dipole-allowed cycling transition and serves as the primary entry point for laser cooling, atomic-beam slowing, and first-stage magneto-optical trapping of Yb atoms. Owing to its large natural linewidth and high scattering rate, this transition enables efficient capture of hot atoms from an atomic source and provides the foundation for all subsequent stages of cooling, spectroscopy, and metastable-state manipulation. Establishing a stable and well-characterized frequency reference at 399 nm is therefore essential for reliable operation of the entire experimental platform.

In addition to its role as cooling light, the 399 nm MTS system functions as a frequency reference that defines the absolute detuning of the first-stage cooling process. In this context, the long-term stability and robustness of the locking point are of critical importance. In particular, systematic shifts of the reference spectrum induced by experimental parameters—such as optical power, beam geometry, and modulation conditions—can directly translate into frequency offsets of the cooling laser, leading to variations in MOT performance and, more importantly, to uncontrolled systematic errors that propagate to later experimental stages.

In this work, we present a systematic experimental study of the 399 nm frequency reference system implemented in our laboratory. Doppler-free spectroscopy of the $^1S_0 \rightarrow ^1P_1$ transition was first performed to resolve the isotopic and hyperfine components and to identify suitable spectral features for laser frequency

stabilization. We then investigated the sensitivity of the ^{171}Yb reference spectrum to experimental parameters, with particular emphasis on the pulling of the locking point induced by variations of the pump-beam power.

The goals of this study are:

- to perform Doppler-free spectroscopy of the $399\text{ nm } ^1\text{S}_0 \rightarrow ^1\text{P}_1$ transition and identify appropriate reference features for laser locking;
- to experimentally investigate the pulling of the ^{171}Yb locking point induced by variations of the pump and probe beam powers, and to determine the optimal operating regime that reduces sensitivity to frequency shifts and maximizes locking-point stability.

6.2 Experimental Setup

The 399 nm light used in this work was generated by a commercial fiber laser system (FL-SF-399-1.2-CW). The output beam was delivered to the spectroscopy platform via an optical fiber and collimated at the fiber launcher. After exiting the fiber, the 399 nm beam was routed through multi-wavelength beam-combining optics, including a dichroic mirror (EO 69-898), which integrates the beam into the shared optical layout.

A polarization-based splitting stage consisting of a polarizing beam splitter (Newport 05FC16PB.3) and an achromatic half-wave plate was used to divide the beam into pump and probe components and to continuously control their relative power. The probe beam propagated directly toward the spectroscopy region, while the pump beam was sent through a phase electro-optic modulator (Qubig PM) driven at 21.4 MHz to generate the frequency sidebands required for modulation transfer spectroscopy.

The pump and probe beams were aligned to be counter-propagating and spatially overlapped through a ytterbium hollow-cathode lamp (Hamamatsu Yb-Ne). The discharge current of the lamp was adjustable from 0.2 mA to 1.8 mA , allowing control of the atomic density and discharge conditions. Careful mode matching and overlap were optimized to maximize the modulation transfer signal and suppress residual Doppler background.

After the interaction region, the transmitted probe beam was separated from other wavelengths by dichroic optics and detected using a silicon photodiode (Thorlabs PDA36A). The photodiode signal was mixed with the reference RF signal using a Mini-Circuits mixer to obtain the dispersive MTS error signal. The mixer output

was low-pass filtered and sent to the data-acquisition system and oscilloscope for real-time monitoring and recording.

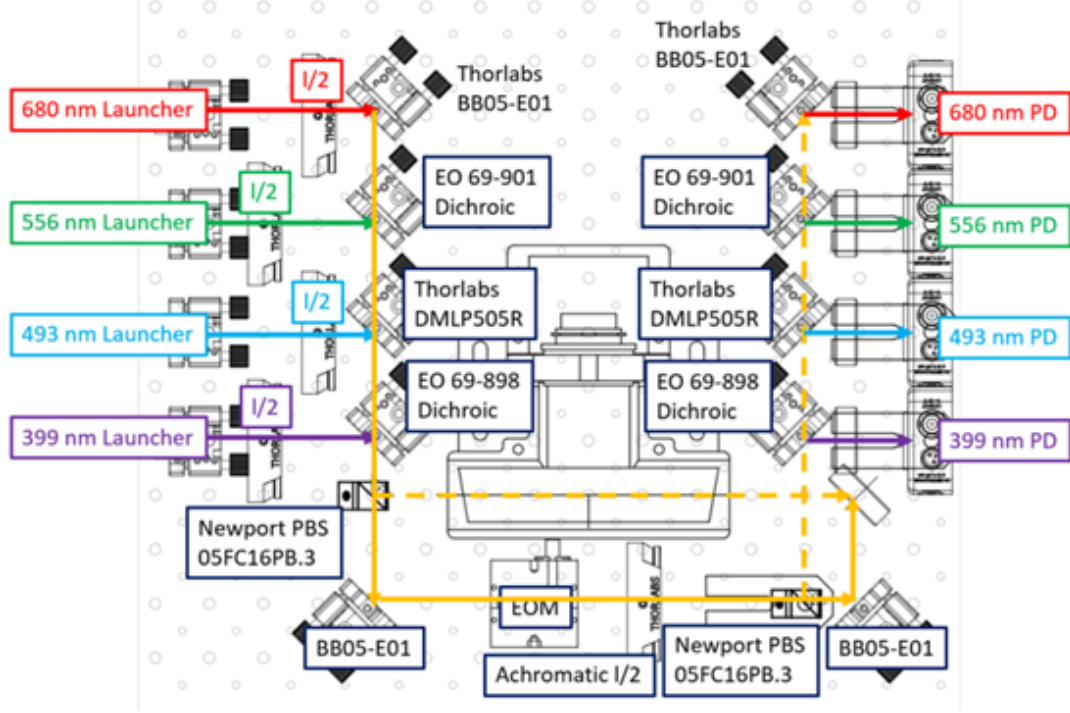


Figure 6.1: 399 nm experiment set up

6.3 399 nm Spectroscopy with Wavemeter-Controlled Frequency Scan

For spectroscopy of the 399 nm $^1S_0 \rightarrow ^1P_1$ transition, the laser frequency was scanned using a wavemeter-based control scheme. The laser was referenced to a HighFinesse wavemeter via its analog control port, allowing the wavemeter to act as an external frequency reference while directly actuating the laser frequency.

A low-frequency triangular waveform was applied to sweep the laser frequency according to

$$f(t) = f_0 + \Delta f [\text{Tri}(\alpha t) - 0.5], \quad (6.1)$$

where f_0 is the center frequency, Δf is the peak-to-peak scan range, and α is the repetition rate of the triangular modulation. Here $\text{Tri}(x)$ denotes a unit triangular waveform ranging from 0 to 1, such that the laser frequency is scanned symmetrically around f_0 over a total range of Δf .

In the experiment the scan parameters were set to

$$f_0 = 187.88166 \text{ THz}, \quad \Delta f = 0.0009 \text{ THz}, \quad \alpha = 0.05 \text{ Hz},$$

corresponding to a symmetric frequency sweep of ± 0.00045 THz around the center frequency.

Because the 399 nm radiation is generated by frequency quadrupling of the infrared source, the effective scan range in the ultraviolet is four times larger, corresponding to ± 1.8 GHz around the 399 nm transition.

During each scan, both the atomic spectroscopy signal and the wavemeter frequency readout were recorded simultaneously using a digital oscilloscope and a data-acquisition system. This allows a direct mapping between the observed spectroscopic signal and the absolute laser frequency.

6.4 Results and Discussion

6.4.1 Experimental MTS Spectra

Figure 6.2 shows a representative modulation transfer spectroscopy (MTS) spectrum of the $^1S_0 \rightarrow ^1P_1$ transition in ytterbium at 399 nm, recorded using the experimental setup described in Sec. 2. The frequency axis is referenced to the ^{174}Yb resonance, which is taken as the zero-detuning point.

Several well-resolved Doppler-free dispersive features are observed across the scanned frequency range. The overall spectral distribution is in good qualitative agreement with the known isotope shifts and hyperfine structure of the 399 nm transition. In particular, the ^{174}Yb line appears as the strongest and most isolated feature, making it a convenient internal frequency reference.

The obtained spectra exhibit a high signal-to-noise ratio and well-defined zero-crossing points, demonstrating that the present MTS system provides a robust Doppler-free signal suitable for quantitative spectral analysis. These data form the basis for the fitting, isotope-shift comparison, and the subsequent investigation of power-induced pulling and locking-point stability discussed in the following sections.

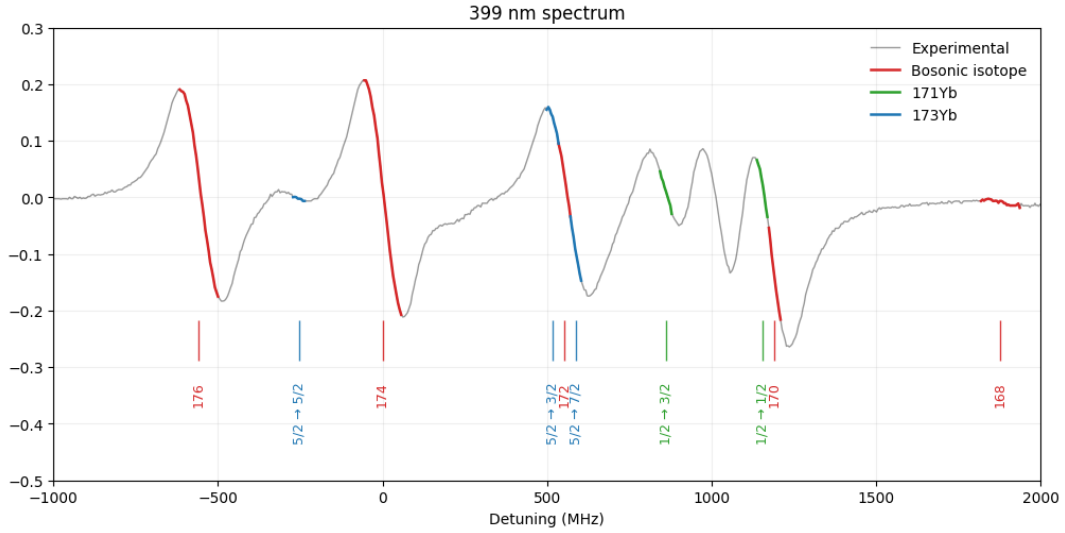


Figure 6.2: Measured MTS dispersive spectrum of the Yb $^1S_0 \rightarrow ^1P_1$ transition at 399 nm. Black curves: experimental signals; frequency axis referenced to the ^{174}Yb line.

6.4.2 Pump-Power-Induced Pulling

To investigate the robustness of the 399 nm frequency reference, we examined the dependence of the Doppler-free spectra on the pump-beam power while keeping the probe-beam power fixed at 0.53 mW and the hollow-cathode lamp (HCL) discharge current fixed at 1.5 mA. Figs. 6.3 show two representative spectra recorded at pump powers of 1.429 mW and 7.8 mW, respectively.

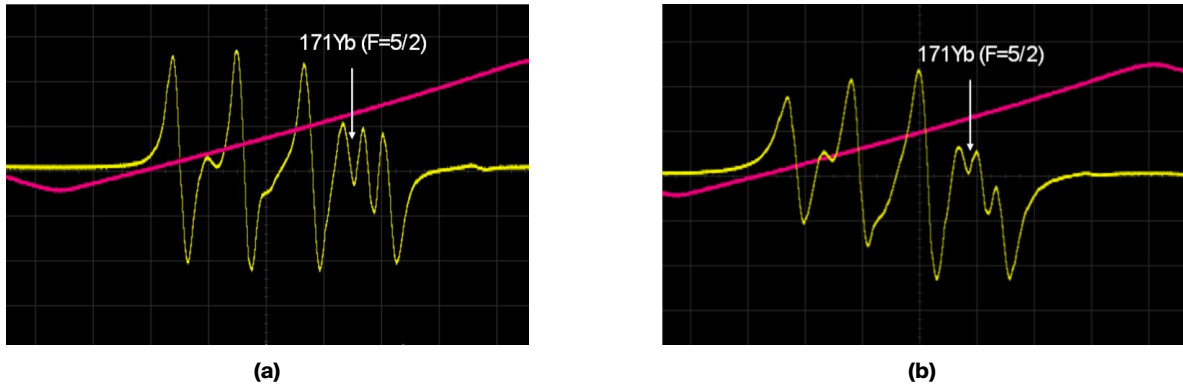


Figure 6.3: Pump-power-induced pulling of the Doppler-free spectra of the 399 nm $^1S_0 \rightarrow ^1P_1$ transition in ytterbium. The probe-beam power was fixed at 0.53 mW in both measurements. (a) Spectrum recorded with a pump power of 1.429 mW, showing well-defined hyperfine features suitable for identifying a locking point. (b) Spectrum recorded with a pump power of 7.8 mW, where strong distortion of the line shape and pronounced shifts of the spectral features are observed. The deformation and apparent displacement of the resonances indicate significant pump-power-induced pulling, which can lead to instability of the locking point if the ^{171}Yb component is used as the frequency reference.

At moderate pump power (1.429 mW), the spectral features associated with the

^{171}Yb hyperfine components are clearly resolved, and the overall line shape remains well defined. The dispersive features exhibit a regular structure suitable for identifying a locking point. In contrast, when the pump power is increased to 7.8 mW, pronounced distortions of the spectral line shape are observed. Several resonances become asymmetric, and the relative amplitudes of neighboring features change significantly. Most importantly, the apparent positions of the spectral features shift, indicating a strong pump-power-induced pulling of the resonance structure.

The pronounced line-shape deformation observed at high pump power indicates that the MTS spectra are sensitive to experimental operating conditions under the present apparatus. While the detailed physical mechanisms underlying this behavior are not addressed here, the strong modification of the spectral profile suggests that variations in pump power and atomic vapor conditions may influence the stability of the dispersive features used for laser locking.

Such power- and discharge-condition-dependent spectral changes raise a practical concern for laser frequency stabilization. If the ^{171}Yb hyperfine component is to be used as a locking reference, it is necessary to experimentally verify whether variations in pump power and hollow-cathode lamp (HCL) discharge current lead to noticeable distortions or shifts of the locking point. The deformation observed at high pump power therefore motivated a systematic experimental investigation of the dependence of the MTS spectra on both optical power and discharge current.

Based on this motivation, we subsequently performed controlled measurements in which the pump and probe beam powers were varied, and additional measurements in which the HCL discharge current was adjusted while scanning the pump power. These experiments were designed to assess the susceptibility of the MTS spectra to pulling effects and, at the same time, to characterize how the spectral line shape evolves with experimentally accessible parameters. The results of this investigation are presented in the following section.

6.4.3 Experimental dependence of the MTS spectra on optical power and discharge current

The pump-power-induced pulling and line-shape distortion observed in the previous section indicate that the 399 nm MTS spectra and the locking-point position are sensitive to the operating parameters of the spectroscopy system. In order to assess the suitability of this system as a frequency reference, it is therefore necessary to experimentally examine how controllable parameters affect the spectral behavior and the robustness of the zero-crossing point.

To this end, two sets of parameter-scan experiments were performed. In the first study, the probe-beam power was varied and Doppler-free MTS spectra were recorded at each setting. The resulting spectra were compared to examine changes in the overall line shape, thereby evaluating the sensitivity of the MTS signal to variations in probe power.

In the second study, the hollow-cathode lamp (HCL) discharge current was adjusted to modify the effective atomic density in the interaction region. For each selected discharge current, the pump-beam power was systematically varied and the corresponding MTS spectra were recorded. The resulting data were used to observe how the spectral line shape and the occurrence of pulling effects depend on the combined choice of pump power and discharge current. Since the internal buffer-gas pressure of the HCL is fixed and not externally tunable, the discharge current serves as the primary experimental control parameter for the vapor environment.

Together, these measurements provide an experimental characterization of the dependence of the 399 nm MTS spectra on probe power, pump power, and discharge current under the operating conditions of the present apparatus, and establish practical constraints for laser frequency stabilization.

Influence of HCL discharge current on the optimal pump power

In the previous section, the effect of pump-beam power on the MTS line shape was investigated while keeping both the probe-beam power and the hollow-cathode lamp (HCL) discharge current fixed. In addition to the optical power, however, another experimentally controllable parameter that can influence the MTS signal is the discharge current of the HCL. Since the gas pressure inside the lamp is fixed by the lamp design and was not intentionally modified in this experiment, the discharge current serves as the primary parameter for controlling the atomic density within the discharge.

To investigate how the discharge conditions influence the MTS spectra, we varied the HCL current while monitoring the signal on the oscilloscope. It was observed that the spectral profile changes noticeably as the discharge current is adjusted. Consequently, under different HCL operating conditions, the pump-beam power must be re-optimized in order to obtain a well-defined MTS signal.

The optimization criterion used in this work follows the guideline proposed by Tanabe *et al.* [20]. In their measurements of the $^1S_0 \rightarrow ^1P_1$ transition in ytterbium at 399 nm, an optimal MTS signal is characterized by three prominent dispersive features whose peak heights are approximately equal. These three features origi-

nate from the carrier and the first-order modulation sidebands. When their amplitudes become comparable, the nonlinear wave-mixing process responsible for the MTS signal is considered to be balanced, and distortions caused by crossover resonances are significantly reduced. In practice, this condition is identified by observing that the tops of the three rightmost dispersive peaks are approximately aligned.

The experimental procedure therefore proceeded as follows. For a given HCL discharge current, the pump-beam power was gradually adjusted while observing the MTS signal on the oscilloscope. The pump power was tuned until the spectral profile satisfied the symmetry criterion described above. The corresponding pump power was then recorded as the optimal value for that particular discharge current.

Figure 6.4 shows representative MTS spectra obtained at different HCL discharge currents, where in each case the pump-beam power was adjusted to satisfy the symmetry condition described above. As the discharge current increases, the overall signal amplitude grows due to the higher atomic density inside the lamp. At the same time, the spectral profile becomes increasingly sensitive to pump power, and noticeable distortions appear if the pump power is not appropriately reduced.

This behavior can be understood from the physical mechanism responsible for the MTS signal. Modulation transfer spectroscopy relies on nonlinear wave-mixing processes in the atomic medium, and the strength of this nonlinear interaction scales with the atomic density inside the lamp. Increasing the discharge current enhances sputtering at the cathode and therefore increases the population of neutral ytterbium atoms in the discharge. As the atomic density rises, the nonlinear susceptibility of the medium becomes larger, allowing strong MTS signals to be generated even at lower optical intensities. Under these conditions, excessive pump power can readily drive the system into a regime where crossover resonances and nonlinear effects distort the spectral line shape and shift the apparent zero-crossing point. Consequently, the pump power must be reduced at higher discharge currents in order to maintain a balanced and symmetric dispersive profile.

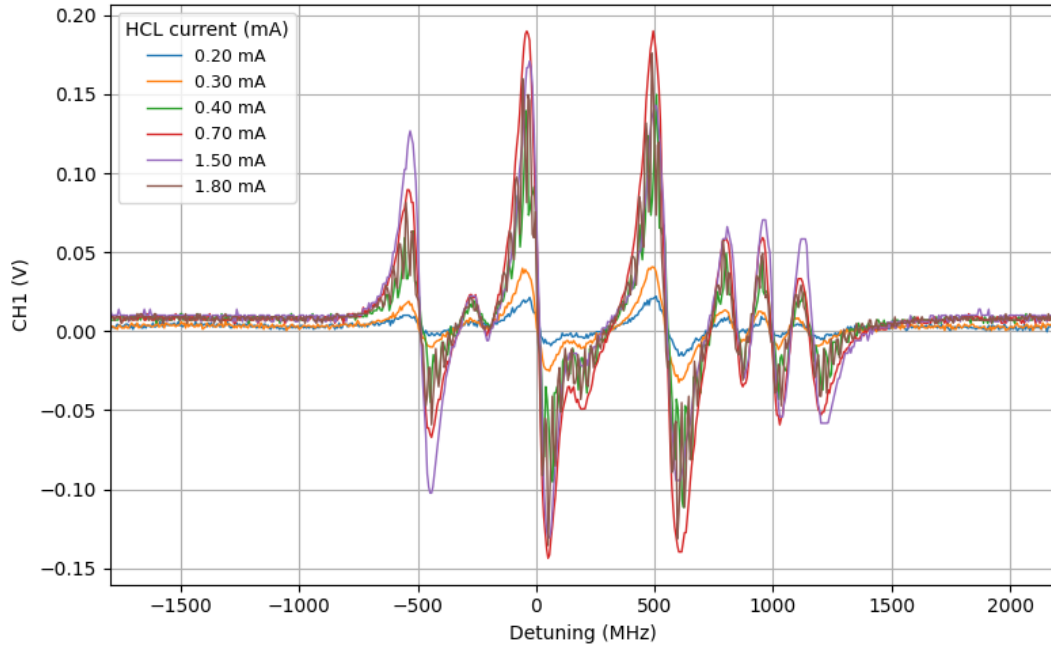


Figure 6.4: Overlay of representative Doppler-free MTS spectra recorded at different hollow-cathode lamp discharge currents. For each current, the pump power was adjusted to obtain a symmetric dispersive profile following the criterion of Tanabe *et al.* [20].

The optimal pump powers obtained for different discharge currents are summarized in Fig. 6.5. A clear inverse relationship is observed: as the HCL discharge current increases, the pump power required to obtain a well-behaved MTS spectrum decreases. This trend is consistent with the increase in atomic density and the resulting enhancement of nonlinear wave-mixing processes inside the lamp.

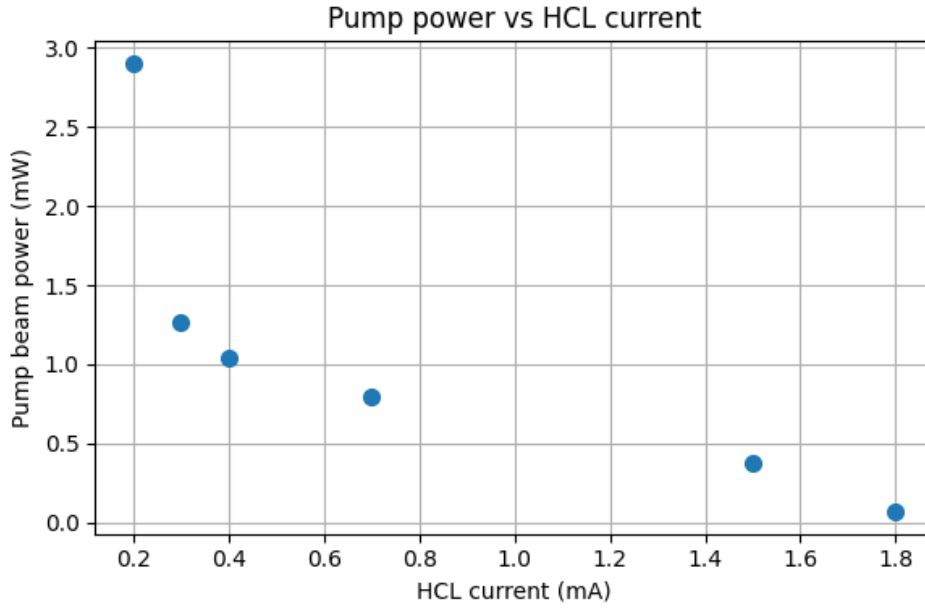


Figure 6.5: Optimal pump-beam power as a function of the hollow-cathode lamp discharge current. The required pump power decreases as the discharge current increases, reflecting the enhanced nonlinear interaction at higher atomic densities.

These results demonstrate that the operating conditions of the hollow-cathode lamp play an important role in determining the optimal pump power for MTS spectroscopy. In practice, whenever the discharge current is changed, the pump-beam power should be re-adjusted to ensure that the spectral profile remains symmetric and minimally affected by crossover-induced distortions. This adjustment is essential for maintaining a stable and reliable frequency reference for subsequent laser-locking experiments.

6.5 Conclusion

In this work, we performed a systematic experimental characterization of the neutral ytterbium $^1S_0 \rightarrow ^1P_1$ transition at 399 nm using a modulation transfer spectroscopy (MTS) system implemented in a hollow-cathode lamp (HCL).

Although the 399 nm transition of ytterbium has been extensively studied and widely used in many laboratories, the primary objective of this work was not to reproduce known results, but rather to establish and validate a frequency reference under the specific optical configuration, discharge conditions, and operating environment of our laboratory. Through systematic measurements and parameter scans, we constructed an experimentally grounded characterization of the 399 nm reference tailored to our platform, providing a reliable foundation for subsequent cooling, trapping, and precision-spectroscopy experiments.

A detailed investigation of pump-power-induced pulling revealed that the MTS spectra are highly sensitive to the operating conditions. At elevated pump powers, pronounced line-shape deformation and apparent shifts of spectral features were observed, indicating strong pulling effects that can compromise the stability of the locking point. Overlay measurements showed that while global spectral distortion increases significantly at high pump powers, the region around the ^{171}Yb hyperfine component is comparatively less affected. Nevertheless, operation in the low pump-power regime yields substantially cleaner and more reproducible dispersive profiles and is therefore preferable for robust long-term laser stabilization.

The dependence of the spectral stability on the HCL discharge current was also systematically examined. An inverse relationship between the optimal pump power and the discharge current was identified: higher discharge currents require lower pump powers to maintain an undistorted spectral profile. This behavior highlights the critical role of atomic density and nonlinear processes in shaping the MTS signal and demonstrates that any change in discharge conditions necessitates a re-optimization of the optical powers to preserve a reliable locking reference.

Overall, this study establishes practical operating regimes for a stable 399 nm ytterbium frequency reference under our laboratory conditions and provides experimental guidelines for minimizing power-induced pulling and spectral distortion. The resulting system not only serves as a reliable reference for first-stage cooling and routine operation, but also constitutes a traceable atomic resonance standard that can be used to calibrate and verify the absolute frequency readout of the wavemeter, thereby improving the long-term consistency and accuracy of the entire experimental platform.

7

Characterization of the Yb $^3P_2 \rightarrow 6s6d\ ^3D_3$ Transition at 493nm

7.1 Introduction

The $6s6p\ ^3P_2 \rightarrow 6s6d\ ^3D_3$ transition in neutral ytterbium (Yb) at 493 nm plays an important role in metastable-manifold detection and manipulation. As discussed in Chapter 2, this transition is suitable for fluorescence detection for imaging atoms prepared in the long-lived 3P_2 state and is therefore relevant to metastable-state cooling, detection, and repumping schemes. In addition to its application in fluorescence imaging, the 493 nm transition is also useful for spectroscopy and state-selective diagnostics within the triplet manifold. Accurate identification of its isotope and hyperfine spectral structure is important for establishing reliable laser-frequency references for future experiments involving metastable ytterbium atoms.

Compared with widely studied ytterbium transitions such as the $^1S_0 \rightarrow ^1P_1$ transition at 399 nm and the $^1S_0 \rightarrow ^3P_1$ intercombination transition at 556 nm, systematic high-resolution spectroscopy of the $^3P_2 \rightarrow 6s6d\ ^3D_3$ transition at 493 nm remains relatively limited. In particular, a complete isotope-resolved Doppler-free spectrum of this transition has not yet been fully established. Nevertheless, hyperfine-structure investigations of the excited $6s6d\ ^3D_3$ state in ^{171}Yb and ^{173}Yb have previously been reported by Baumann and Braun [21]. Their work provided the magnetic dipole (A) and electric quadrupole (B) hyperfine coefficients of the excited $6s6d$ states, which serve as important references for estimating the expected transition frequencies. Together with previously reported hyperfine constants of the 3P_2 state, these known coefficients provide a basis for identifying

the observed spectral features.

Following the analysis procedure developed in previous chapters, theoretical hyperfine transition patterns can therefore be compared with the measured spectra to perform preliminary spectral assignment and transition identification. The present analysis is intended primarily for transition identification and experimentally useful frequency referencing, rather than precision determination of hyperfine coefficients.

The objectives of this work are as follows:

1. To investigate the isotope and hyperfine spectral structure of the transition using modulation transfer spectroscopy.
2. To perform preliminary spectral identification based on previously reported hyperfine coefficients.
3. To provide experimentally useful frequency references for future metastable-state imaging and cooling experiments.

7.2 Experimental Setup

The 493 nm light used in this work was generated using a Moglab external-cavity diode laser (ECDL). The laser output was delivered to the spectroscopy platform through an optical fiber and collimated at the fiber launcher. After exiting the fiber, the 493 nm beam was routed through the shared optical layout using multi-wavelength beam-combining optics, including dichroic mirrors for wavelength separation and recombination.

A polarization-based beam-splitting stage consisting of a polarizing beam splitter (Newport 05FC16PB.3) and an achromatic half-wave plate was used to divide the beam into pump and probe components while continuously controlling their relative optical powers. The probe beam propagated directly toward the spectroscopy region, while the pump beam was sent through a phase electro-optic modulator (Qubig PM) driven at 21.4 MHz to generate the modulation sidebands required for modulation transfer spectroscopy (MTS).

The pump and probe beams were aligned in a counter-propagating geometry and spatially overlapped within a ytterbium hollow-cathode lamp (Hamamatsu Yb-Ne). The hollow-cathode lamp discharge current was adjusted according to the observed signal amplitude and signal-to-noise ratio, typically within the range of 11 mA to 16 mA. Careful beam alignment and mode matching were optimized to maximize the MTS signal while suppressing residual Doppler background and unwanted interference signals.

After passing through the interaction region, the transmitted probe beam was separated from other wavelengths using dichroic optics and detected with a silicon photodiode (Thorlabs PDA36A). The photodiode output was mixed with the reference RF signal using a Mini-Circuits mixer to generate the dispersive MTS error signal. The mixer output was subsequently low-pass filtered and recorded using an oscilloscope and data-acquisition system for real-time monitoring and further spectral analysis.

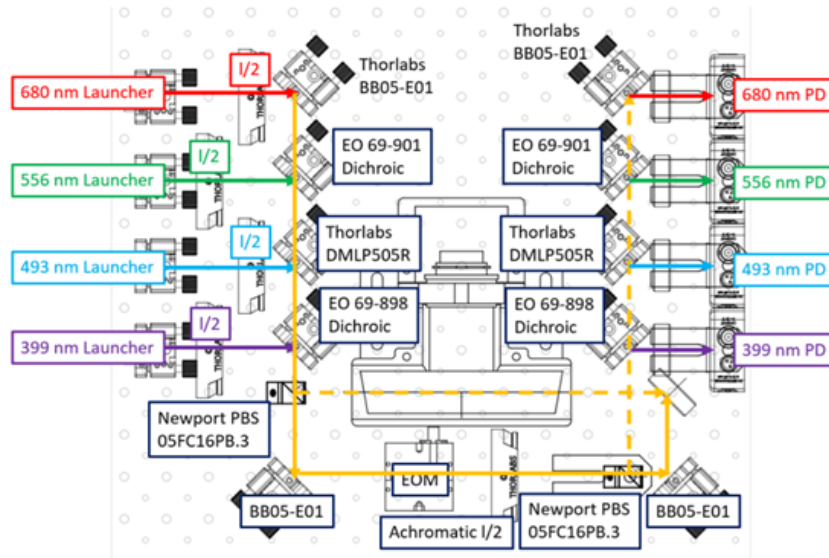


Figure 7.1: 493 nm experiment set up

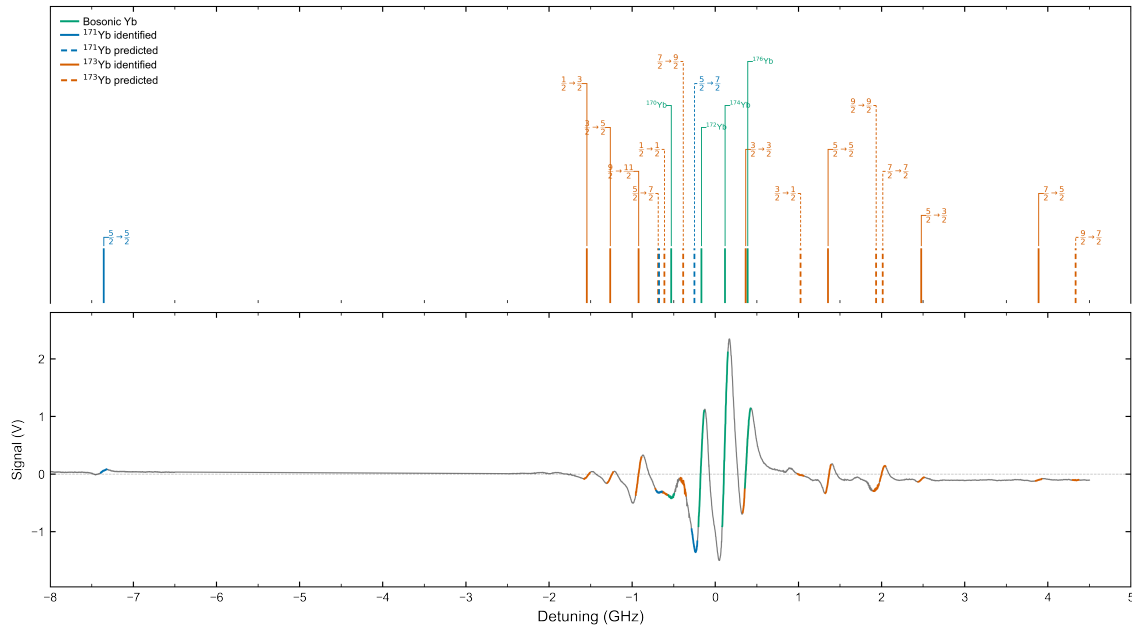
7.3 Experimental Spectra

The experimentally measured modulation transfer spectroscopy (MTS) signal of the $^3P_2 \rightarrow 6s6d, ^3D_3$ transition at 493 nm is shown in Fig. 7.2. The spectrum contains both bosonic and fermionic spectral features distributed across the scanning range. Compared with the relatively isolated spectral structures observed in some previously studied ytterbium transitions, the present 493 nm spectrum exhibits significantly stronger spectral congestion and overlap. In particular, the fermionic hyperfine components of ^{171}Yb and ^{173}Yb partially overlap with neighboring bosonic isotope transitions, making direct spectral assignment difficult.

Nevertheless, several major spectral features can still be identified qualitatively. The strongest unresolved feature is expected to correspond primarily to the ^{174}Yb transition because of its highest natural abundance among the stable ytterbium isotopes. Additional weaker structures are attributed to overlapping fermionic hyperfine transitions and neighboring bosonic isotopes.

The observed spectrum also exhibits residual asymmetry and broadened spectral structures, which are likely caused by a combination of residual amplitude modulation (RAM), unresolved hyperfine overlap, discharge broadening inside the hollow-cathode lamp, and imperfect pump-probe mode matching. These effects further increase the difficulty of reliable centroid extraction and isotope identification.

Because of the complexity of the spectral structure, the present work focuses primarily on preliminary transition identification and consistency analysis rather than precision extraction of hyperfine coefficients.



ing the centroids of ^{171}Yb and ^{173}Yb , the small unresolved feature located on the far-right side of the spectrum was tentatively assigned to ^{176}Yb . The assignments of ^{170}Yb and ^{172}Yb were significantly more ambiguous because of spectral overlap and insufficient fitting constraints. Therefore, additional consistency checks were required.

7.4.1 Ambiguity in the ^{172}Yb Assignment

The first hypothesis was that the small peak located between the ^{171}Yb and ^{173}Yb centroids corresponded to the ^{172}Yb transition. However, according to the hyperfine fitting results obtained using the reported A and B coefficients, this spectral feature was also predicted to correspond to the $^{173}\text{Yb } F = 7/2 \rightarrow F' = 9/2$ transition. To evaluate this possibility, the relative transition strength was estimated using Wigner-coefficient calculations. The calculated relative intensity of the $^{173}\text{Yb } 7/2 \rightarrow 9/2$ transition was approximately 19.4 %, suggesting that the observed feature is likely dominated by the expected fermionic hyperfine transition rather than an isolated bosonic resonance. A second interpretation was therefore considered, namely that the ^{172}Yb transition partially overlaps with the $^{173}\text{Yb } 7/2 \rightarrow 9/2$ hyperfine component. Such overlap is physically plausible because spectral overlap between bosonic and fermionic isotope transitions is commonly observed in ytterbium spectroscopy. For example, partial overlap between ^{172}Yb and ^{173}Yb hyperfine components has also been observed in the 399 nm transition. Under this assumption, the spectral feature located between the ^{171}Yb and ^{173}Yb centroids was used as a tentative estimate for the ^{172}Yb transition frequency. However, the hyperfine fitting results further suggest that this feature may also contain contributions from the $^{171}\text{Yb } F = 5/2 \rightarrow F' = 7/2$ transition.

Consequently, the fitted center should be regarded as a potentially unresolved mixture of ^{172}Yb and neighboring fermionic transitions, rather than a uniquely identified bosonic resonance.

7.4.2 King-Plot Consistency Check

To further evaluate the validity of the tentative assignment, preliminary King-plot analysis was performed using both ^{172}Yb and ^{174}Yb as reference isotopes. The resulting predictions for the expected ^{170}Yb position did not converge consistently. This indicates that the uncertainty associated with the present ^{172}Yb assignment remains significant. In addition, the instability of the King-plot prediction is also affected by the uncertainty in the ^{173}Yb centroid determination. Because the

fermionic spectra contain partially unresolved hyperfine structures, the extracted centroid depends strongly on the fitting procedure and transition assignment. Small centroid shifts can significantly affect the modified isotope shifts used in the King plot, leading to large variations in both the fitted slope and the extrapolated ^{170}Yb position. These results suggest that a more reliable determination of the ^{172}Yb transition frequency is required before meaningful King-plot predictions can be obtained.

7.5 Isotope-Selective Optical Pumping

As discussed in the previous sections, the assignment of the ^{172}Yb transition remains uncertain because of substantial spectral overlap with neighboring fermionic hyperfine components. Consequently, the preliminary King-plot analysis is also unable to provide a definitive validation of the proposed assignment, since the predicted isotope shifts depend sensitively on the assumed ^{172}Yb transition frequency. An independent experimental approach that does not rely on spectral fitting or King-plot analysis is therefore desirable.

To address this issue, isotope-selective optical pumping was employed. As described in Chapter 5, this technique selectively increases the population of a chosen isotope in the metastable 3P_2 state. By tuning the 556 nm and 680 nm repumping lasers to a specific isotope, only the addressed isotope experiences a significant increase in the metastable-state population. If a spectral feature indeed originates from that isotope, the corresponding 493 nm transition is expected to exhibit a measurable enhancement, whereas neighboring isotope transitions remain largely unaffected. The same repumping concept is adopted here to provide an independent experimental verification of the tentative bosonic-isotope assignments in the 493 nm spectrum.

Atoms are first excited from the ground state to the 3P_1 state by the 556 nm transition and are subsequently transferred to the metastable 3P_2 state through the 680 nm transition. In principle, the same method can also be applied to the fermionic isotopes. However, this would require repeatedly relocking the repumping lasers to different hyperfine transitions. Because the wavemeter-based frequency-locking system used in the present 493 nm experiment was not well suited to frequent relocking, the isotope-selective repumping measurements were restricted to the four bosonic isotopes.

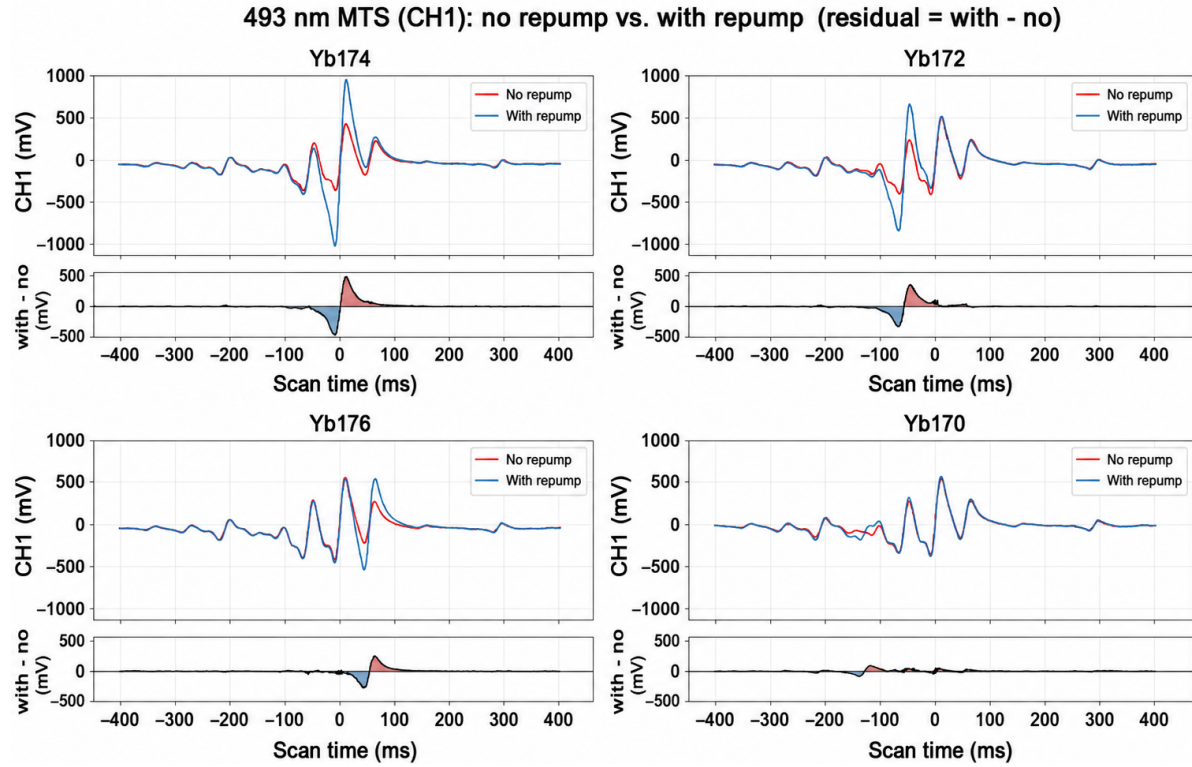


Figure 7.3: Comparison of 493 nm modulation-transfer spectroscopy signals measured with and without isotope-selective optical pumping for the four bosonic isotopes ^{170}Yb , ^{172}Yb , ^{174}Yb , and ^{176}Yb . The red traces correspond to measurements without optical pumping, while the blue traces were obtained with isotope-selective optical pumping. The lower panels show the residual signal calculated as (with pumping) minus (without pumping). Significant enhancement is observed for ^{170}Yb , ^{172}Yb , ^{174}Yb , and ^{176}Yb after isotope-selective repumping is applied.

For each isotope, two spectra were recorded under otherwise identical experimental conditions: one with the repumping lasers blocked and one with the isotope-selective repumping lasers applied. To facilitate direct comparison of the signal amplitudes, the spectra were aligned using cross-correlation analysis, and the residual signal was obtained by subtracting the spectrum recorded without repumping from that recorded with repumping. The resulting spectra are summarized in Fig. 7.3. The upper panels show the measured MTS spectra, while the lower panels display the residual enhancement induced by isotope-selective repumping.

Clear enhancement of the MTS signal is observed for all four bosonic isotopes after isotope-selective repumping is applied. In each case, the residual spectrum exhibits a characteristic dispersive profile centered on the resonance position, indicating that the increased population transferred into the metastable 3P_2 state contributes directly to the observed 493 nm transition signal. As expected, the enhancement is strongest for ^{174}Yb , consistent with its highest natural abundance among the stable yttrium isotopes, whereas the enhancement observed for

^{170}Yb is considerably weaker owing to its much lower abundance.

More importantly, the isotope-selective enhancement provides an independent means of identifying the origin of partially overlapped spectral features. When the repumping lasers are tuned to a particular isotope, only the corresponding resonance exhibits a significant increase in signal amplitude, while neighboring resonances remain nearly unchanged. Consequently, the observed enhancement directly confirms that the addressed spectral feature contains a contribution from the selected isotope, independent of the spectral fitting procedure.

For the particularly ambiguous ^{172}Yb resonance discussed in the previous sections, a clear enhancement is observed when the ^{172}Yb repumping scheme is applied. This result provides independent experimental evidence that the corresponding spectral feature indeed contains a substantial contribution from the ^{172}Yb transition. Although partial overlap with neighboring fermionic hyperfine components cannot be completely excluded, the isotope-selective response significantly strengthens the confidence in the tentative assignment.

Overall, isotope-selective repumping provides an independent verification of the tentative bosonic-isotope assignments in the 493 nm spectrum. The isotope-dependent enhancement observed in the repumped spectra confirms that the addressed resonances contain contributions from the selected isotopes, thereby providing additional confidence in the proposed assignments despite the substantial spectral overlap.

8

King-Plot Analysis

8.1 Motivation

The purpose of the King-plot analysis is to exploit the linear relationship between isotope shifts measured in two different atomic transitions. This linearity provides a powerful tool to separate mass-shift and field-shift contributions, to test the internal consistency of isotope-shift datasets, and to probe possible nonlinearities that may arise from higher-order nuclear or electronic effects or from interactions beyond the Standard Model. In this work, King plots are used primarily as a cross-transition consistency test and as a method to infer the unknown frequency position of the ^{168}Yb transition.

8.2 Isotope shift, mass shift, field shift, and modified isotope shift

For a given atomic transition t , the isotope shift between two isotopes A and A_0 is defined as the frequency difference

$$\delta\nu_t^{(A,A_0)} = \nu_t(A) - \nu_t(A_0). \quad (8.1)$$

Physically, the isotope shift originates from two main effects: (i) the dependence of the electronic motion on the finite nuclear mass, and (ii) the modification of the electronic wavefunctions due to changes in the nuclear charge distribution. To leading order, the isotope shift can therefore be expressed as the sum of a mass

shift (MS) and a field shift (FS) contribution,

$$\delta\nu_t^{(A,A_0)} = K_t \delta\left(\frac{1}{M}\right)^{(A,A_0)} + F_t \delta\langle r^2 \rangle^{(A,A_0)}. \quad (8.2)$$

Here K_t is the mass-shift coefficient, which contains both the normal mass shift and the specific mass shift. The coefficient F_t is the field-shift coefficient, which quantifies the sensitivity of the electronic transition to changes in the nuclear charge radius. The quantity $\delta\langle r^2 \rangle^{(A,A_0)}$ denotes the difference in the mean-square nuclear charge radius between isotopes A and A_0 . The mass-dependent factor is given by

$$\delta\left(\frac{1}{M}\right)^{(A,A_0)} = \frac{1}{m_A} - \frac{1}{m_{A_0}} = \frac{m_{A_0} - m_A}{m_A m_{A_0}}. \quad (8.3)$$

To remove this nonlinear mass dependence and place isotope shifts from different isotope pairs on a common linear footing, one defines the mass factor

$$\mu^{(A,A_0)} = \frac{m_A m_{A_0}}{|m_A - m_{A_0}|}, \quad (8.4)$$

and introduces the modified isotope shift

$$\text{mIS}_t^{(A,A_0)} = \mu^{(A,A_0)} \delta\nu_t^{(A,A_0)}. \quad (8.5)$$

Substituting the leading-order isotope-shift expression yields

$$\text{mIS}_t = -K_t + F_t \mu^{(A,A_0)} \delta\langle r^2 \rangle^{(A,A_0)}. \quad (8.6)$$

For two different transitions i and j measured on the same set of isotope pairs, the nuclear parameter $\mu^{(A,A_0)} \delta\langle r^2 \rangle^{(A,A_0)}$ is common. Eliminating it leads directly to the linear King-plot relation

$$\text{mIS}_j = a + b \text{mIS}_i, \quad (8.7)$$

with

$$b = \frac{F_j}{F_i}, \quad a = -K_j + \frac{F_j}{F_i} K_i. \quad (8.8)$$

Thus, in a King plot of mIS_j versus mIS_i , all isotope pairs are expected to lie on a straight line within the leading-order approximation. Any statistically significant deviation from linearity would indicate the presence of higher-order effects beyond the simple MS+FS model, such as higher-order nuclear moments [22], relativistic corrections [18], or possible new-physics contributions [23, 24, 25].

8.3 Experimental and Literature Datasets

First, long-term frequency-locking measurements were performed for both the 680 nm and 1798 nm transitions. The locked frequencies were averaged to obtain stable isotope-dependent absolute frequencies. The corresponding isotope shifts relative to ^{172}Yb were then extracted and converted into modified isotope shifts.

Second, high-precision absolute-frequency Yb neutral atom atomic transition data at 578 nm, 431 nm, and 770 nm were taken from the literature [26, 27, 28]. From these values, isotope shifts relative to ^{172}Yb were calculated and converted into modified isotope shifts in the same manner.

In addition, literature isotope-shift data for Yb trapped ion atomic transition at the 411 nm and 467 nm transitions [29] were incorporated to extend the King-plot comparison network and provide additional consistency checks among different transitions. All datasets were expressed in terms of modified isotope shifts using a common reference isotope and subsequently used to construct the King plots presented in this chapter.

8.4 King plots of the 680 nm transition

King plots were constructed to examine the consistency of the measured isotope shifts of the 680 nm transition with previously reported transitions in both neutral and singly ionized ytterbium. Modified isotope shifts were calculated using ^{174}Yb as the reference isotope.

Five independent comparisons were performed using literature isotope-shift data at 770 nm, 431 nm, 578 nm, 411 nm, 467 nm, and the present 493 nm measurement. The 770 nm, 431 nm, and 578 nm transitions correspond to neutral ytterbium, whereas the 411 nm and 467 nm transitions are the electric-quadrupole (E2) and electric-octupole (E3) clock transitions of Yb^+ reported in Ref. [30]. The resulting King plots are summarized in Fig. 8.1.

For each transition pair, the modified isotope shifts are found to follow a nearly linear relationship, indicating that the present 680 nm isotope-shift measurements are consistent with previous data within the current experimental precision. In particular, the 680–770 nm comparison exhibits a slope close to unity, reflecting the similar electronic sensitivities of these two repumping transitions.

The King plots involving the ionic 411 nm and 467 nm clock transitions further provide a cross-species consistency check between neutral Yb and Yb^+ . The overall linear behavior observed in all comparisons supports the validity of the King-plot

formalism and confirms the reliability of the measured isotope shifts at 680 nm.

Although the 680–493 nm King plot also exhibits an approximately linear trend, noticeable deviations are observed for the fermionic isotopes ^{171}Yb and ^{173}Yb . These deviations are likely associated with unresolved hyperfine components and the comparatively larger uncertainties of the 493 nm transition measurements.

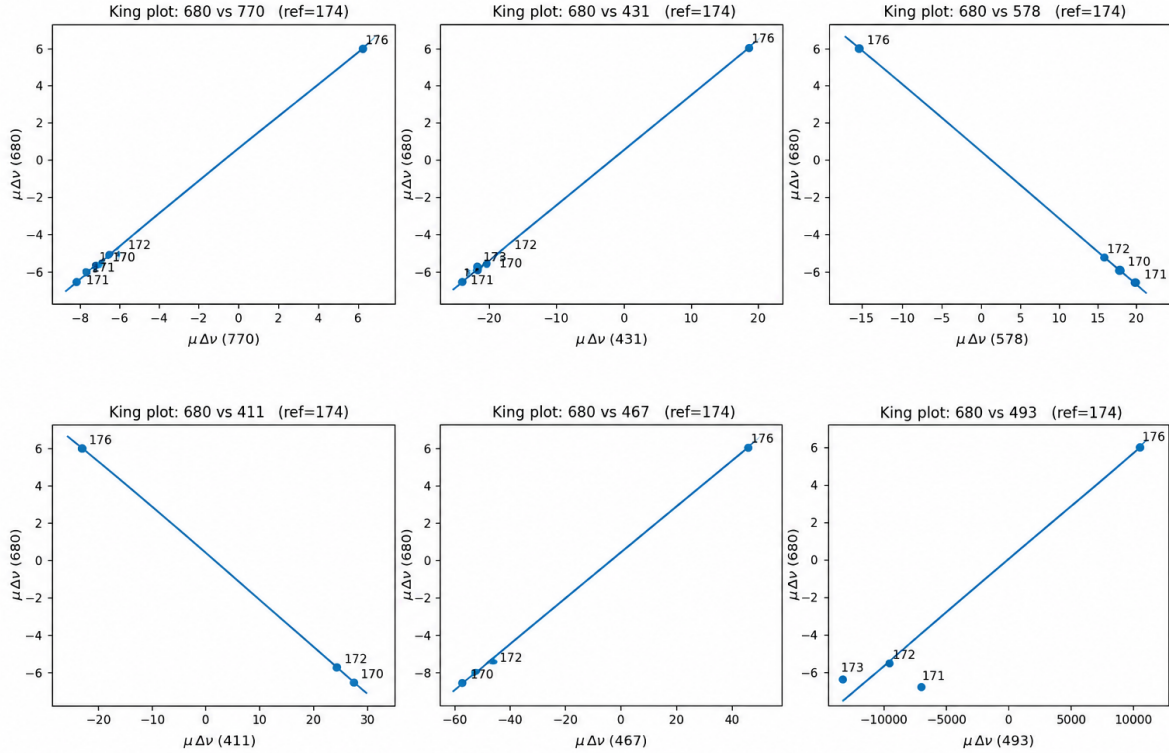


Figure 8.1: Summary of King plots involving the 680 nm transition. Modified isotope shifts relative to ^{174}Yb are compared with those of the 770 nm, 431 nm, 578 nm, 411 nm, 467 nm, and 493 nm transitions. Solid lines represent linear least-squares fits. The overall linear correlations demonstrate the consistency of the present 680 nm isotope-shift measurements with previously reported transitions in neutral and singly ionized ytterbium. Deviations observed in the 680–493 nm comparison are attributed to larger uncertainties of the 493 nm transition measurements.

8.5 Consistency checks among literature transitions

King plots were constructed using isotope-shift data taken entirely from the literature for the 411 nm, 431 nm, 467 nm, 578 nm, and 770 nm transitions. Four independent comparisons were performed: 431–578 nm, 770–578 nm, 770–431 nm, and 467–411 nm. As shown in Fig. 8.2, all four literature–literature King plots exhibit clear linear correlations, and all isotope pairs lie on single straight lines within

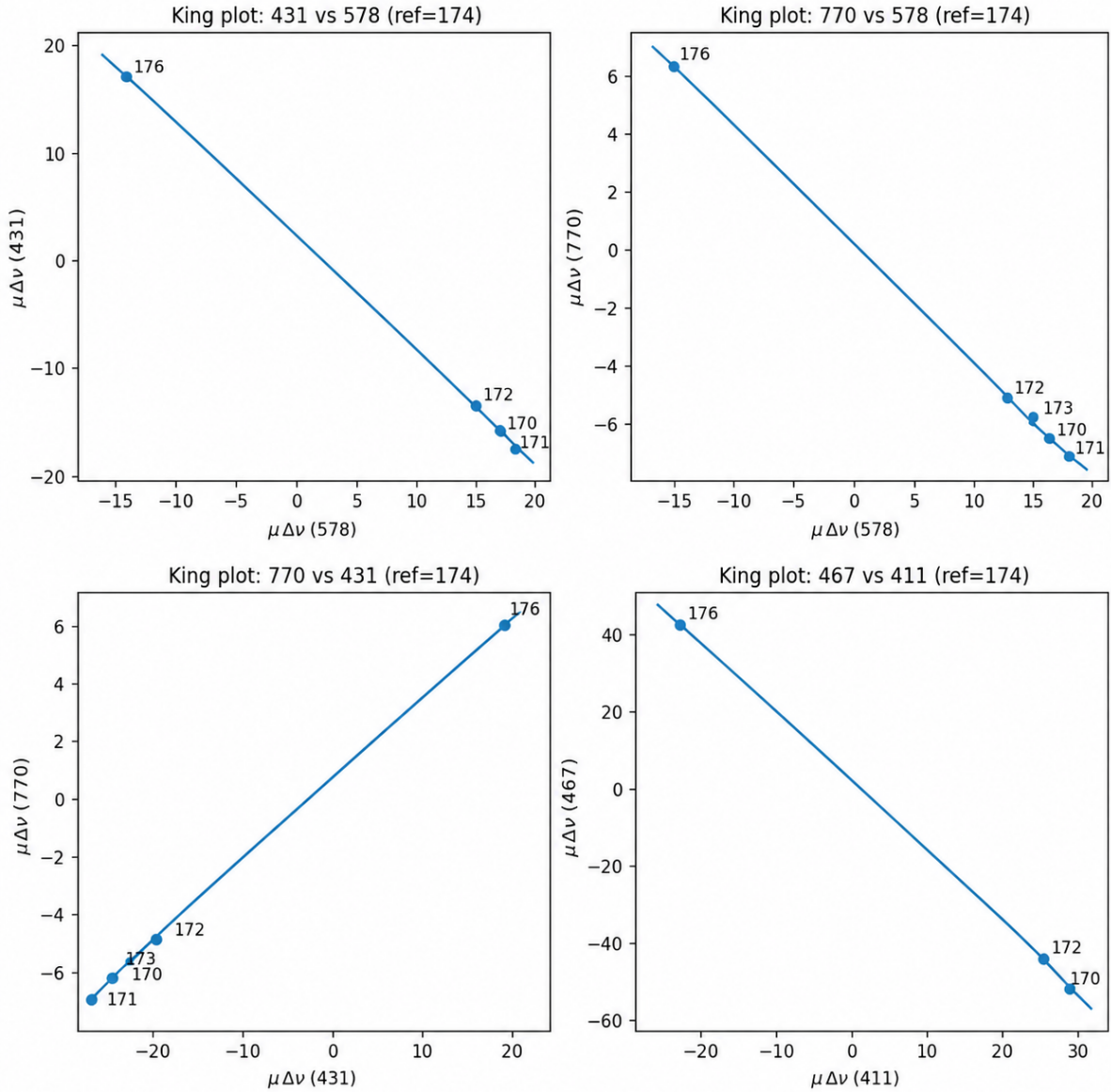


Figure 8.2: King plots constructed exclusively from literature isotope-shift data for the 431–578 nm, 770–578 nm, 770–431 nm, and 467–411 nm transition pairs. Modified isotope shifts are referenced to ^{174}Yb . The solid lines represent linear least-squares fits. All four comparisons exhibit excellent linearity, confirming the mutual consistency of the literature isotope-shift data used in the present work.

the reported experimental uncertainties. No significant deviation from linearity is observed in any of the four comparisons. These results provide an internal consistency check for the literature reference transitions employed in the present work and demonstrate that the isotope-shift data at 411 nm, 431 nm, 467 nm, 578 nm, and 770 nm are mutually consistent within the standard King-plot framework. Together with the 680 nm results presented in the previous section, the observed agreement further confirms the overall consistency of the isotope-shift data and

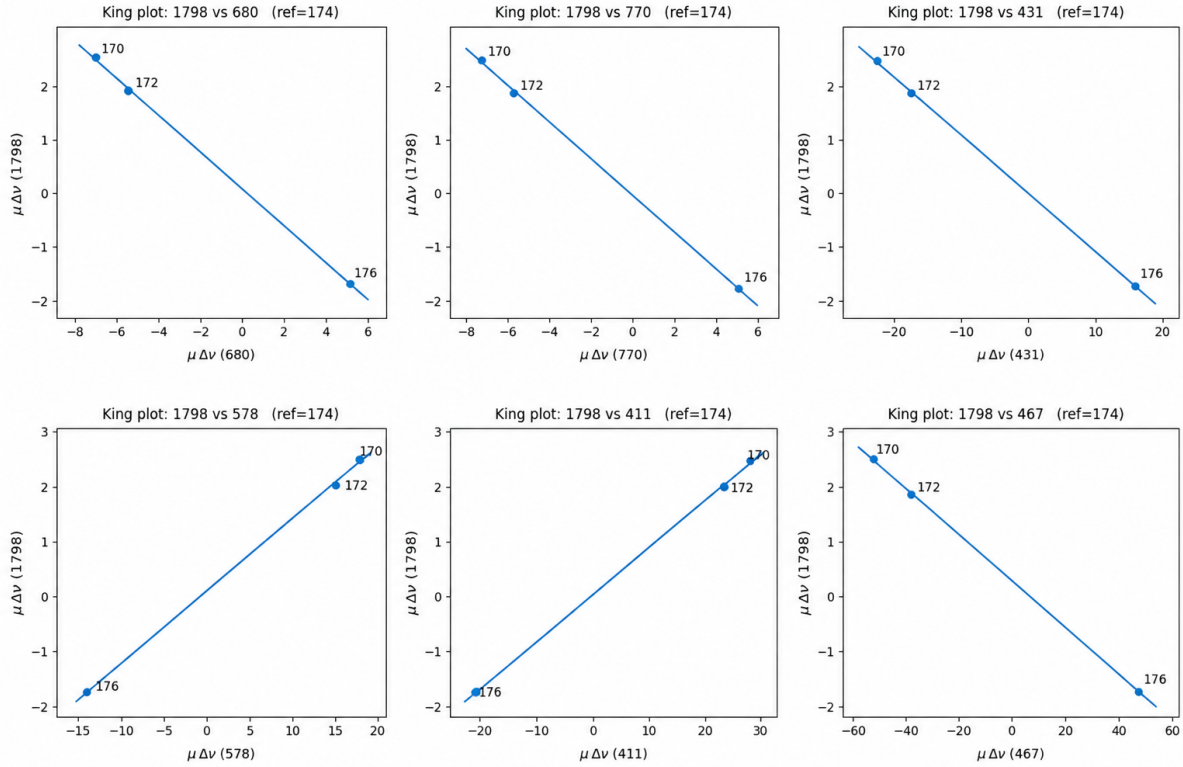


Figure 8.3: King plots of the 1798 nm transition and six reference transitions: 1798–680 nm, 1798–770 nm, 1798–431 nm, 1798–578 nm, 1798–411 nm, and 1798–467 nm. Modified isotope shifts are referenced to ^{174}Yb . The solid lines represent linear least-squares fits. All six comparisons exhibit good linearity, demonstrating the consistency of the isotope shifts extracted from the present 1798 nm spectroscopy with both literature data and laboratory measurements.

supports the validity of the King-plot analysis performed in this work.

8.6 King plots of the 1798 nm transition

To further examine the consistency of the isotope shifts extracted from the present 1798 nm spectroscopy measurements, additional King plots were constructed between the 1798 nm transition and several independent reference transitions. The reference isotope-shift data at 431 nm, 578 nm, 411 nm, and 467 nm were taken from the literature, whereas the 680 nm and 770 nm data were obtained from measurements performed in our laboratory. Six transition pairs were investigated: 1798–680 nm, 1798–770 nm, 1798–431 nm, 1798–578 nm, 1798–411 nm, and 1798–467 nm. The corresponding King plots are shown in Fig. 8.3. Clear linear correlations are observed for all six comparisons, and the isotope pairs lie close to single straight lines within the present experimental uncertainties. The agreement among these independent transition pairs indicates that the isotope shifts extracted from the 1798 nm transition are consistent with both literature data

and previous measurements performed in our laboratory. These results further support the reliability of the isotope shifts obtained from the present 1798 nm spectroscopy and confirm the overall consistency of the isotope-shift network considered in this work.

8.7 Prediction of the ^{168}Yb Resonance Position

The isotope-shift measurements presented above provide resonance positions for ^{170}Yb , ^{171}Yb , ^{172}Yb , ^{173}Yb , ^{174}Yb , and ^{176}Yb . However, no clear spectral feature corresponding to ^{168}Yb was observed in the present measurements. This is expected because the natural abundance of ^{168}Yb is only 0.13%, making its signal significantly weaker than those of the other stable isotopes.

For a given transition i , the isotope shift can be expressed as

$$\delta\nu_i^{A,174} = F_i \delta\langle r^2 \rangle^{A,174} + K_i \mu^{A,174}, \quad (8.9)$$

where F_i and K_i denote the field-shift and mass-shift constants, respectively, and

$$\mu^{A,174} = \frac{1}{m_A} - \frac{1}{m_{174}}. \quad (8.10)$$

Using the measured isotope shifts of ^{170}Yb , ^{172}Yb , and ^{176}Yb , the corresponding field-shift and mass-shift contributions were determined and subsequently extrapolated to ^{168}Yb . In addition, the King-plot relations between the 1798 nm transition and the reference transitions were employed to obtain independent estimates. The predicted values from the six King plots fall within the range of 262–273 MHz relative to ^{174}Yb , yielding an average value of

$$\delta\nu_{168,174}(1798 \text{ nm}) = +0.27 \text{ GHz}. \quad (8.11)$$

Figure 8.4 shows the predicted position of the ^{168}Yb resonance in the 1798 nm spectrum. The predicted resonance lies on the high-frequency side of ^{170}Yb and continues the systematic trend established by the even isotopes. Owing to the extremely low abundance of ^{168}Yb , no corresponding spectral feature is clearly visible in the measured trace.

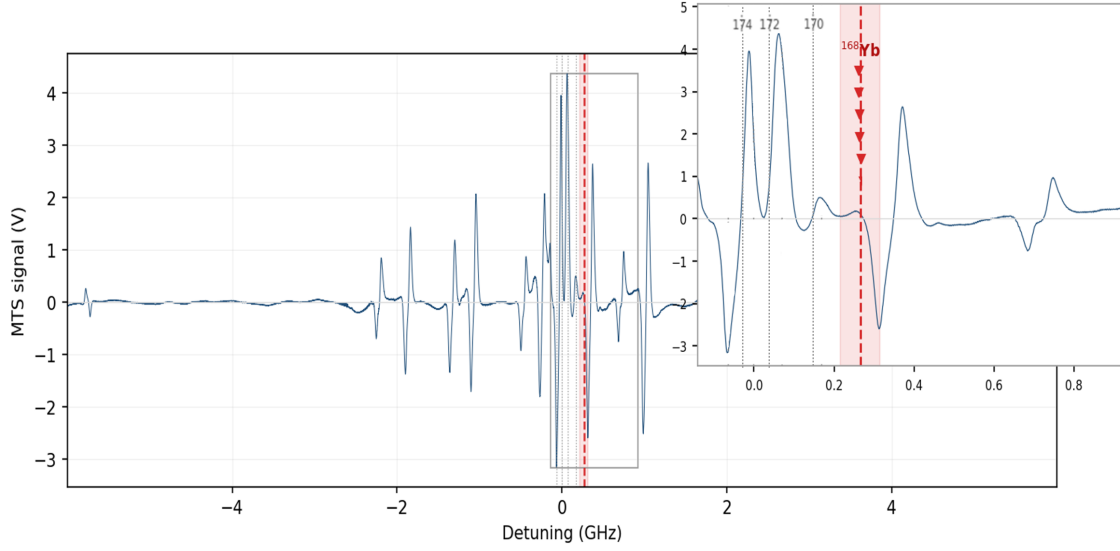


Figure 8.4: Prediction of the ^{168}Yb resonance position in the 1798 nm spectrum. The measured even-isotope resonances are indicated by gray dotted lines, while the red dashed line denotes the average predicted position of ^{168}Yb . The shaded region represents the estimated uncertainty of the prediction. The red arrows correspond to the ^{168}Yb positions independently inferred from the five King-plot analyses using different reference transitions. Their small spread indicates good consistency among the predictions.

As an additional consistency check, the same extrapolation procedure was also applied to the 680 nm transition. Since the methodology is identical to that described above, it will not be repeated here. Using the measured isotope shifts of the even isotopes and the corresponding King-plot relations, the predicted ^{168}Yb resonance position was found to be approximately

$$\delta\nu_{168,174}(680 \text{ nm}) = -1.53 \text{ GHz}, \quad (8.12)$$

relative to ^{174}Yb . Figure 8.5 shows the resulting prediction. Similar to the 1798 nm case, the predicted resonance lies beyond the ^{170}Yb resonance and continues the trend established by the even isotopes. The close agreement between the independently obtained predictions from the 1798 nm and 680 nm transitions provides additional confidence in the extrapolated position of the ^{168}Yb resonance and suggests that the absence of an observable signal is mainly a consequence of its extremely low natural abundance rather than an inconsistency in the isotope-shift analysis.

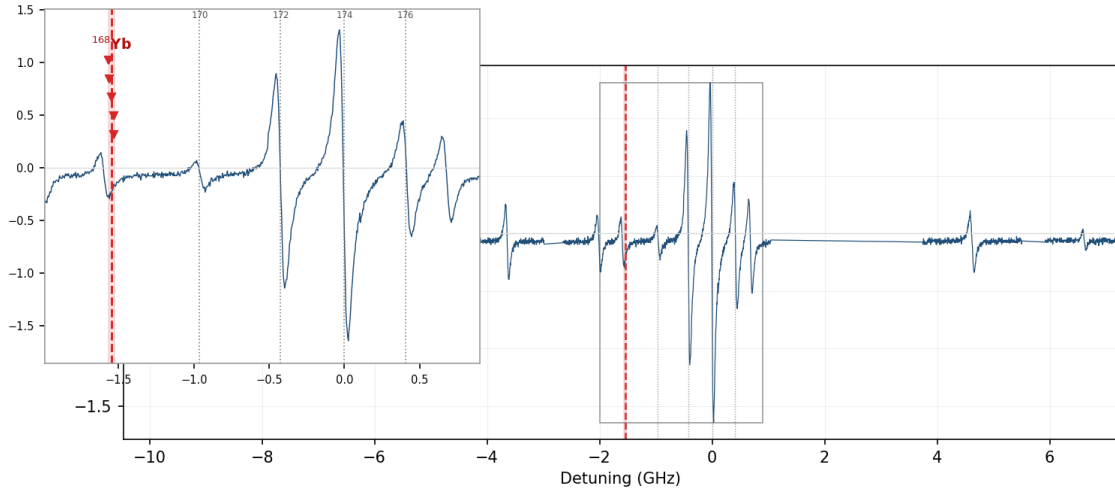


Figure 8.5: Predicted position of the ^{168}Yb resonance in the 680 nm spectrum. The red dashed line indicates the average predicted position of ^{168}Yb , and the shaded region represents the estimated uncertainty. The red arrows correspond to the individual ^{168}Yb positions independently inferred from the six King-plot analyses using different reference transitions. Their small spread demonstrates the consistency of the extrapolation.

8.8 Interactive King-Plot Analysis Tool

To facilitate systematic comparison of isotope-shift data, an interactive web-based analysis platform was developed and deployed. The platform integrates both the experimental results obtained in this work and reference data from the literature into a unified framework.

Users can select a reference transition and a target transition, upon which the corresponding King plot is generated in real time. This enables rapid visualization of linear correlations, direct comparison across different transition pairs, and efficient validation of isotope-shift consistency. In addition, the platform provides a convenient interface for exploring different combinations of transitions, making it a useful tool for identifying systematic trends and potential deviations from King linearity. The platform is publicly accessible at: <https://s20000125-alt.github.io/IAMS110-kingplot/>.

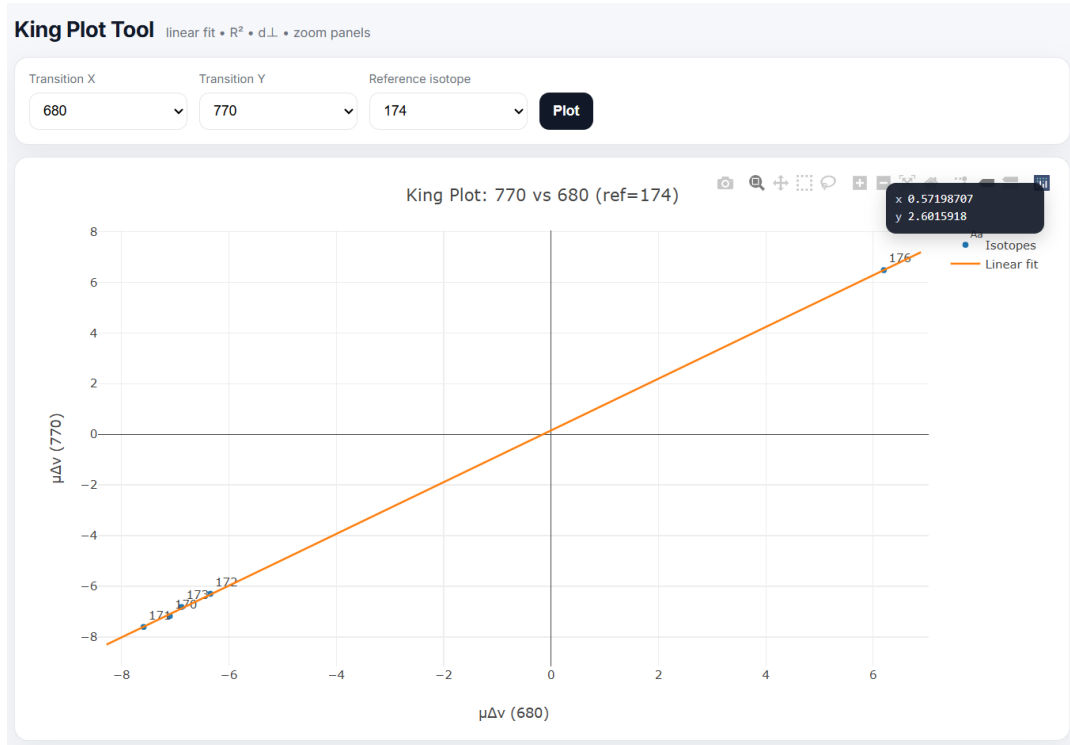


Figure 8.6: Screenshot of the interactive King-plot analysis platform developed in this work. The interface allows users to select a reference transition and a comparison transition, automatically generating the corresponding King plot for both experimental and literature data.

8.9 Conclusion

In this work, King-plot analysis was employed to investigate the consistency of the isotope-shift measurements obtained for both the 680 nm and 1798 nm transitions through comparison with several well-established reference transitions. King plots constructed from the present measurements exhibit excellent linearity within the experimental uncertainty, demonstrating that the isotope shifts extracted from both transitions are fully consistent with the existing ytterbium isotope-shift database. The fitted slopes represent ratios of field-shift coefficients, whereas the intercepts correspond to the associated combinations of mass-shift contributions. No statistically significant deviation from linearity is observed, indicating that higher-order isotope-shift effects remain below the present experimental sensitivity.

For the 680 nm transition, five independent comparisons with the 578 nm, 431 nm, 770 nm, 411 nm, and 467 nm reference transitions all exhibit excellent linear correlations. Likewise, the 1798 nm transition also shows consistent linearity when compared with multiple independent reference transitions. Together, these results demonstrate that the isotope shifts measured in the present work are mutu-

ally consistent with previously reported experimental data and validate both newly established datasets within a common King-plot framework.

Additional King plots constructed entirely from literature isotope-shift data, including the 431–578, 431–770, 411–467, and 578–770 combinations, also display excellent linear correlations. These literature–literature comparisons provide an independent validation of the conventional isotope-shift formalism and confirm that the reference transitions adopted in the present analysis form a mutually self-consistent dataset.

More importantly, the combination of the present measurements and literature data establishes a connected King-plot comparison network, illustrated in Fig. 8.7. Rather than relying on a single pair of transitions, this network enables consistency checks among multiple independent spectroscopic measurements. If a newly measured transition fails to produce linear King plots with several mutually consistent reference transitions, the discrepancy may indicate systematic errors or incorrect transition assignments. Consequently, the King-plot network serves not only as a tool for extracting isotope-shift parameters but also as a powerful framework for cross-validating independent spectroscopic measurements.

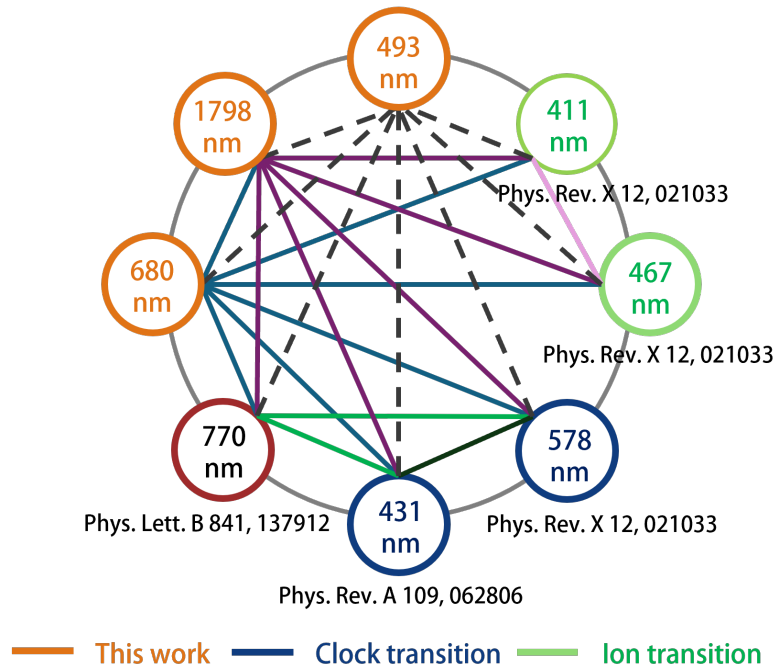


Figure 8.7: Schematic of the King-plot comparison network among the 411 nm, 431 nm, 467 nm, 578 nm, 680 nm, 770 nm, 493 nm, and 1798 nm transitions. Solid lines denote transition pairs included in the present analysis, whereas dashed lines indicate potential future comparisons. The network provides a framework for consistency checks among isotope-shift measurements.

Beyond providing consistency checks, the established King-plot network also

enables predictive applications. In the present work, this capability was employed to estimate the resonance position of the extremely weak ^{168}Yb isotope. Owing to its natural abundance of only 0.13%, no identifiable ^{168}Yb resonance was directly observed in either the 680 nm or the 1798 nm spectra. By combining the measured isotope shifts with literature values of the nuclear charge-radius differences, the King plots enabled extrapolation of the expected resonance positions. Independent predictions obtained from six reference transitions converge to approximately +0.27 GHz relative to ^{174}Yb for the 1798 nm transition and approximately -1.53 GHz for the 680 nm transition. The excellent agreement among these independent extrapolations further supports the internal consistency of the King-plot network and provides practical frequency regions for future searches for the ^{168}Yb resonance.

The predictive capability of the King-plot network extends beyond the specific case of ^{168}Yb . Because the network contains multiple interconnected transitions, newly measured isotope shifts can be independently verified through several King-plot comparisons rather than a single reference transition. As additional isotope-shift measurements become available, particularly for the 493 nm transition and other unexplored transitions, new connections can be incorporated into the existing network to further improve its reliability, redundancy, and sensitivity to possible deviations from conventional King-plot linearity.

For completeness, an interactive analysis tool has also been developed to facilitate systematic exploration of different transition pairs and visualization of the corresponding King plots, as described in Sec. 8.8. The flexibility of this platform allows newly measured transitions to be readily incorporated into the existing network and provides a convenient framework for future isotope-shift studies.

In summary, the present work demonstrates that isotope shifts measured from both the 680 nm and 1798 nm transitions can be independently validated within a common King-plot framework. By combining these measurements with well-established literature data, a connected King-plot network has been established that enables cross-validation among different spectroscopic experiments while simultaneously providing predictive capability for previously unobserved isotope resonances such as ^{168}Yb . The resulting network not only confirms the reliability and internal consistency of the isotope shifts determined in the present work, but also provides a robust framework for future precision spectroscopy, expansion of isotope-shift databases, and increasingly stringent investigations of higher-order isotope-shift effects and possible deviations from conventional King-plot linearity.

9

Interactive Optical System Modeling Platform

9.1 Motivation

Optical system construction in atomic physics experiments is inherently intuitive and experience-driven. Even for researchers with a solid theoretical background—including Gaussian beam optics, ABCD matrix formalism, and component-level physical models—it remains difficult to translate abstract equations into a concrete understanding of how light behaves in real space. In practice, beam propagation, divergence, spatial filtering, polarization, and alignment constraints act simultaneously and are strongly coupled. Predicting the outcome of moving a lens by a few millimeters, or inserting a waveplate at a particular orientation, requires tracking many interdependent quantities at once. For beginners, this creates a steep learning curve that is not easily overcome by reading the literature alone.

Existing optical design tools such as ZEMAX or Oslo provide highly accurate simulation capabilities, but are optimized for professional engineering workflows. Their interaction models require users to specify systems in a structured, parameter-driven format rather than through spatial intuition. Real-time visual feedback—seeing where the beam goes as a component is moved—is largely absent. These tools are therefore less effective for developing the kind of physical intuition that is essential for day-to-day experimental work.

Motivated by these limitations, this work develops an interactive optical modeling platform, *PathDesigner*, that emphasizes physical realism, direct visualization,

and real-time feedback, including three-dimensional visualization of beam propagation. Rather than replacing high-precision engineering tools, PathDesigner serves as a complementary environment in which users can directly observe how light propagates and responds to changes in the optical layout. A central design goal is that every displayed quantity—beam diameter, frequency content, coupling efficiency, and diffraction angle—is derived from the same underlying physical model, ensuring that the simulation remains self-consistent as the user manipulates the system.¹

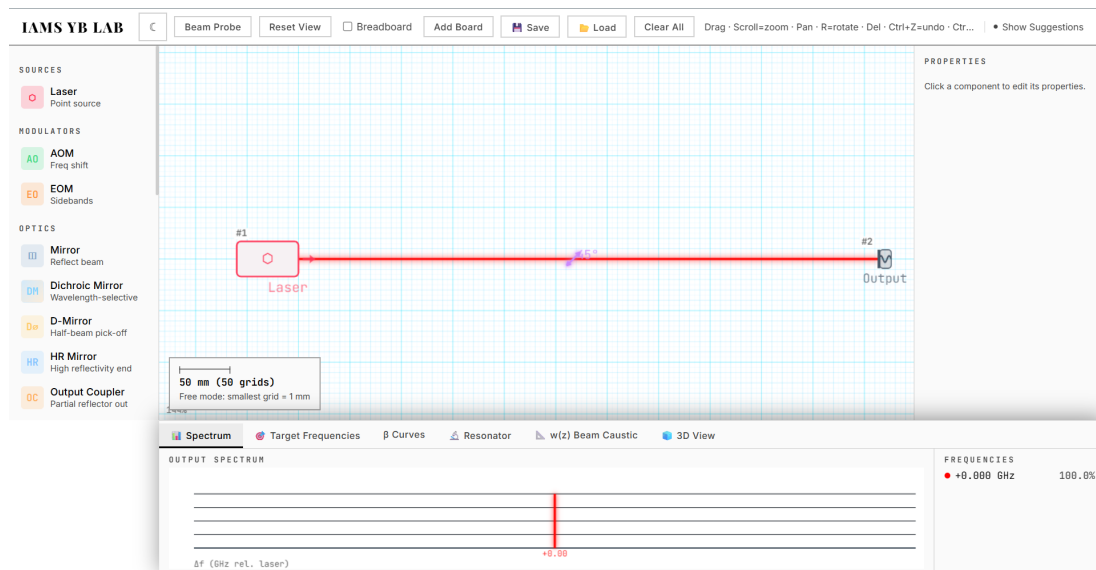


Figure 9.1: Overview of the PathDesigner user interface. The platform integrates interactive component placement, real-time beam tracing, and parameter inspection within a single browser-based environment.

The platform also originates from practical experimental needs in the construction of laser systems for atomic physics, including AOM double-pass frequency shifting, single-mode fiber coupling, isotope-selective EOM addressing, and cavity stability analysis. By encoding these common configurations as physically motivated, automatically placed building blocks, PathDesigner reduces setup time and lowers the barrier for newcomers to develop concrete experimental intuition.

9.2 System Architecture

PathDesigner is implemented as a self-contained, browser-based interactive application requiring no installation. The entire simulation engine, rendering

¹https://s20000125-alt.github.io/frequency_shift_simulator/

pipeline, and user interface are contained within a single HTML document, making it immediately portable to any modern browser without server-side dependencies.

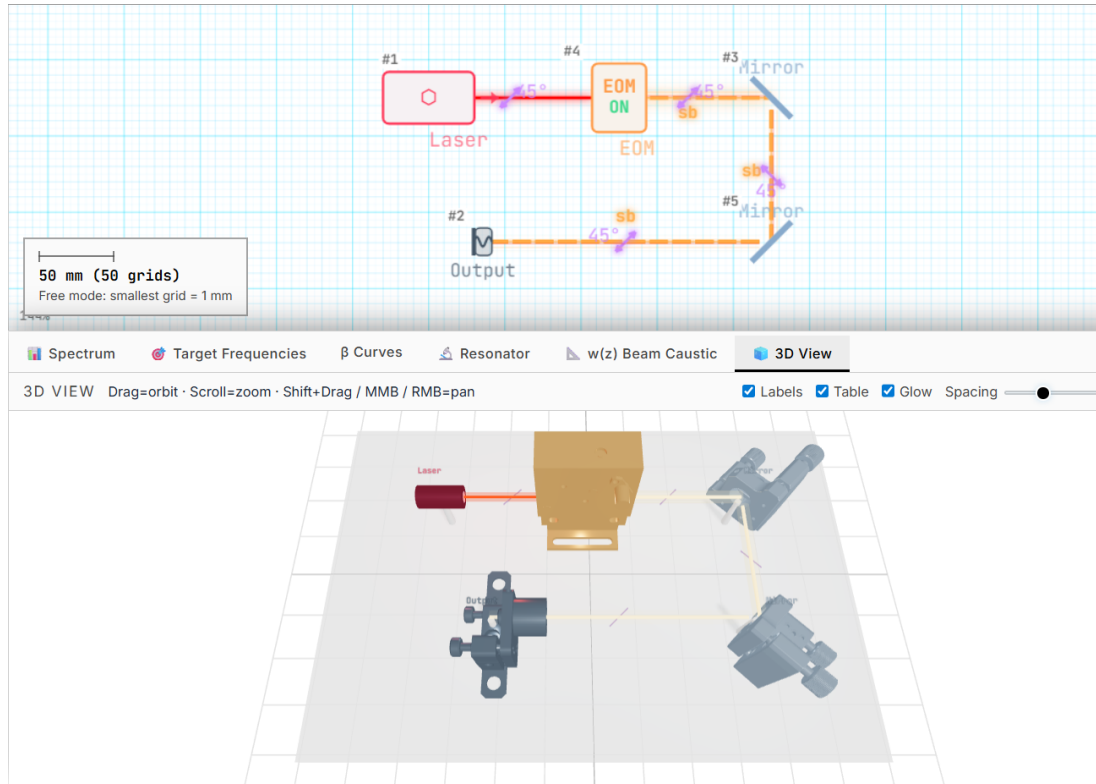


Figure 9.2: Three-dimensional visualization of the optical system in PathDesigner. The 3D rendering provides spatial insight into beam propagation and component alignment, complementing the 2D layout view.

9.3 Gaussian Beam Module

Gaussian beam propagation is implemented using the complex beam parameter q , defined such that the beam radius $w(z)$ and wavefront radius of curvature $R(z)$ satisfy

$$\frac{1}{q(z)} = \frac{1}{R(z)} - i \frac{\lambda}{\pi w(z)^2}. \quad (9.1)$$

Free-space propagation over distance d and passage through a thin lens of focal length f are applied via the ABCD transformations $q' = q + d$ and $1/q' = 1/q - 1/f$, respectively. The resulting beam radius at every point in the layout is displayed graphically as the beam width and numerically in the properties panel for real-time diagnostics.

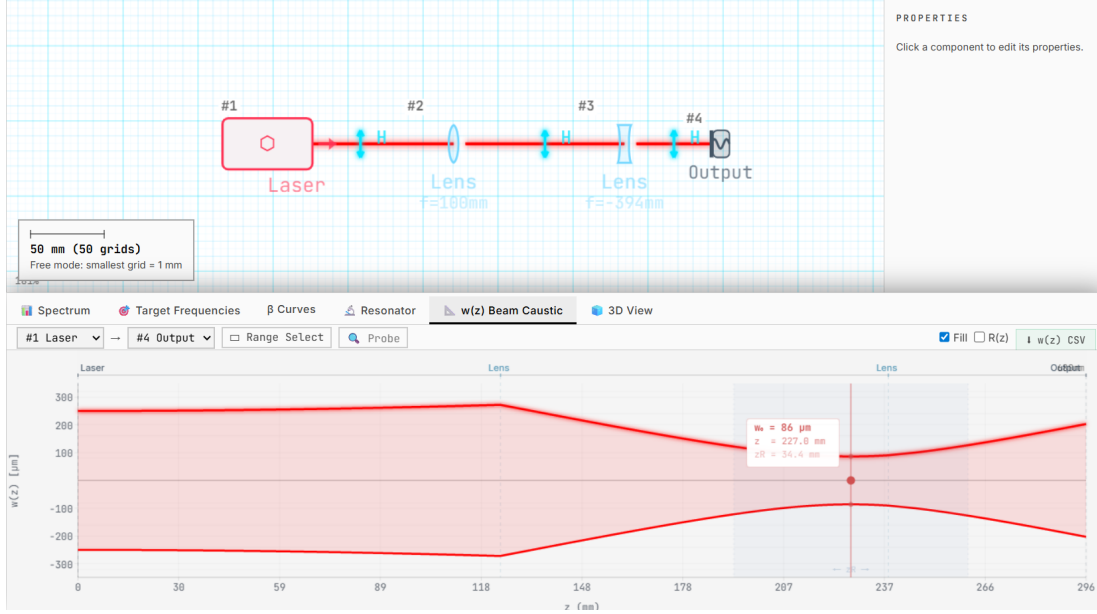


Figure 9.3: Visualization of Gaussian beam propagation in PathDesigner. The displayed beam width is computed from the propagated complex beam parameter q and updated continuously along the optical path.

9.4 Acousto-Optic Modulator Model

9.4.1 Single-Pass Diffraction

The AOM model is based on Bragg diffraction geometry. For an RF drive frequency f_{RF} and acoustic velocity v_a in the crystal, the Bragg angle is

$$\theta_B = \frac{\lambda f_{\text{RF}}}{2v_a}, \quad (9.2)$$

and the output frequency of the n -th diffraction order is $f_{\text{out}} = f_{\text{in}} + n f_{\text{RF}}$. All orders from $n = -2$ to $n = +2$ are rendered as a geometrically consistent fan, with deflection angles computed from the actual wavelength, acoustic velocity v_a , and crystal refractive index n_{AOM} , ensuring that the internal propagation geometry is correctly represented.

9.4.2 Double-Pass Configuration

The double-pass AOM configuration is implemented as a single-button automated setup whose physical steps are:

1. The first pass deflects the selected order and shifts the frequency by $+n f_{\text{RF}}$.
2. A spatial aperture selects the desired order, blocking all others.

3. A quarter-wave plate (QWP) converts the polarization from linear to circular.
4. A cat's-eye retroreflector (focusing lens at distance f from the AOM, flat mirror at distance f from the lens) reflects the beam back, ensuring that the return beam re-enters the AOM at the correct position independent of RF frequency tuning.
5. The second pass adds a further $+n f_{\text{RF}}$ shift; the QWP restores linear polarization, now orthogonal to the input, allowing the PBS upstream to separate the double-pass output.
6. A second aperture on the input side passes only the double-pass output (aom_dp_out, net shift $+2n f_{\text{RF}}$), blocking the return-path 0th order.

Before placing components, the platform checks whether the current beam diameter at the AOM falls within the manufacturer-specified recommended range. If not (**Case A**), a focus lens is automatically placed upstream and a matching collimation lens downstream; if the beam already fits (**Case B**), only the QWP, retro mirror, and apertures are placed.

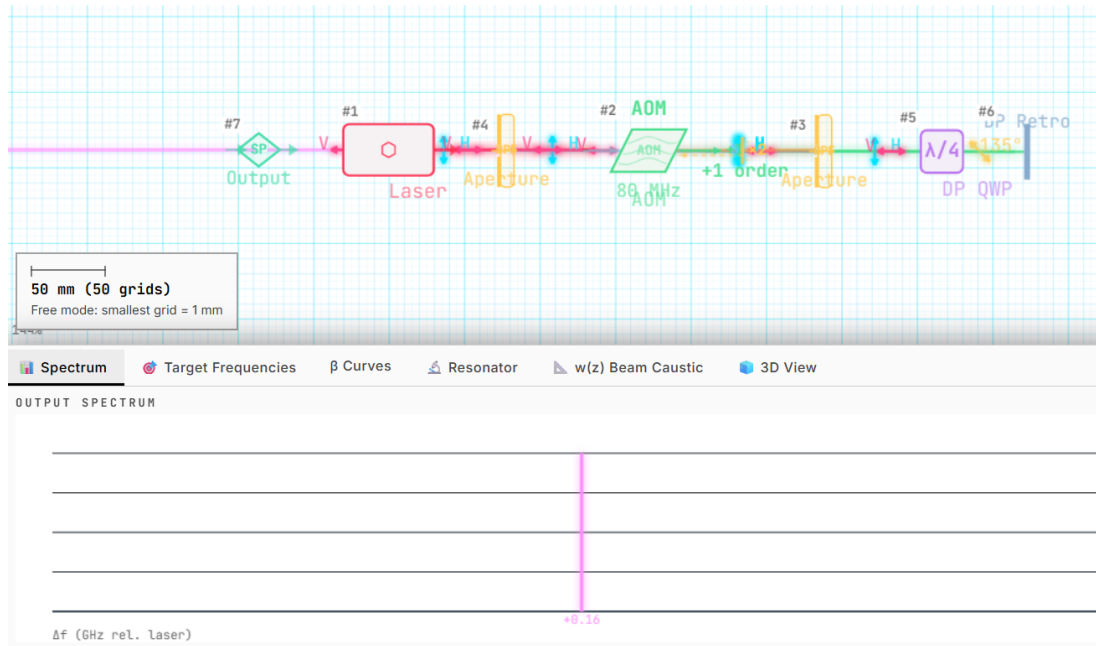


Figure 9.4: Automated double-pass AOM configuration generated by PathDesigner. The platform places the QWP, apertures, and cat's-eye retroreflector to realize the frequency-doubled return path and output separation.

9.4.3 Efficiency Model and Datasheet Calibration

Accurate prediction of diffraction efficiency requires a model that captures the dependence on both beam diameter and RF power. The total efficiency is decomposed as

$$\eta(d, \lambda, P) = \eta_{\max}(\lambda) \cdot f_{\text{dia}}(d, \lambda) \cdot \eta_{\text{RF}}(P, \lambda), \quad (9.3)$$

where $\eta_{\max}(\lambda)$ is the plateau efficiency at large beam size and saturated RF, f_{dia} is a dimensionless beam-size factor, and η_{RF} captures the RF-power dependence.

Beam-size factor. The factor f_{dia} combines a rising term (Bragg acceptance improves as beam divergence decreases) and a slowly falling term (acoustic overlap decreases for very large beams):

$$f_{\text{dia}}(d, \lambda) = \frac{1}{\mathcal{N}} \left[1 - e^{-d/(k_s(\lambda) d_{\text{opt}})} \right] \cdot \frac{1}{1 + (d/(k_l(\lambda) d_{\text{opt}}))^2}, \quad (9.4)$$

where $\mathcal{N}$ normalises the peak to unity, d_{opt} is a curve-shape scale parameter, and k_s, k_l control the rise and roll-off rates, respectively.

RF-power model. At moderate to high RF drive, TeO_2 AOMs exhibit a characteristic efficiency peak followed by a roll-off as higher-order acoustic harmonics degrade the Bragg condition. This behaviour is captured by a Bragg-motivated sinusoidal model:

$$\eta_{\text{RF}}(P, \lambda) = \eta_{\max, \text{RF}}(\lambda) \sin^2(c(\lambda) \cdot P^{\alpha(\lambda)}), \quad (9.5)$$

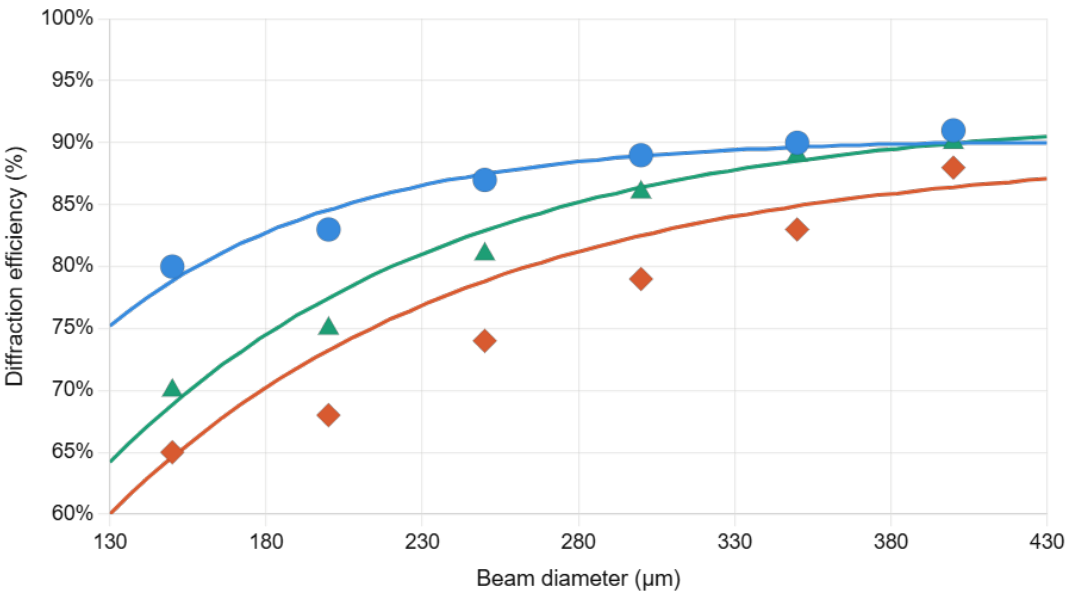
where $c(\lambda)$ and $\alpha(\lambda)$ are per-wavelength shape parameters and $\eta_{\max, \text{RF}}$ is the RF-saturation ceiling. The peak RF power is $P_{\text{peak}} = (\pi/2c)^{1/\alpha}$.

Calibration against the G&H 3080-120 datasheet. The device currently implemented in PathDesigner is the G&H 3080-120 acousto-optic modulator (TeO_2 , longitudinal mode, 80 MHz centre frequency, 440–900 nm). Two sets of manufacturer theoretical curves served as calibration targets:

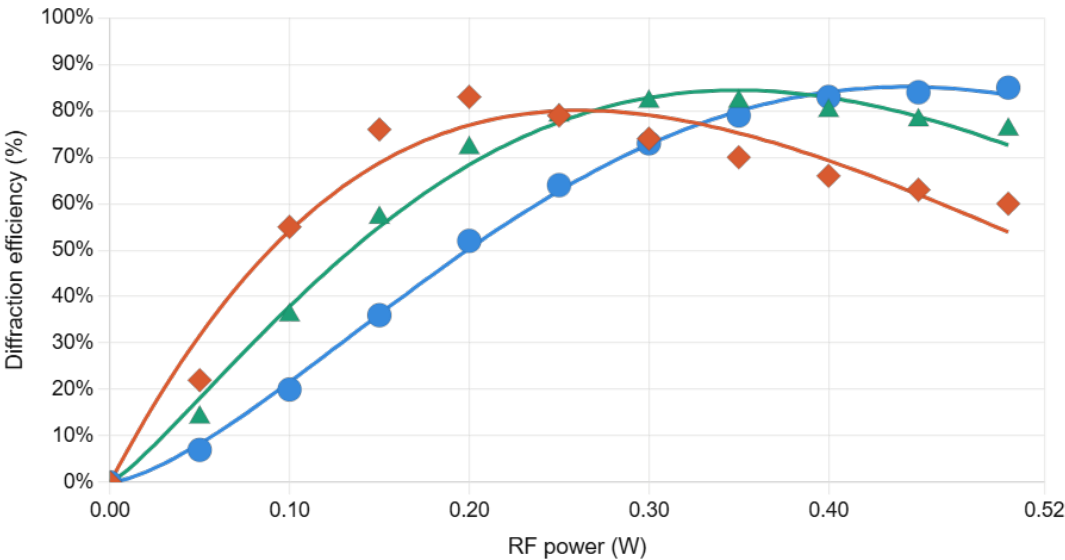
- *Efficiency vs. beam diameter* at saturated RF drive, provided for 442 nm, 515 nm, and 633 nm. Because RF is saturated, $\eta_{\text{RF}} \approx 1$, so fitting isolates $\eta_{\max}(\lambda)$ and the shape parameters k_s, k_l , and d_{opt} in Eq. (9.4).
- *Efficiency vs. RF power* at a fixed beam diameter of 195 μm , again at three wavelengths. Fitting this curve yields $c(\lambda)$, $\alpha(\lambda)$, and $\eta_{\max, \text{RF}}(\lambda)$ in Eq. (9.5).

The three RF-power curves correspond to 442 nm (solid line, peaks at ≈ 0.26 W), 515 nm (long dash, peaks at ≈ 0.35 W), and 633 nm (dot-dash, peaks at ≈ 0.44 W), reflecting the wavelength-dependent Bragg efficiency of TeO_2 . Both

sets of calibration results are shown in Fig. 9.5, and the fitted parameters are collected in Table 9.1.



(a) Diffraction efficiency vs. beam diameter at saturated RF drive.



(b) Diffraction efficiency vs. RF power at fixed beam diameter.

Figure 9.5: Calibration of the G&H 3080-120 AOM efficiency model against manufacturer datasheet curves at 442 nm, 515 nm, and 633 nm. Filled markers indicate datasheet values, and solid lines represent the fitted model (Eq. (9.4) and Eq. (9.5)). The RF power curves peak at approximately 0.26 W (442 nm), 0.35 W (515 nm), and 0.44 W (633 nm).

Table 9.1: G&H 3080-120 efficiency model parameters. Physical device parameters are taken directly from the manufacturer datasheet; model parameters are obtained by least-squares fitting to the published theoretical curves.

Parameter	633 nm	515 nm	442 nm
<i>Physical device parameters (datasheet spec)</i>			
Centre frequency f_{RF}	80 MHz		
Acoustic medium	TeO ₂ , longitudinal mode		
Acoustic velocity v_a	4.2 mm/ μ s		
Refractive index n	2.26 (TeO ₂ at 633 nm)		
Active aperture	1.0 mm		
Interaction length L	12 mm (TeO ₂ typical)		
<i>Beam-size model (Eq. 9.4), fitted from efficiency vs. diameter curves</i>			
η_{max}	0.900	0.920	0.890
k_s	0.050	0.075	0.080
k_l	10.0	20.0	20.0
d_{opt} (all λ)	1.46 mm		
RMS residual	0.93%	1.26%	2.40%
<i>RF-power model (Eq. 9.5), fitted from efficiency vs. RF curves</i>			
$\eta_{\text{max,RF}}$	0.852	0.844	0.801
c	2.847	2.981	3.096
α	0.732	0.609	0.506
P_{peak}	0.44 W	0.35 W	0.26 W
RMS residual	1.2%	2.5%	5.1%

A key finding from the beam-diameter calibration is that $d_{\text{opt}} \approx 1.46$ mm for all three wavelengths, which is substantially larger than the device's active aperture of 1.0 mm. This means efficiency is still rising across the entire usable beam-size range, with no plateau visible within the aperture. The parameter d_{opt} therefore functions purely as a curve-shape parameter and is stored separately from the physically meaningful recommended beam-size range used for alignment warnings.

The 442 nm RF-power fit carries the largest residual (5.1%), concentrated at the smallest measured power (0.05 W). This reflects a structural limitation of the $\sin^2$ model: it cannot simultaneously achieve a low value at 0.05 W and a peak at 0.26 W without error at one of these points. For all three wavelengths, the model correctly reproduces the qualitative behaviour of TeO₂ under high RF drive: efficiency rises to a peak and then rolls off as higher-order acoustic harmonics degrade the Bragg condition, in contrast to the monotonically saturating model that is commonly assumed.

Workflow and outlook. The calibration above establishes a repeatable workflow for incorporating additional components as the platform expands. For each new device, the relevant datasheet curves are digitised, model parameters are obtained

by least-squares minimisation, and the values are stored as device defaults. The long-term objective of PathDesigner is for every component in its library to carry parameters directly traceable to its manufacturer datasheet, so that a simulation built in PathDesigner can serve as a quantitatively reliable prediction of what will be observed on the optical table. This ongoing calibration effort—comparing simulation predictions against real device specifications and, ultimately, direct experimental measurements—is central to the platform’s design philosophy and will continue as new optics are added to the system.

9.5 Electro-Optic Modulator Model

The EOM model generates frequency sidebands using the standard Bessel function expansion of a phase-modulated field. For modulation index β , the power in the n -th sideband is

$$P_n = P_0 J_n^2(\beta). \quad (9.6)$$

The platform supports single-tone and two-tone RF driving. Power conservation is enforced: the total power across all tracked sidebands is normalised to the input beam power, preventing unphysical accumulation in cascaded modulator configurations.

9.6 Fiber Coupling Model

Single-mode fiber couplers are modeled using the Gaussian overlap integral. For an incident beam with complex parameter q and a fiber mode of waist radius w_f , the power coupling efficiency is

$$\eta = \left| \frac{2}{\frac{w}{w_f} + \frac{w_f}{w} + i \left(\frac{z_R}{R} - \frac{z_{Rf}}{R_f} \right)} \right|^2, \quad (9.7)$$

where w , R , and z_R are the beam radius, wavefront curvature, and Rayleigh range at the fiber facet, and w_f , z_{Rf} are the corresponding fiber-mode quantities. This reduces to $\eta = 1$ when $w = w_f$ and the wavefronts are matched, corresponding to perfect mode matching. The fiber output always emits the fiber’s own Gaussian eigenmode, correctly modelling the mode-cleaning function of single-mode fibers.

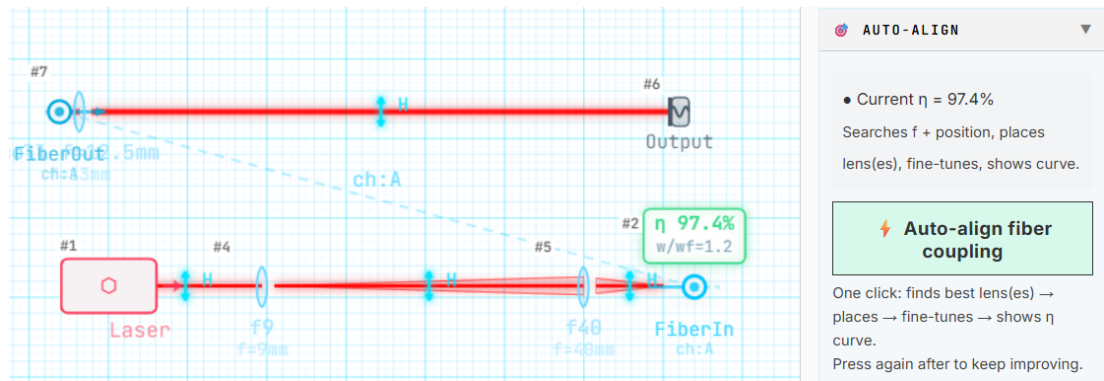


Figure 9.6: Fiber-coupling analysis in PathDesigner. The interface visualizes the dependence of coupling efficiency on beam-size and wavefront matching at the fiber facet.

9.7 Frequency-Domain Spectrum Analysis

The spectrum module evaluates optical frequency content only at designated output components. The display shows a bar chart of power vs. frequency offset, together with user-defined target frequencies and a pass/fail indicator for each. If no output component is placed, the spectrum is empty—an absence of signal that itself conveys meaningful information about the system configuration. This output-gated approach ensures that all displayed frequency content represents physically measurable quantities.

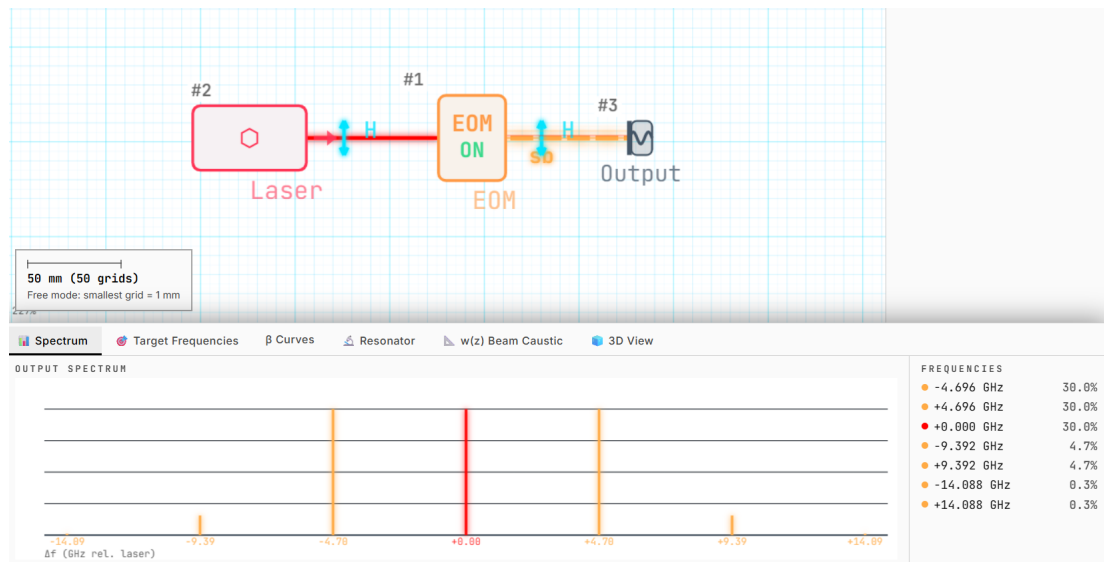


Figure 9.7: Frequency-domain spectrum module in PathDesigner. Only light reaching designated output components contributes to the displayed spectrum, ensuring that the result corresponds to a physically measurable signal.

9.8 Physical Realism as a Design Principle

All quantities displayed in PathDesigner are derived from a self-consistent physical model. Several specific design choices reinforce this principle:

1. **Beam width is computed, not assumed.** The displayed beam width at every point is derived from the q parameter propagated through the full ABCD chain.
2. **Diffraction angles are exact.** AOM deflection angles are computed from the Bragg condition using the actual wavelength, acoustic velocity, and crystal refractive index.
3. **Polarization is tracked.** Waveplates apply full Jones matrices; PBS splitting ratios are computed from the incident Jones vector, so an incorrectly oriented QWP produces a visually incorrect splitting ratio.
4. **Frequency content is propagated.** EOM sidebands are tracked as distinct beams with correct Bessel-function amplitudes through all downstream elements.
5. **Aperture geometry is computed from the trace.** The positions of automatically placed double-pass apertures are derived from actual beam intersection points, not from the nominal component centre.
6. **Component parameters are traceable to manufacturer datasheets.** Rather than using generic assumed values, device models carry parameters fitted directly from published performance curves. For the G&H 3080-120 AOM, the efficiency model parameters were obtained by fitting to both the beam-diameter and RF-power datasheet curves at three wavelengths (Section 9.4.3). This datasheet-driven calibration workflow will be applied to every new component added to the library, progressively closing the gap between simulation output and experimental measurement.

If the simulation is not physically self-consistent, users may develop incorrect intuitions that are actively harmful when transferred to the optical table. The ongoing calibration against real device datasheets reflects the broader ambition to make PathDesigner not merely a qualitative visualisation tool, but a quantitatively reliable predictor of experimental behaviour.

9.9 Summary

PathDesigner provides a physically consistent, real-time interactive framework for optical system design and education. Its core contributions are a unified ray-

tracing engine that simultaneously propagates beam direction, Gaussian q parameter, optical frequency, and polarization Jones vector; physically accurate models of AOMs, EOMs, waveplates, and fiber couplers; automated single-button setup for common complex configurations; and detector-based frequency spectrum analysis. Component parameters are calibrated against manufacturer datasheets—most fully for the G&H 3080-120 AOM, where both the beam-size and RF-power efficiency curves have been fitted at three wavelengths—with the aim of making every simulation result directly comparable to a physical measurement on the optical table. As the component library grows, each new device will be characterised using the same fitting workflow, making PathDesigner an increasingly reliable tool for predictive experimental design in atomic physics laser systems.

10

Development of an AI-Assisted Analysis Agent for Modulation Transfer Spectroscopy

10.1 Motivation

The analysis workflow of modulation transfer spectroscopy (MTS) measurements is often highly repetitive and follows a relatively standardized sequence of procedures. After data acquisition from the oscilloscope, the experimental traces typically require frequency-axis reconstruction, zero-crossing identification, spectral-feature extraction, and comparison with theoretically predicted hyperfine transition patterns.

In the present work, these analysis steps were repeatedly applied to multiple ytterbium transitions, including the 680 nm and 493 nm spectroscopy measurements discussed in previous chapters. Although the detailed spectral structures differ between transitions, the overall analysis logic remains largely similar. This observation motivated the development of a semi-automated analysis framework capable of assisting the spectroscopy workflow.

A particularly important step in MTS analysis is the identification of dispersive zero-crossing positions, which are commonly used as references for transition frequencies and laser-frequency stabilization. In addition, when approximate hyperfine coefficients are available from previous studies, theoretical transition frequencies can be calculated and compared with the measured spectra to perform preliminary spectral assignment.

However, practical MTS spectra frequently exhibit challenges such as overlapping resonances, weak fermionic signals, residual Doppler background, scan distortions, and baseline fluctuations. As a result, fully automatic fitting using a single global model is often unreliable. Instead, experimental analysis usually relies on iterative reasoning and pattern recognition performed by experienced spectroscopists.

The purpose of the present work is therefore not to replace physical interpretation, but rather to reformulate part of the spectroscopy reasoning workflow into a modular computational framework. By combining rule-based candidate detection, residual-based iterative fitting, and hyperfine-assisted spectral matching, the analysis process can be partially automated while preserving experimental interpretability and manual validation capability.

10.2 Concept of the Analysis Agent

The analysis agent developed in this work was designed to emulate part of the reasoning process commonly performed during manual spectroscopy analysis. Rather than relying on a single fitting routine, the workflow was divided into multiple independent modules responsible for candidate detection, fitting validation, residual analysis, and spectral assignment.

The general workflow of the agent is summarized as follows:

1. Reconstruction of the frequency axis from oscilloscope traces and wavelength-meter references.
2. Detection of candidate dispersive features using multiple independent strategies.
3. Extraction and fitting of local spectral regions.
4. Identification and rejection of pathological or duplicated fits.
5. Comparison between measured spectral features and theoretically predicted hyperfine transition patterns.
6. Iterative residual analysis to identify weak or previously missed spectral features.

The overall workflow therefore combines deterministic fitting procedures with rule-based decision logic inspired by practical spectroscopy analysis.

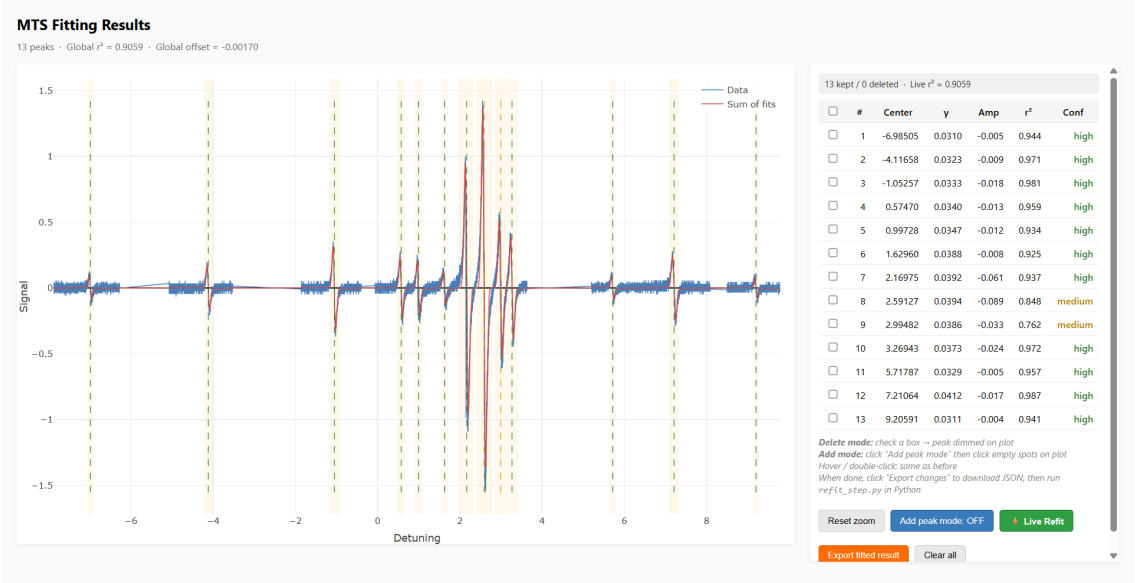


Figure 10.1: Screenshot of the AI-assisted MTS analysis interface. The interface visualizes the measured spectrum, fitted curves, detected zero crossings, fitting confidence levels, and interactive candidate-selection tools used for semi-automated spectroscopy analysis.

Figure 10.1 shows the graphical interface developed for the analysis agent. The interface allows the user to visualize candidate zero crossings, fitting residuals, and local spectral structures interactively. Compared with previous fully manual workflows based on repeatedly modifying Python fitting parameters, the present interface provides a significantly more intuitive analysis environment.

Instead of manually adjusting large numbers of fitting thresholds and fitting windows through trial-and-error, the user can directly inspect candidate transitions generated by the agent and evaluate their physical validity visually. This substantially improves workflow efficiency and reduces repetitive manual fitting procedures.

10.3 Multi-Channel Candidate Detection

A major difficulty in MTS analysis arises from the identification of physically meaningful zero crossings. In experimental spectra, residual amplitude modulation (RAM), electronic noise, baseline fluctuations, and overlapping transitions can produce false zero crossings that do not correspond to actual atomic resonances.

To improve robustness, the present agent employs multiple independent candidate-detection strategies rather than relying on a single criterion.

The first strategy identifies candidate transitions directly from sign changes in the dispersive signal after smoothing. The second strategy searches for extrema pairs corresponding to the positive and negative lobes of dispersive lineshapes.

The third strategy analyzes extrema in the derivative of the signal to identify rapidly changing spectral regions.

Candidates detected by different methods are subsequently cross-validated and merged. Features identified simultaneously by multiple strategies are assigned higher confidence levels, while isolated weak candidates are treated more cautiously.

This multi-channel approach improves robustness against noise and reduces false-positive detections caused by local fluctuations or modulation artifacts.

10.4 Scoring and Confidence Evaluation Mechanism

To improve the reliability of automatic candidate selection, the agent incorporates a rule-based scoring and confidence-evaluation mechanism inspired by practical spectroscopy analysis.

After local fitting is performed, each candidate transition is evaluated according to multiple criteria, including:

- coefficient of determination (R^2),
- local signal amplitude,
- linewidth consistency,
- symmetry of the dispersive structure,
- agreement between different detection strategies,
- residual structure after fitting,
- and proximity to neighboring peaks.

Candidates satisfying multiple criteria simultaneously are assigned higher confidence levels, while suspicious or weak candidates receive lower confidence scores. The confidence level is displayed directly in the graphical interface to assist user validation.

For example, transitions exhibiting stable fitting convergence, physically reasonable linewidths, and consistent detection across multiple strategies are typically labeled as high-confidence candidates. In contrast, peaks associated with poor fitting residuals, strong overlap, or unstable linewidths are categorized as medium- or low-confidence features.

This scoring mechanism does not represent a machine-learning probability model. Instead, it serves as a structured rule-based evaluation framework

designed to emulate the reasoning process commonly used during manual spectroscopy analysis.

10.5 Iterative Fitting and Self-Critique Mechanism

After candidate detection, local spectral regions are fitted using derivative-type Lorentzian models. Multiple fitting strategies with different fitting windows and constraints are tested iteratively in order to improve convergence stability.

To prevent overfitting and unphysical solutions, the agent incorporates a self-critique mechanism inspired by manual spectroscopy analysis. The fitting results are automatically evaluated according to several criteria, including:

- coefficient of determination (R^2),
- fitted linewidth range,
- duplicated peak positions,
- unrealistically large amplitudes,
- pathological cancellation between neighboring peaks,
- and residual structures remaining after fitting.

When suspicious fits are detected, the corresponding peaks can be rejected or merged automatically, followed by an additional global refitting step.

The residual signal between the measured spectrum and the fitted model is also analyzed iteratively. Regions containing significant residual structures are rescanned computationally in order to identify weak or previously missed transitions.

This iterative procedure allows the agent to refine the fitting results progressively rather than relying on a single-pass optimization.

10.6 Hyperfine-Assisted Spectral Assignment

When approximate hyperfine coefficients are available from previous studies, theoretical transition frequencies can be calculated using the standard hyperfine Hamiltonian. The agent therefore generates theoretical stick spectra based on the known magnetic dipole (A) and electric quadrupole (B) coefficients together with a global frequency offset.

The measured spectral features are subsequently compared with the predicted transition patterns to perform preliminary spectral assignment. This

procedure is particularly useful for partially resolved fermionic spectra in which direct identification from the experimental signal alone is difficult.

In the present implementation, the hyperfine coefficients and frequency offset can be adjusted interactively to improve agreement between theory and experiment. The fitting procedure is intended primarily for transition identification and consistency checks rather than precision determination of hyperfine constants.

10.7 Current Limitations

At the current stage, the analysis agent should be regarded as a semi-automated assistant rather than a fully autonomous fitting system. For clean spectra with well-isolated dispersive features, the agent can reliably identify candidate zero crossings and perform preliminary fitting. However, for spectra with strong residual amplitude modulation (RAM), baseline fluctuations, low signal-to-noise ratio, or severe spectral overlap, the automatic decision process may become ambiguous.

In particular, the present zero-crossing detection agent is still not capable of perfectly identifying all physically meaningful spectral features in complicated spectra. For spectra containing strong overlap, residual amplitude modulation, or distorted baselines, certain candidate zero crossings still require manual selection and validation by the user.

Nevertheless, compared with the previous fully manual workflow – in which fitting parameters, detection thresholds, and scan regions were adjusted repeatedly through trial-and-error in Python scripts – the present agent-assisted framework provides a significantly more intuitive and efficient analysis process.

Instead of manually tuning large numbers of fitting parameters, the user can directly inspect candidate transitions, residual structures, and possible spectral assignments generated by the agent. This substantially reduces repetitive fitting work and improves the interpretability of the overall spectroscopy analysis workflow.

Therefore, although the current implementation still requires partial human supervision, the agent already serves as a practical tool for accelerating spectroscopy analysis and assisting preliminary transition identification.

Two approaches can improve the reliability of the analysis. First, the experimental spectra should be optimized before being processed by the agent. In practice, this requires reducing RAM, improving pump–probe alignment, suppressing electronic noise, and recording spectra with sufficient signal-to-noise ratio. Providing cleaner spectra reduces the number of false zero crossings and improves the

stability of the fitting procedure.

Second, the agent can be improved by incorporating more spectra into the training or rule-refinement process. By analyzing a larger variety of spectra, including clean, noisy, overlapping, and weak-signal cases, the decision rules for candidate selection and peak rejection can be further optimized.

The present implementation also shows limitations in hyperfine-assisted spectral assignment. Although known A and B hyperfine coefficients can be used to generate theoretical stick spectra, the automatic optimization of the coefficients and global offset does not always converge to a unique physical solution, especially when the experimental spectrum contains unresolved or missing transitions.

Therefore, a more practical strategy is to allow the agent to generate several physically plausible assignment candidates rather than forcing a single automatic decision. In this human-in-the-loop approach, the agent provides multiple possible spectral assignments together with fitting scores, residuals, and predicted transition labels. The user can then select the most physically reasonable solution based on experimental knowledge, after which the agent performs the final fitting and export.

This strategy preserves the efficiency of automated analysis while avoiding over-reliance on ambiguous automatic decisions.

10.8 Conclusion

In this work, an AI-assisted analysis framework for modulation transfer spectroscopy was developed to assist repetitive spectroscopy analysis tasks. The framework reformulates part of the spectroscopy reasoning process into modular computational procedures including candidate detection, iterative fitting, residual analysis, confidence evaluation, and hyperfine-assisted spectral assignment.

Rather than replacing physical interpretation, the present implementation is intended to function as a semi-automated analysis assistant capable of accelerating preliminary transition identification while preserving experimental interpretability and manual validation.

The present results demonstrate that spectroscopy analysis workflows can be partially formalized into structured computational pipelines. Future improvements may include probabilistic assignment methods, Bayesian inference, machine-learning-assisted spectral classification, and multi-transition global fitting across different ytterbium spectroscopy datasets.

Conclusion and outlook

In this thesis, precision spectroscopic techniques and related analysis tools were developed to support the development of an ultracold metastable-state ytterbium platform. By employing modulation transfer spectroscopy (MTS), high-resolution and long-term stable spectroscopic measurements were performed on several optical transitions relevant to laser cooling, repumping, and imaging applications, providing reliable frequency references for ytterbium experiments.

Experimental investigations were carried out on the 680 nm, 1798 nm, 399 nm, and 493 nm transitions. Through systematic measurements and analyses, hyperfine structures, isotope shifts, and spectroscopic characteristics were studied. In particular, isotope-selective optical pumping was employed in the studies of the 1798 nm and 493 nm transitions to distinguish bosonic and fermionic resonances and to improve the reliability of spectral assignments. Long-term stabilized measurements further enabled accurate determination of transition frequencies and hyperfine information.

By combining the measured isotope shifts with previously reported data, King-plot analyses were performed to examine the consistency among different transitions. Besides validating the measured results, these analyses allowed the transition frequency of the rare isotope ^{168}Yb to be extrapolated and established a network connecting multiple transitions relevant to laser cooling and manipulation of ytterbium atoms. These results provide additional spectroscopic references and contribute to a better understanding of isotope-dependent effects in ytterbium.

In addition to spectroscopic measurements, several approaches for improving the efficiency and versatility of the experimental platform were explored. An interactive optical-system modeling platform was developed to facilitate optical design

and component analysis. Furthermore, an AI-assisted analysis agent for modulation transfer spectroscopy was proposed to automate zero-crossing detection, spectral identification, and fitting procedures. These developments demonstrate how computational methods and artificial intelligence can complement conventional spectroscopy and enhance experimental efficiency.

Overall, the results presented in this thesis provide new spectroscopic references and practical analysis tools for ytterbium experiments. The combination of precision spectroscopy, isotope-selective optical pumping, and computational techniques establishes a foundation for future studies involving laser cooling, optical frequency standards, and quantum technologies. In particular, the methodologies developed in this work offer a flexible framework for characterizing atomic transitions and analyzing complex spectroscopic signals, thereby extending the capabilities of ytterbium-based experimental platforms.

Future work may focus on improving the signal-to-noise ratio and frequency calibration accuracy to further reduce the uncertainties of isotope-shift measurements. Additional optical transitions may be incorporated into the King-plot network to investigate higher-order effects and possible deviations from linearity. Further studies on metastable-state cooling and imaging schemes may also provide more efficient approaches for atom manipulation. Moreover, the AI-assisted spectroscopy framework may be extended to more complicated spectra and integrated with automated experimental control systems.

Through the integration of experimental techniques and intelligent analysis tools, the ytterbium platform developed in this work is expected to provide a versatile foundation for future advances in precision spectroscopy, atomic physics, and quantum technologies. It is hoped that the spectroscopic references, analysis methods, and computational tools presented in this thesis will contribute to the continuing development of ytterbium-based experiments and inspire further applications of intelligent methodologies in precision measurement research.

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